Linear Algebra II

Lecture notes, Chapters 9–16

Based on Lee In-Suk, *Linear Algebra and Groups*

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Chapter 9. Rigid Motions of Euclidean Space

Rigid motions of Euclidean space motivate the inner product spaces, groups and forms of later chapters.

9.1 The dot product on $\mathbb{R}^n$

Motivation. The dot product gives $\mathbb{R}^n$ length, distance, angle and perpendicularity.

Definition 9.1.1 (Dot product)

For $X = (a_1, \ldots, a_n)^t$, $Y = (b_1, \ldots, b_n)^t \in \mathbb{R}^n$,

$$\langle X, Y \rangle = X^t \cdot Y = \sum_{i=1}^{n} a_i b_i$$

is the **dot product** on $\mathbb{R}^n$; $\mathbb{R}^n$ with the dot product is **Euclidean space**.

Exercise 9.1.2

(a) $(X, Y) \mapsto \langle X, Y \rangle$ is a bilinear form on $\mathbb{R}^n$ (Definition 6.1.1).
(b) $\langle X, Y \rangle = \langle Y, X \rangle$.
(c) $\langle X, X \rangle \geq 0$, with equality iff $X = 0$.

Solution.

(a) $X^t(aY + bZ) = aX^tY + bX^tZ$, and similarly in X.

(b) $\sum a_i b_i = \sum b_i a_i$.

(c) $\langle X, X \rangle = \sum a_i^2 \geq 0$, and a sum of squares of reals vanishes iff every $a_i = 0$. □

Exercise 9.1.3

If $\langle Y, X \rangle = 0$ for all $Y \in \mathbb{R}^n$, then $X = 0$.

Solution. Take $Y = X$ and use Exercise 9.1.2(c). □

Definition 9.1.4 (Norm)

$\|X\| = \sqrt{\langle X, X \rangle}$ is the (**Euclidean**) **norm**, or length, of X.

Exercise 9.1.5

For $X \in \mathbb{R}^n$, $c \in \mathbb{R}$:
(a) $\|X\| \geq 0$, and $\|X\| = 0$ iff $X = 0$.
(b) $\|cX\| = |c|\|X\|$.
(c) If $X \neq 0$, then $\left\| \frac{1}{\|X\|} X \right\| = 1$. A vector of norm 1 is a **unit vector**.

Solution.

(a) Exercise 9.1.2(c).

(b) $\|cX\|^2 = c^2 \langle X, X \rangle$.

(c) By (b) with $c = 1/\|X\|$. □

Exercise 9.1.6 (Parallelogram law)

$\|X+Y\|^2 + \|X-Y\|^2 = 2(\|X\|^2 + \|Y\|^2)$.

Solution. Expand: $\|X \pm Y\|^2 = \|X\|^2 \pm 2\langle X, Y\rangle + \|Y\|^2$, and add. □

The angle $\theta \in [0, \pi]$ between X and Y is measured in the plane $\langle X\rangle \oplus \langle Y\rangle \cong \mathbb{R}^2$, where

$$\langle X, Y\rangle = \|X\|\|Y\| \cos\theta. \qquad (\angle)$$

Exercise 9.1.7

(a) (Cauchy–Schwarz) $|\langle X, Y\rangle| \le \|X\|\|Y\|$.
(b) (Triangle inequality) $\|X+Y\| \le \|X\| + \|Y\|$.

Solution.

(a) Trivial if $Y = 0$. Otherwise $0 \le \|X - tY\|^2 = \|X\|^2 - 2t\langle X, Y\rangle + t^2\|Y\|^2$ for all t, so the discriminant gives $\langle X, Y\rangle^2 \le \|X\|^2\|Y\|^2$; equality iff X, Y are dependent.

(b) $\|X+Y\|^2 = \|X\|^2 + 2\langle X, Y\rangle + \|Y\|^2 \le (\|X\| + \|Y\|)^2$ by (a). □

So $\|X - Y\|$ is a distance (metric) on $\mathbb{R}^n$:

Exercise 9.1.8

(a) $\|X - X\| = 0$.
(b) $\|X - Y\| = \|Y - X\|$.
(c) $\|X - Y\| \le \|X - Z\| + \|Z - Y\|$.

Solution.

(a) Clear.

(b) Exercise 9.1.5(b) with $c = -1$.

(c) Exercise 9.1.7(b) applied to $(X - Z) + (Z - Y)$. □

Definition 9.1.9 (Perpendicularity)

(a) $X \perp Y$ (X, Y are **perpendicular**, **orthogonal**) if $\langle X, Y\rangle = 0$.
(b) For $S, T \subseteq \mathbb{R}^n$, $S \perp T$ if $\langle X, Y\rangle = 0$ for all $X \in S$, $Y \in T$.

Exercise 9.1.10

The following are equivalent: (1) $X \perp Y$; (2) $\|X+Y\|^2 = \|X\|^2 + \|Y\|^2$ (Pythagoras); (3) $\|X+Y\| = \|X-Y\|$.

Solution. $\|X \pm Y\|^2 = \|X\|^2 + \|Y\|^2 \pm 2\langle X, Y\rangle$. So (2) and (3) both say $\langle X, Y\rangle = 0$. □

Definition 9.1.11 (Orthogonal and orthonormal sets)

(a) Nonzero $X_1, \ldots, X_m \in \mathbb{R}^n$ with $X_i \perp X_j$ for $i \ne j$ form an **orthogonal subset**; if moreover all are unit vectors, an **orthonormal subset**.

(b) For $W \leq \mathbb{R}^n$, a basis $\mathfrak{B}$ of W is an **orthogonal basis** if it is an orthogonal subset, and an **orthonormal basis** if it is orthonormal. Normalizing (Exercise 9.1.5(c)) turns an orthogonal basis into an orthonormal one.

Exercise 9.1.12

(a) An orthogonal subset $\{X_1, \ldots, X_m\}$ of $\mathbb{R}^n$ is linearly independent.
(b) For $W \leq \mathbb{R}^n$, a basis $\{X_1, \ldots, X_m\}$ of W is orthonormal iff $\langle X_i, X_j \rangle = \delta_{ij}$.

Solution.

(a) If $\sum c_i X_i = 0$, then $0 = \langle \sum c_i X_i, X_j \rangle = c_j \|X_j\|^2$, so $c_j = 0$.

(b) This is the definition. □

Exercise 9.1.13

The following are orthonormal bases:
(a) the standard basis $\mathcal{E}$ of $\mathbb{R}^n$;
(b) $\{(1,0)^t, (0,-1)^t\}$ of $\mathbb{R}^2$;
(c) $\{(\cos\theta, \sin\theta)^t, (-\sin\theta, \cos\theta)^t\}$ of $\mathbb{R}^2$;
(d) $\{(\cos\theta, \sin\theta, 0)^t, (-\sin\theta, \cos\theta, 0)^t, (0,0,1)^t\}$ of $\mathbb{R}^3$;
(e) $\{\frac{1}{\sqrt{5}}(2,1)^t, \frac{1}{\sqrt{5}}(-1,2)^t\}$ of $\mathbb{R}^2$;
(f) $\{\frac{1}{\sqrt{2}}(-1,1,0)^t, \frac{1}{\sqrt{3}}(1,1,1)^t, \frac{1}{\sqrt{6}}(1,1,-2)^t\}$ of $\mathbb{R}^3$;
(g) $\{(\cos\theta, \sin\theta, 0, 0)^t, (-\sin\theta, \cos\theta, 0, 0)^t, (0, 0, \cos\eta, \sin\eta)^t, (0, 0, -\sin\eta, \cos\eta)^t\}$ of $\mathbb{R}^4$.

Solution. Distinct vectors have dot product 0 and each has norm 1 ($\cos^2 + \sin^2 = 1$); n orthonormal vectors are independent (Exercise 9.1.12(a)), hence a basis. □

Definition 9.1.14 (Orthogonal complement)

For $S \subseteq \mathbb{R}^n$, $S^\perp = \{X \in \mathbb{R}^n : Y \perp X \text{ for all } Y \in S\}$ ("S perp"). For $W \leq \mathbb{R}^n$, $W^\perp$ is the **orthogonal complement** of W.

Observation 9.1.15

Let $S \subseteq \mathbb{R}^n$, $W \leq \mathbb{R}^n$, and $\mathfrak{C}$ a basis of W.
(a) $S^\perp$ is a subspace of $\mathbb{R}^n$.
(b) $S^\perp = \langle S \rangle^\perp$.
(c) $W^\perp = \mathfrak{C}^\perp$.

Proof.

(a) If $S \perp X$ and $S \perp Y$, then $S \perp X + Y$ and $S \perp cX$.

(b) $S \subseteq \langle S \rangle$ gives $\langle S \rangle^\perp \subseteq S^\perp$; conversely $\langle S \rangle$ consists of linear combinations of S (Proposition 3.1.6).

(c) The case $S = \mathfrak{C}$ of (b). ■

Exercise 9.1.16

Let $\mathfrak{B}$ be a basis of $\mathbb{R}^n$ and $W \leq \mathbb{R}^n$.
(a) $0^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \mathfrak{B}^\perp = 0$.
(b) $W \cap W^\perp = 0$.

(c) $W \leq (W^\perp)^\perp$.

Solution.

(a) Every X is perpendicular to 0; $(\mathbb{R}^n)^\perp = 0$ by Exercise 9.1.3, and $\mathfrak{B}^\perp = (\mathbb{R}^n)^\perp$ by Observation 9.1.15(c).

(b) $X \in W \cap W^\perp$ gives $\langle X, X\rangle = 0$.

(c) Each $X \in W$ is perpendicular to every vector of $W^\perp$. □

Question 9.1.17

If $W \leq \mathbb{R}^n$, is $W = (W^\perp)^\perp$?

There is no direct proof of $(W^\perp)^\perp \leq W$; we compute $\dim W^\perp$.

Example 9.1.18

For $0 \neq Y \in \mathbb{R}^n$, the linear map $\varphi_Y : \mathbb{R}^n \to \mathbb{R}$, $\varphi_Y(X) = \langle Y, X\rangle = Y^tX = L_{Y^t}(X)$, is surjective, since $\varphi_Y(Y) \neq 0$. By the dimension theorem,

$$\dim\langle Y\rangle^\perp = \dim\ker\varphi_Y = n - \dim\operatorname{im}\varphi_Y = n - 1.$$

So $\dim W^\perp = n - \dim W$ holds when $\dim W = 1$.

If $\dim W^\perp = n - \dim W$ for all W, then $\dim(W^\perp)^\perp = \dim W$, so $W = (W^\perp)^\perp$ by Exercise 9.1.16(c) and dimension count, and $\mathbb{R}^n = W \oplus W^\perp$ by Exercise 9.1.16(b) and Observation 7.6.4.

Question 9.1.19

If $W \leq \mathbb{R}^n$, do we always have $\dim W^\perp = n - \dim W$ and $\mathbb{R}^n = W \oplus W^\perp$?

Yes, by the rank theorem (§10.3).

9.2 Rigid motions of $\mathbb{R}^n$

Motivation. Every rigid motion is a translation composed with a *linear* rigid motion, hence a bijection.

Definition 9.2.1 (Rigid motion)

$M : \mathbb{R}^n \to \mathbb{R}^n$ is a **rigid motion** (**isometry**) of $\mathbb{R}^n$ if $\|M(X) - M(Y)\| = \|X - Y\|$ for all $X, Y \in \mathbb{R}^n$.

Example 9.2.2

For $Y \in \mathbb{R}^n$, the **translation** by Y, $T_Y(X) = X + Y$, is a rigid motion; T_Y is a bijection with $(T_Y)^{-1} = T_{-Y}$, and $T_0 = I$.

Example 9.2.3

For $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, the **rotation** L_{R_θ} by θ about the origin is a linear rigid motion of $\mathbb{R}^2$ (Exercise 4.3.5); write $L_{R_\theta} = R_\theta$, a bijection with $(R_\theta)^{-1} = R_{-\theta}$, $R_0 = I$.

Example 9.2.4

$S = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ gives a linear rigid motion $L_S = S$ of $\mathbb{R}^2$ (Exercise 4.3.6), a bijection with $S^{-1} = S$. (It is a reflection; see §9.4.)

Observation 9.2.5

A rigid motion M is injective.

Proof. $M(X) = M(Y)$ gives $\|X - Y\| = 0$, so $X = Y$. ■

Question 9.2.6

Is every rigid motion of $\mathbb{R}^n$ a bijection?

Yes: Corollary 9.2.12.

Exercise 9.2.7

(a) A composite of rigid motions is a rigid motion.
(b) The inverse of a bijective rigid motion is a rigid motion.

Solution.

(a) $\|MN(X) - MN(Y)\| = \|N(X) - N(Y)\| = \|X - Y\|$.

(b) Put $X = M(X')$, $Y = M(Y')$: $\|M^{-1}X - M^{-1}Y\| = \|X' - Y'\| = \|MX' - MY'\| = \|X - Y\|$. □

For a rigid motion M, $L = T_{-M(0)} \circ M$ is a rigid motion with $L(0) = 0$ and $M = T_{M(0)} \circ L$.

Observation 9.2.8

A rigid motion L of $\mathbb{R}^n$ with $L(0) = 0$ satisfies:
(a) $\|L(X)\| = \|X\|$;
(b) $\langle L(X), L(X)\rangle = \langle X, X\rangle$;
(c) $\langle L(X), L(Y)\rangle = \langle X, Y\rangle$;
(d) if $\mathfrak{B}$ is an orthonormal basis of $\mathbb{R}^n$, so is $L(\mathfrak{B})$.

Proof.

(a) Put $Y = 0$ in $\|L(X) - L(Y)\| = \|X - Y\|$.

(b) Square (a).

(c) Square $\|L(X) - L(Y)\| = \|X - Y\|$ and expand: $\langle LX, LX\rangle - 2\langle LX, LY\rangle + \langle LY, LY\rangle = \langle X, X\rangle - 2\langle X, Y\rangle + \langle Y, Y\rangle$; by (b), $\langle LX, LY\rangle = \langle X, Y\rangle$.

(d) If $\mathfrak{B} = \{X_i\}$, then $\langle L(X_i), L(X_j)\rangle = \langle X_i, X_j\rangle = \delta_{ij}$ by (c), so $L(\mathfrak{B})$ is an orthonormal set of n vectors, hence a basis (Exercise 9.1.12). ■

Observation 9.2.9

If the columns $[A]^1, \ldots, [A]^n$ of $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ form an orthonormal basis of $\mathbb{R}^n$, then L_A is a rigid motion.

Proof. For $X = \sum a_i e_i$, $AX = \sum a_i Ae_i = \sum a_i [A]^i$, so $\|AX\|^2 = \sum_{i,j} a_i a_j \delta_{ij} = \sum a_i^2 = \|X\|^2$. Replacing X by $X - Y$ gives $\|AX - AY\| = \|X - Y\|$. ■

Observation 9.2.10

If a rigid motion L with $L(0) = 0$ fixes every vector of a basis $\mathfrak{B} = \{X_i\}$ of $\mathbb{R}^n$, then $L = I$.

Proof. For $X \in \mathbb{R}^n$, $\langle X_i, L(X)\rangle = \langle L(X_i), L(X)\rangle = \langle X_i, X\rangle$ by Observation 9.2.8(c). So $L(X) - X \in \mathfrak{B}^\perp = 0$ (Exercise 9.1.16(a)). ■

Theorem 9.2.11

(a) A rigid motion L of $\mathbb{R}^n$ with $L(0) = 0$ is linear.
(b) Hence every rigid motion M of $\mathbb{R}^n$ is a translation composed with a linear rigid motion.

Proof.

(a) Put $X_j = L(e_j)$, $A = (X_1, \ldots, X_n)$. By Observation 9.2.8(d) $\{X_j\}$ is an orthonormal basis, so A is invertible (Chapter 5). Then $M' = (L_A)^{-1} \circ L$ is a rigid motion (Observation 9.2.9, Exercise 9.2.7) with $M'(0) = 0$, $M'(e_j) = A^{-1} X_j = e_j$, so $M' = I$ by Observation 9.2.10: $L = L_A$.

(b) With $L = T_{-M(0)} \circ M$, L is linear by (a) and $M = T_{M(0)} \circ L$. ■

Second proof of (a). By Observation 9.2.8(c), for all j,

$$\langle L(e_j), L(X + Y)\rangle = \langle e_j, X + Y\rangle = \langle e_j, X\rangle + \langle e_j, Y\rangle = \langle L(e_j), L(X) + L(Y)\rangle.$$

Since $L(\mathcal{E})$ is a basis (Observation 9.2.8(d)), $L(X + Y) - L(X) - L(Y) \in L(\mathcal{E})^\perp = 0$. Likewise $\langle L(e_j), L(cX)\rangle = c\langle e_j, X\rangle = \langle L(e_j), cL(X)\rangle$ gives $L(cX) = cL(X)$. ■

Corollary 9.2.12

Every rigid motion of $\mathbb{R}^n$ is a bijection.

Proof. A linear rigid motion is injective (Observation 9.2.5), hence bijective (dimension theorem); combine with Theorem 9.2.11(b). ■

Exercise 9.2.13

Let $\mathcal{A}(\mathbb{R}^n)$ be the set of rigid motions of $\mathbb{R}^n$. (a) If $L, M \in \mathcal{A}(\mathbb{R}^n)$, then $L \circ M \in \mathcal{A}(\mathbb{R}^n)$.
(b) If $L \in \mathcal{A}(\mathbb{R}^n)$, then $L^{-1} \in \mathcal{A}(\mathbb{R}^n)$.

Solution. Exercise 9.2.7, with Corollary 9.2.12 for (b). □

So $\mathcal{A}(\mathbb{R}^n)$ is a **group** (Chapter 11).

Exercise 9.2.14

Let $L_A, L_B \in \mathcal{A}(\mathbb{R}^n)$ and $Y, Z \in \mathbb{R}^n$.
(a) If $T_Y \circ L_A = T_Z \circ L_B$, then $A = B$ and $Y = Z$. (So every element of $\mathcal{A}(\mathbb{R}^n)$ is uniquely

$T_Y \circ L_A$.)
(b) $L_A \circ T_Y = T_{AY} \circ L_A$.

Solution.

(a) Evaluate at 0: $Y = Z$. Then $L_A = L_B$, so $A = B$.

(b) $A(X+Y) = AX + AY$. □

9.3 Orthogonal operators and orthogonal matrices

Motivation. Linear rigid motions are the orthogonal operators, with matrices characterised by $A^tA = I$.

Definition 9.3.1 (Orthogonal operators and matrices)

(a) A *linear* rigid motion of $\mathbb{R}^n$ is a (real) **orthogonal operator**; if $L = L_A$ is one, the real $n \times n$ matrix A is an **orthogonal matrix**.
(b) The **orthogonal group** on $\mathbb{R}^n$ is $\mathbf{O}(\mathbb{R}^n) = \{L \in \mathfrak{L}(\mathbb{R}^n, \mathbb{R}^n) : \|L(X) - L(Y)\| = \|X - Y\| \ \forall X, Y\}$, and

$$\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : \|AX - AY\| = \|X - Y\| \ \forall X, Y \in \mathbb{R}^n\}.$$

Corollary 9.3.2

For every rigid motion M of $\mathbb{R}^n$ there are unique $A \in \mathbf{O}(n)$, $B \in \mathbb{R}^n$ with $M(X) = AX + B$.

Proof. In Theorem 9.2.11, $M = T_{M(0)} \circ L$ with $L = L_A$ linear; put $B = M(0)$. Uniqueness is Exercise 9.2.14(a). ■

Exercise 9.3.3

(a) If $L, M \in \mathbf{O}(\mathbb{R}^n)$, then $L \circ M \in \mathbf{O}(\mathbb{R}^n)$; if $A, B \in \mathbf{O}(n)$, then $AB \in \mathbf{O}(n)$.
(b) If $L \in \mathbf{O}(\mathbb{R}^n)$, then $L^{-1} \in \mathbf{O}(\mathbb{R}^n)$; if $A \in \mathbf{O}(n)$, then $A^{-1} \in \mathbf{O}(n)$.

Solution. Composites and inverses of linear rigid motions are linear rigid motions (Exercise 9.2.13); $L_AL_B = L_{AB}$ and $(L_A)^{-1} = L_{A^{-1}}$. □

Observation 9.3.4

For linear $L : \mathbb{R}^n \to \mathbb{R}^n$ the following are equivalent:
(1) $\|LX - LY\| = \|X - Y\|$ for all X, Y (i.e. $L \in \mathbf{O}(\mathbb{R}^n)$);
(2) $\|LX\| = \|X\|$ for all X;
(3) $\langle LX, LX\rangle = \langle X, X\rangle$ for all X;
(4) $\langle LX, LY\rangle = \langle X, Y\rangle$ for all X, Y;
(5) if $\mathfrak{B}$ is an orthonormal basis of $\mathbb{R}^n$, so is $L(\mathfrak{B})$.

Proof. (1)⇒(2): $Y = 0$. (2)⇒(1): replace X by $X - Y$, using $L(X - Y) = LX - LY$. (2)⇔(3): square. (3)⇒(4): replace X by $X - Y$ (or $X + Y$) and expand, as in Observation 9.2.8(c). (4)⇒(3): clear. (4)⇒(5): $\langle LX_i, LX_j\rangle = \langle X_i, X_j\rangle = \delta_{ij}$. (5)⇒(2): for an orthonormal basis $\mathfrak{B} = \{X_i\}$ and $X = \sum a_iX_i$, $\|LX\|^2 = \|\sum a_iLX_i\|^2 = \sum a_i^2 = \|X\|^2$. ■

Exercise 9.3.5

The conditions of Observation 9.3.4 are also equivalent to
$(2)'$ if $\|X\| = 1$, then $\|LX\| = 1$ (L preserves unit vectors);
$(5)'$ some orthonormal basis $\mathfrak{B}$ has $L\mathfrak{B}$ orthonormal.

Solution. (2)⇒$(2)'$ and (5)⇒$(5)'$ are clear. $(2)'$ ⇒(2): for $X \neq 0$, $\|LX\| = \|X\| \left\|L(\frac{X}{\|X\|})\right\| = \|X\|$.
$(5)'$ ⇒(2): the last step of the proof of Observation 9.3.4 uses only one orthonormal basis. □

Observation 9.3.6

For $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ the following are equivalent:
(1) $\|AX - AY\| = \|X - Y\|$ for all X, Y (i.e. $A \in \mathbf{O}(n)$);
(2) $\|AX\| = \|X\|$;
(3) $\langle AX, AX\rangle = \langle X, X\rangle$;
(4) $\langle AX, AY\rangle = \langle X, Y\rangle$;
(5) the columns $\{[A]^1, \ldots, [A]^n\}$ form an orthonormal basis of $\mathbb{R}^n$;
(6) $A^tA = I = AA^t$, i.e. $A^{-1} = A^t$;
(7) $A^t \in \mathbf{O}(n)$;
(8) the rows $\{[A]_1, \ldots, [A]_n\}$ form an orthonormal basis of $\mathfrak{M}_{1,n}(\mathbb{R})$.

Proof. (1)⇔(2)⇔(3)⇔(4)⇔(5): Observation 9.3.4 for $L = L_A$ with $\mathfrak{B} = \mathcal{E}$, using $[A]^i = Ae_i$.

(5)⇔(6): $(A^tA)_{ij} = \langle [A]^i, [A]^j\rangle$; for square A, $A^tA = I$ implies $AA^t = I$.

(6)⇒(7) [Supplement]: $(A^t)^tA^t = AA^t = I$, so A^t satisfies (6). (7)⇒(6): apply (6) to A^t.

(7)⇔(8) [Supplement]: the rows of A are the transposed columns of A^t; apply (5) to A^t. ■

Second proof of (4)⇔(6). (4) says $X^tY = (AX)^tAY = X^t(A^tA)Y = X^tIY$ for all X, Y, which by Exercise 9.3.7 means $A^tA = I$. The converse reads the computation backwards. ■

Exercise 9.3.7

If $B, C \in \mathfrak{M}_{m,n}(F)$ and $X^tBY = X^tCY$ for all $X \in F^m$, $Y \in F^n$, then $B = C$.

Solution. $e_i^tBe_j = B_{ij}$; take $X = e_i$, $Y = e_j$. □

Hence

$$\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : A^tA = I\},$$

and with $(L_A)^t = L_{A^t}$, $\mathbf{O}(\mathbb{R}^n) = \{L : L^t \circ L = I\}$.

Exercise 9.3.8

If $A \in \mathbf{O}(n)$, then $\det A = \pm 1$.

Solution. $1 = \det(A^tA) = \det(A^t) \det A = (\det A)^2$. □

Definition 9.3.9 (Special orthogonal group)

$\mathbf{SO}(n) = \{A \in \mathbf{O}(n) : \det A = 1\}$ and $\mathbf{SO}(\mathbb{R}^n) = \{L \in \mathbf{O}(\mathbb{R}^n) : \det L = 1\}$.

L_A is **orientation preserving** if $A \in \mathbf{SO}(n)$, **orientation reversing** if $A \in \mathbf{O}(n) - \mathbf{SO}(n)$.

Exercise 9.3.10

(a) $I \in \mathbf{SO}(n)$.
(b) If $A, B \in \mathbf{SO}(n)$, then $AB \in \mathbf{SO}(n)$.
(c) If $A \in \mathbf{SO}(n)$, then $A^{-1} \in \mathbf{SO}(n)$.

Solution. Exercise 9.3.3 together with $\det(AB) = \det A \det B$ and $\det A^{-1} = (\det A)^{-1}$. □

Exercise 9.3.11

Fix $S \in \mathbf{O}(n) - \mathbf{SO}(n)$.
(a) Indeed $\mathbf{O}(n) - \mathbf{SO}(n) \neq \emptyset$.
(b) $\lambda_S : \mathbf{SO}(n) \to S \cdot \mathbf{SO}(n) = \{SA : A \in \mathbf{SO}(n)\}$, $\lambda_S(A) = SA$, is a bijection.
(c) $\mathbf{O}(n) - \mathbf{SO}(n) = S \cdot \mathbf{SO}(n)$.
(d) $\mathbf{O}(n) = \mathbf{SO}(n) \amalg (S \cdot \mathbf{SO}(n))$ (disjoint union).

Solution.

(a) $\operatorname{diag}(-1, 1, \ldots, 1)$.

(b) $SA = SB$ implies $A = B$; surjective by definition.

(c) If $R \in \mathbf{O}(n) - \mathbf{SO}(n)$, then $S^{-1}R \in \mathbf{SO}(n)$ ($\det = (-1)(-1)$), so $R = S(S^{-1}R)$; conversely $\det(SA) = -1$.

(d) Combine (c) with $\mathbf{SO}(n) \cap (\mathbf{O}(n) - \mathbf{SO}(n)) = \emptyset$. □

9.4 Reflections

Motivation. In an orthonormal frame starting with $Y/\|Y\|$, the reflection in $\langle Y \rangle^\perp$ is $\operatorname{diag}(-1, 1, \ldots, 1)$.

Example 9.4.1

For $X, Y \in \mathbb{R}^3$, $Y \neq 0$, write $X = aY + (X - aY)$ with $(X - aY) \perp Y$; then $a\langle Y, Y \rangle = \langle X, Y \rangle$, so $a = \frac{\langle X,Y \rangle}{\langle Y,Y \rangle}$. Reflecting X in the plane $\langle Y \rangle^\perp$ gives $X - \frac{2\langle X,Y \rangle}{\langle Y,Y \rangle} Y$.

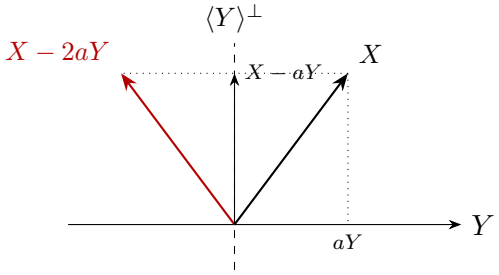


Figure. *Reflection of X in the hyperplane $\langle Y \rangle^\perp$.*

Definition 9.4.2 (Reflection)

For $0 \neq Y \in \mathbb{R}^n$, $S_Y : \mathbb{R}^n \to \mathbb{R}^n$,

$$S_Y(X) = X - \frac{2\langle X, Y \rangle}{\langle Y, Y \rangle} Y,$$

is the **reflection** (**symmetry**) with respect to Y.

Remark 9.4.3

The "reflection in the x-axis" of $\mathbb{R}^2$ is $S_{(0,1)^t}$ in this terminology.

Exercise 9.4.4

For $0 \neq Y \in \mathbb{R}^n$:
(a) $S_Y = S_{cY}$ for $0 \neq c \in \mathbb{R}$;
(b) $S_Y(Y) = -Y$;
(c) if $X \perp Y$, i.e. $X \in \langle Y \rangle^\perp$, then $S_Y(X) = X$.

Solution.
(a) $\frac{\langle X, cY \rangle}{\langle cY, cY \rangle} cY = \frac{c^2 \langle X, Y \rangle}{c^2 \langle Y, Y \rangle} Y$.
(b) $Y - 2Y = -Y$.
(c) $\langle X, Y \rangle = 0$. □

Exercise 9.4.5

$S_Y \in \mathbf{O}(\mathbb{R}^n)$.

Solution. S_Y is linear because $X \mapsto \langle X, Y \rangle$ is. And $\|S_Y X\|^2 = \|X\|^2 - \frac{4\langle X,Y\rangle^2}{\langle Y,Y\rangle} + \frac{4\langle X,Y\rangle^2}{\langle Y,Y\rangle^2}\langle Y, Y\rangle = \|X\|^2$. Apply Observation 9.3.4. □

By Example 9.1.18 and Observation 7.6.4, $\mathbb{R}^n = \langle Y \rangle \oplus \langle Y \rangle^\perp$. For a basis $\mathfrak{C} = \{X_2, \ldots, X_n\}$ of $\langle Y \rangle^\perp$, $\mathfrak{B} = \{Y, X_2, \ldots, X_n\}$ is a basis of $\mathbb{R}^n$, and Exercise 9.4.4(b),(c) give

$$[S_Y]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, 1, \ldots, 1).$$

Exercise 9.4.6

For $0 \neq Y \in \mathbb{R}^n$, $(S_Y)^2 = I$ and $(S_Y)^{-1} = S_Y$.

Solution. $[S_Y^2]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, 1, \ldots, 1)^2 = I$. □

Exercise 9.4.7

For $0 \neq Y \in \mathbb{R}^n$, $\det(S_Y) = -1$.

Solution. $\det S_Y = \det [S_Y]_{\mathfrak{B}}^{\mathfrak{B}} = \det \operatorname{diag}(-1, 1, \ldots, 1) = -1$. □

As a matrix, $S_Y = L_A$ for a unique $A \in \mathbf{O}(n)$, and

$$A = [S_Y]_{\mathcal{E}}^{\mathcal{E}} = [I]_{\mathcal{E}}^{\mathfrak{B}} \cdot [S_Y]_{\mathfrak{B}}^{\mathfrak{B}} \cdot [I]_{\mathfrak{B}}^{\mathcal{E}} = [I]_{\mathcal{E}}^{\mathfrak{B}} \cdot \operatorname{diag}(-1, 1, \ldots, 1) \cdot [I]_{\mathfrak{B}}^{\mathcal{E}}.$$

$[I]_{\mathfrak{B}}^{\mathcal{E}} \in \mathbf{O}(n)$ iff $\mathfrak{B}$ is orthonormal (Observation 9.3.6(5)); with $\|Y\| = 1$ (Exercise 9.4.4(a)) this means $\mathfrak{C}$ is orthonormal.

Question 9.4.8

Does every $W \leq \mathbb{R}^n$ have an orthonormal basis?

Yes (Gram–Schmidt, next chapter). Then all three factors lie in $\mathbf{O}(n)$ ($B = U^{-1}AU$ with $U \in \mathbf{O}(n)$: *orthogonally similar*): in the frame $\mathfrak{B}$, S_Y flips the Y-axis and fixes the others.

9.5 $\mathbf{O}(2)$ and $\mathbf{SO}(2)$

Motivation. Every orthogonal operator of the plane is a rotation or a reflection, according to its determinant.

Example 9.5.1

$\mathbf{O}(1) = \{1, -1\}$ and $\mathbf{SO}(1) = \{1\}$.

Theorem 9.5.2

The elements of $\mathbf{SO}(2)$ are exactly the 2-dimensional rotations:

$$\mathbf{SO}(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} : 0 \leq \theta < 2\pi \right\} = \left\{ \begin{pmatrix} x & -y \\ y & x \end{pmatrix} : x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\}.$$

High-school proof. Let $A = \left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \in \mathbf{SO}(2)$: $x^2 + y^2 = 1$, $z^2 + w^2 = 1$, $xz + yw = 0$, $xw - yz = 1$. Write $(x, y) = (\cos\theta, \sin\theta)$ and $(z, w) = (\sin\eta, \cos\eta)$. Then $\sin(\theta + \eta) = xz + yw = 0$ and $\cos(\theta + \eta) = xw - yz = 1$, so $\theta + \eta \in 2\pi\mathbb{Z}$. Hence $z = \sin\eta = -\sin\theta$ and $w = \cos\eta = \cos\theta$. ■

Freshman's proof. If $y = 0$, then $x = \pm 1$; $xz = 0$ gives $z = 0$, and $xw = 1$ gives $w = x$: $A = \pm I$, of the stated form. If $y \neq 0$, $w = -\frac{xz}{y}$, and $1 = z^2 + w^2 = \frac{z^2(y^2+x^2)}{y^2} = \frac{z^2}{y^2}$, so $z = \pm y$. If $z = y$, then $w = -x$ and $xw - yz = -(x^2 + y^2) = -1$, impossible; so $z = -y$ and $w = x$. ■

Sophomore's proof. The columns are an orthonormal basis (Observation 9.3.6(5)): $(\cos\theta, \sin\theta)^t$ and then $(-\sin\theta, \cos\theta)^t$ (determinant 1) or $(\sin\theta, -\cos\theta)^t$ (determinant -1). ■

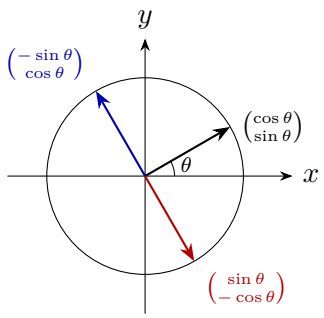


Figure. *The two choices of second basis vector: determinant $+1$ or -1.*

Corollary 9.5.3

$$\mathbf{O}(2) = \mathbf{SO}(2) \amalg \left\{ \begin{pmatrix} x & y \\ y & -x \end{pmatrix} : x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\}.$$

Proof. By Exercise 9.3.11, $\mathbf{O}(2)-\mathbf{SO}(2) = S\cdot\mathbf{SO}(2)$ for any $S\in\mathbf{O}(2)-\mathbf{SO}(2)$. With $S = \left(\begin{smallmatrix}-1 & 0\\ 0 & 1\end{smallmatrix}\right)$, the elements are $S\left(\begin{smallmatrix}x & -y\\ y & x\end{smallmatrix}\right) = \left(\begin{smallmatrix}-x & y\\ y & x\end{smallmatrix}\right)$; replace x by $-x$. ■

Write $R_\theta = \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$ and $S_\theta = \begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{pmatrix}$.

Question 9.5.4

Is S_θ a reflection of $\mathbb{R}^2$?

Question 9.5.5

For reflections S_Y, S_Z of $\mathbb{R}^2$, $\det(S_YS_Z) = (-1)(-1) = 1$, so $S_Y\circ S_Z = R_\theta$ for some θ (Theorem 9.5.2). How do we find θ?

Observation 9.5.6

For $\theta\in\mathbb{R}$ and $Y_\theta = (-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^t$, $S_\theta = S_{Y_\theta}$. In particular S_θ is a reflection.

Proof. Compare on e_1, e_2, with $s = \sin\frac{\theta}{2}$, $c = \cos\frac{\theta}{2}$: $S_{Y_\theta}(e_1) = e_1 - 2\langle e_1, Y_\theta\rangle Y_\theta = e_1 + 2s(-s,c)^t = (1-2s^2, 2sc)^t = (\cos\theta, \sin\theta)^t = S_\theta e_1$, and $S_{Y_\theta}(e_2) = e_2 - 2c(-s,c)^t = (2sc, 1-2c^2)^t = (\sin\theta, -\cos\theta)^t = S_\theta e_2$ (half-angle formulas). ■

Theorem 9.5.7

An orthogonal operator L of $\mathbb{R}^2$ is a rotation or a reflection: a rotation if $\det L = 1$, a reflection if $\det L = -1$.

Proof. $\mathbf{O}(2) = \mathbf{SO}(2)\amalg(\mathbf{O}(2)-\mathbf{SO}(2)) = \{R_\theta\}\amalg\{S_\theta\}$ (Corollary 9.5.3), and each S_θ is a reflection (Observation 9.5.6). ■

Example 9.5.8

(a) For a rotation R and a reflection S of $\mathbb{R}^2$, $R\circ S$ and $S\circ R$ are reflections (determinant -1).
(b) For reflections S, S', $S\circ S'$ is a rotation (determinant 1).

$R_\theta\circ R_\eta = R_{\theta+\eta}$, $(R_\theta)^{-1} = R_{-\theta}$, and $(S_\theta)^2 = I$ (Exercise 9.4.6).

Observation 9.5.9

For a reflection S of $\mathbb{R}^2$ and $\theta\in\mathbb{R}$: $S^{-1}\circ R_\theta\circ S = (R_\theta)^{-1} = R_{-\theta}$, i.e. $R_\theta\circ S = S\circ R_{-\theta}$, or $(S\circ R_\theta)^2 = I$.

Proof. $S\circ R_\theta$ is a reflection (Example 9.5.8), so $(S\circ R_\theta)^2 = I$ by Exercise 9.4.6. Thus $SR_\theta SR_\theta = I$, i.e. $SR_\theta S = (R_\theta)^{-1} = R_{-\theta}$, and $S = S^{-1}$. ■

Example 9.5.10

Fix a reflection S of $\mathbb{R}^2$. By Exercise 9.3.11, $\mathbf{O}(2)-\mathbf{SO}(2) = S\cdot\mathbf{SO}(2)$, so every reflection is uniquely $S\circ R_\theta$, and Observation 9.5.9 moves S past rotations in any product; e.g. $(S\circ R_\theta)\circ(S\circ R_\eta) = S\circ(S\circ R_{-\theta})\circ R_\eta = R_{\eta-\theta}$.

Example 9.5.11

Answer to Question 9.5.5: find θ, η with $S_Y = S_\theta$, $S_Z = S_\eta$, i.e. $\frac{1}{\|Y\|}Y = \pm(-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^t$ and $\frac{1}{\|Z\|}Z = \pm(-\sin\frac{\eta}{2}, \cos\frac{\eta}{2})^t$ (Observation 9.5.6, Exercise 9.4.4(a)). With $S = \left(\begin{smallmatrix}-1 & 0\\ 0 & 1\end{smallmatrix}\right) = S_\pi$ we have $S \circ R_{\pi-\theta} = S_\theta$, so by Observation 9.5.9

$$S_\theta \circ S_\eta = (S \circ R_{\pi-\theta}) \circ (S \circ R_{\pi-\eta}) = S^2 \circ R_{\theta-\pi} \circ R_{\pi-\eta} = R_{\theta-\eta}.$$

Exercise 9.5.12

For $Y = (-1, 1)^t$, $Z = (-1, \sqrt{3})^t$, find θ with $S_Y \circ S_Z = R_\theta$.

Solution. $\frac{1}{\|Y\|}Y = (-\sin\frac{\pi}{4}, \cos\frac{\pi}{4})^t$ and $\frac{1}{\|Z\|}Z = (-\frac{1}{2}, \frac{\sqrt{3}}{2})^t = (-\sin\frac{\pi}{6}, \cos\frac{\pi}{6})^t$, so $S_Y = S_{\pi/2}$, $S_Z = S_{\pi/3}$ and $S_Y \circ S_Z = R_{\pi/2-\pi/3} = R_{\pi/6}$. □

Exercise 9.5.13

$R_\theta \circ S_\eta = S_{\theta+\eta}$ and $S_\eta \circ R_\theta = S_{\eta-\theta}$.

Solution. Addition formulas, column by column: $R_\theta S_\eta e_1 = R_\theta(\cos\eta, \sin\eta)^t = (\cos(\theta+\eta), \sin(\theta+\eta))^t$ and $S_\eta R_\theta e_1 = S_\eta(\cos\theta, \sin\theta)^t = (\cos(\eta-\theta), \sin(\eta-\theta))^t$; likewise for e_2. □

Exercise 9.5.14

For a reflection S of $\mathbb{R}^2$, compute $(S \circ R_{\pi/6} \circ S)^{2000}$.

Solution. $S \circ R_{\pi/6} \circ S = R_{-\pi/6}$ (Observation 9.5.9, $S = S^{-1}$). So the power is $R_{-2000\pi/6} = R_{-1000\pi/3}$, and $-\frac{1000\pi}{3} \equiv \frac{2\pi}{3} \pmod{2\pi}$ ($1000 = 3 \cdot 334 - 2$). The answer is $R_{2\pi/3}$. □

Exercise 9.5.15

Every element of $\mathbf{SO}(2)$ is a product of two reflections, but not uniquely.

Solution. By Example 9.5.11, $S_{\theta+\alpha} \circ S_\alpha = R_\theta$ for every $\alpha \in \mathbb{R}$; different $\alpha \pmod{2\pi}$ give different factorizations. □

9.6 $\mathbf{SO}(3)$ and $\mathbf{SO}(n)$

Motivation. Every element of $\mathbf{SO}(3)$ fixes an axis and rotates the perpendicular plane.

Definition 9.6.1 (Rotation of $\mathbb{R}^3$)

Let $Z \in \mathbb{R}^3$ be a unit vector and $\mathfrak{B} = \{Z, X, Y\}$ an orthonormal basis of $\mathbb{R}^3$ (so $\{X, Y\}$ is an orthonormal basis of $\langle Z\rangle^\perp$). Then $(Z, X, Y) \in \mathbf{O}(3)$ has determinant ± 1; order X, Y so that $\det(Z, X, Y) = \det(X, Y, Z) = 1$. The **rotation** $R_{Z,\theta} : \mathbb{R}^3 \to \mathbb{R}^3$ is the linear map with

$$[R_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \operatorname{diag}(1, R_\theta);$$

Z is the **axis**, $\langle X\rangle \oplus \langle Y\rangle$ the **plane of rotation**, θ the **angle**.

It does not depend on $\{X, Y\}$ (Exercise 9.6.6). As in §9.4,

$$[R_{Z,\theta}]_{\mathcal{E}}^{\mathcal{E}} = [I]_{\mathcal{E}}^{\mathfrak{B}} \cdot \operatorname{diag}(1, R_\theta) \cdot [I]_{\mathfrak{B}}^{\mathcal{E}},$$

where $\operatorname{diag}(1, R_\theta) \in \mathbf{SO}(3)$ and $[I]_{\mathfrak{B}}^{\mathcal{E}} = (Z, X, Y) \in \mathbf{SO}(3)$.

Observation 9.6.2

Rotations $R_{Z,\theta}$ of $\mathbb{R}^3$ belong to $\mathbf{SO}(\mathbb{R}^3)$.

Proof. By the displayed product all three factors lie in $\mathbf{SO}(3)$ (Exercise 9.3.10). ■

Question 9.6.3

Is the composite of two rotations of $\mathbb{R}^3$ (with different axes) a rotation?

Theorem 9.6.4

Every element of $\mathbf{SO}(\mathbb{R}^3)$ is a rotation of $\mathbb{R}^3$.

Proof. (i) *An axis.* Let $L_A \in \mathbf{SO}(\mathbb{R}^3)$. We need a unit vector Z with $AZ = Z$, i.e. $\det(A - I) = 0$. Since $\det A = \det A^t = 1$ and $A^t A = I$,

$$\begin{aligned}\det(A - I) &= \det(A^t)\det(A - I) = \det(A^t(A - I)) = \det(I - A^t) \\ &= \det((I - A)^t) = \det(I - A) = (-1)^3 \det(A - I),\end{aligned}$$

so $\det(A - I) = 0$.

(ii) *The plane.* For an orthonormal $\mathfrak{B} = \{Z, X, Y\}$ with $\det(Z, X, Y) = 1$, $[L_A]_{\mathfrak{B}}^{\mathfrak{B}} = [I]_{\mathcal{E}}^{\mathfrak{B}}\, A\, [I]_{\mathfrak{B}}^{\mathcal{E}} = \left(\begin{smallmatrix} 1 & a & b \\ 0 & & \\ 0 & & B \end{smallmatrix}\right) \in \mathbf{SO}(3)$. Orthonormal columns force $a = b = 0$, $B \in \mathbf{O}(2)$, and $\det B = 1$, so $B = R_\theta$ (Theorem 9.5.2). ■

Second proof of (i). If $A \in \mathbf{O}(3)$ has a real eigenvalue λ with eigenvector X, then $\langle X, X\rangle = \langle AX, AX\rangle = \lambda^2\langle X, X\rangle$, so $\lambda = \pm 1$. Triangularizing A over $\mathbb{C}$ (§7.3), its eigenvalues satisfy $\lambda_1\lambda_2\lambda_3 = \det A = 1$ (Corollary 7.3.3). A real cubic has a real root, say $\lambda_1 = \pm 1$. If $\lambda_2, \lambda_3 \notin \mathbb{R}$, then $\lambda_3 = \bar{\lambda}_2$ and $\lambda_2\lambda_3 = |\lambda_2|^2 > 0$, so $\lambda_1 = 1$. If all are real, they are ± 1 with product 1, so one of them is 1. ■

Corollary 9.6.5

The composite of two rotations of $\mathbb{R}^3$ is a rotation.

Proof. By Observation 9.6.2 and Exercise 9.3.10 the composite lies in $\mathbf{SO}(\mathbb{R}^3)$; apply Theorem 9.6.4. ■

The axis of $L_{A_1} \circ L_{A_2}$ is the unit eigenvector of $A_1 A_2$ for eigenvalue 1.

Exercise 9.6.6

Let $\mathfrak{B}' = \{Z, X', Y'\}$ be another orthonormal basis with $\det(Z, X', Y') = 1$, and $R'_{Z,\theta}$ the rotation defined from it. Show $R_{Z,\theta} = R'_{Z,\theta}$:
(a) since $\langle X\rangle \oplus \langle Y\rangle = \langle X'\rangle \oplus \langle Y'\rangle = \langle Z\rangle^\perp$, $[I]_{\mathfrak{B}'}^{\mathfrak{B}} = \operatorname{diag}(1, U)$ for some $U \in \mathfrak{M}_{2,2}(\mathbb{R})$;
(b) $[I]_{\mathfrak{B}'}^{\mathfrak{B}} = [I]_{\mathcal{E}}^{\mathfrak{B}}[I]_{\mathfrak{B}'}^{\mathcal{E}} \in \mathbf{SO}(3)$, hence $U \in \mathbf{SO}(2)$;
(c) $[R'_{Z,\theta}]_{\mathfrak{B}}^{\mathfrak{B}} = [I]_{\mathfrak{B}'}^{\mathfrak{B}} \operatorname{diag}(1, R_\theta)[I]_{\mathfrak{B}}^{\mathfrak{B}'} = \operatorname{diag}(1, R_\theta) = [R_{Z,\theta}]_{\mathfrak{B}}^{\mathfrak{B}}$.

Solution.

(a) Z is fixed and $X', Y' \in \langle X, Y\rangle$, so the change of coordinates has this block form.

(b) Both factors are in $\mathbf{SO}(3)$ (columns orthonormal, determinant 1), so $\operatorname{diag}(1, U) \in \mathbf{SO}(3)$, whence $U \in \mathbf{O}(2)$ with $\det U = 1$.

(c) $\operatorname{diag}(1, U)\operatorname{diag}(1, R_\theta)\operatorname{diag}(1, U^{-1}) = \operatorname{diag}(1, UR_\theta U^{-1})$, and $UR_\theta U^{-1} = R_\theta$ because $\mathbf{SO}(2)$ is commutative ($R_\alpha R_\theta = R_{\alpha+\theta}$). □

Example 9.6.7

Unlike Theorem 9.5.7: for $R_\theta \neq I$, $\operatorname{diag}(-1, R_\theta) \in \mathbf{O}(3)$ has determinant -1 (not a rotation) and no 2-dimensional 1-eigenspace (not a reflection).

Exercise 9.6.8

For $L \in \mathbf{O}(\mathbb{R}^3) - \mathbf{SO}(\mathbb{R}^3)$:
(a) L has eigenvalue -1;
(b) $[L]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(-1, R_\theta)$ for some orthonormal basis $\mathfrak{B}$ and $\theta \in \mathbb{R}$.

Solution.

(a) With $L = L_A$, $\det A = -1$: $\det(A + I) = -\det(A^t)\det(A + I) = -\det(I + A^t) = -\det(A + I)$, so $\det(A + I) = 0$.

(b) Take a unit eigenvector Z for -1 and an orthonormal basis $\{Z, X, Y\}$. As in Theorem 9.6.4(ii), $[L]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(-1, B)$ with $B \in \mathbf{O}(2)$ and $\det B = 1$, so $B = R_\theta$. □

So in a suitable orthonormal frame every $L \in \mathbf{O}(\mathbb{R}^3)$ is $\operatorname{diag}(1, R_\theta)$ (rotation), $\operatorname{diag}(-1, 1, 1)$ (reflection) or $\operatorname{diag}(-1, R_\eta)$, $R_\eta \neq I$.

Definition 9.6.9 (Rotations of $\mathbb{R}^n$)

For every n, an element of $\mathbf{SO}(\mathbb{R}^n)$ (or of $\mathbf{SO}(n)$) is called a **rotation** of $\mathbb{R}^n$.

Example 9.6.10

The 4-dimensional rotation $\operatorname{diag}(R_\theta, R_\eta)$ has two "planes of rotation" with two angles.

All of $\mathbf{O}(\mathbb{R}^n)$ is described in Chapter 15.

Exercise 9.6.11

(a) $L \in \mathbf{O}(\mathbb{R}^n)$ iff $[L]^{\mathfrak{B}}_{\mathfrak{B}} \in \mathbf{O}(n)$ for some orthonormal basis $\mathfrak{B}$ of $\mathbb{R}^n$.
(b) The same holds with $\mathbf{SO}$ in place of $\mathbf{O}$.

Solution.

(a) $P = [I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{O}(n)$ and $[L]^{\mathfrak{B}}_{\mathfrak{B}} = P^{-1}[L]^{\mathcal{E}}_{\mathcal{E}}P$; $\mathbf{O}(n)$ is closed under products and inverses (Exercise 9.3.3).

(b) Additionally $\det[L]^{\mathfrak{B}}_{\mathfrak{B}} = \det L$. □

Chapter 10. Inner Product Spaces

10.1 Inner product spaces

Motivation. The four properties of the dot product used in Chapter 9 define an inner product on any real or complex space.

Definition 10.1.1 (Hermitian dot product)

For $X = (a_1, \ldots, a_n)^t$, $Y = (b_1, \ldots, b_n)^t \in \mathbb{C}^n$,

$$\langle X, Y \rangle = X^t \cdot \bar{Y} = \sum_{i=1}^{n} a_i \bar{b}_i$$

is the (**Hermitian**) **dot product** of $\mathbb{C}^n$.

Remark 10.1.2

Without conjugation, $(1+i, 1-i)^t$ would have "length" 0 ($(1+i)^2 + (1-i)^2 = 0$) and $(i, 0)^t$ an imaginary "length".

Exercise 10.1.3

For $X, Y, Z \in \mathbb{C}^n$, $c \in \mathbb{C}$: (a) $\langle X+Y, Z\rangle = \langle X, Z\rangle + \langle Y, Z\rangle$, $\langle cX, Y\rangle = c\langle X, Y\rangle$;
(b) $\langle X, Y+Z\rangle = \langle X, Y\rangle + \langle X, Z\rangle$, $\langle X, cY\rangle = \bar{c}\langle X, Y\rangle$;
(c) $\langle X, Y\rangle = \overline{\langle Y, X\rangle}$;
(d) if $X \neq 0$, then $\langle X, X\rangle \in \mathbb{R}$ and $\langle X, X\rangle > 0$.

Solution. (a), (b) from $\overline{b+c} = \bar{b} + \bar{c}$, $\overline{cb} = \bar{c}\bar{b}$.

(c) $\overline{\sum b_i \bar{a}_i} = \sum \bar{b}_i a_i$.

(d) $\langle X, X\rangle = \sum |a_i|^2$. □

Definition 10.1.4 (Inner product on F^n)

($F = \mathbb{R}$ or $\mathbb{C}$.) A function $(X, Y) \mapsto \langle X, Y\rangle$, $F^n \times F^n \to F$, is an **inner product** on F^n if for all $X, Y, Z \in F^n$, $c \in F$: (1) $\langle X+Y, Z\rangle = \langle X, Z\rangle + \langle Y, Z\rangle$; (2) $\langle cX, Y\rangle = c\langle X, Y\rangle$; (3) $\langle X, Y\rangle = \overline{\langle Y, X\rangle}$; (4) if $X \neq 0$, then $\langle X, X\rangle \in \mathbb{R}$ and $\langle X, X\rangle > 0$. (For $F = \mathbb{C}$, also a **Hermitian inner product**.)

Observation 10.1.5

If $\langle \cdot, \cdot \rangle$ is an inner product on F^n: (a) $\langle X, Y+Z\rangle = \langle X, Y\rangle + \langle X, Z\rangle$;
(b) $\langle X, cY\rangle = \bar{c}\langle X, Y\rangle$.

Proof. By (3), $\langle X, Y+Z\rangle = \overline{\langle Y+Z, X\rangle} = \overline{\langle Y, X\rangle} + \overline{\langle Z, X\rangle} = \langle X, Y\rangle + \langle X, Z\rangle$, and $\langle X, cY\rangle = \overline{c\langle Y, X\rangle} = \bar{c}\,\overline{\langle Y, X\rangle} = \bar{c}\langle X, Y\rangle$. ■

Definition 10.1.6 (Inner product space)

Let V be an F-vector space (possibly infinite-dimensional). $\langle\cdot,\cdot\rangle : V \times V \to F$ is an **inner product** if for all $u, v, w \in V$, $c \in F$: (1) $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$; (2) $\langle cv, w\rangle = c\langle v, w\rangle$; (3) $\langle v, w\rangle = \overline{\langle w, v\rangle}$; (4) if $v \neq 0$, then $\langle v, v\rangle \in \mathbb{R}$ and $\langle v, v\rangle > 0$ (*positive definiteness*). V with an inner product is an **inner product space**; for $F = \mathbb{C}$, $\langle\cdot,\cdot\rangle$ is also called **Hermitian**.

Remark 10.1.7

(a) Observation 10.1.5 still holds.
(b) For $F = \mathbb{R}$, conjugation does nothing, so a real inner product is a *symmetric* bilinear form.
(c) For $F = \mathbb{C}$, an inner product is linear in the first variable and **conjugate linear** (anti-linear) in the second.
(d) Every F-subspace of an inner product space is an inner product space.

Exercise 10.1.8

$\langle X, Y\rangle = \sum_{j=1}^{n} j\, a_j b_j$ is an inner product on $\mathbb{R}^n$.

Solution. (1)–(3) are clear; $\langle X, X\rangle = \sum j a_j^2 > 0$ for $X \neq 0$. □

Exercise 10.1.9

$\left\langle \binom{a}{b}, \binom{c}{d} \right\rangle = 2ac - bc - ad + 2bd$ is an inner product on $\mathbb{R}^2$.

Solution. It is bilinear and symmetric, and $\langle (a,b),(a,b)\rangle = 2a^2 - 2ab + 2b^2 = a^2 + b^2 + (a-b)^2 > 0$ for $(a,b) \neq 0$. □

Exercise 10.1.10

If $\langle\cdot,\cdot\rangle_1, \langle\cdot,\cdot\rangle_2$ are inner products on V, so is $\langle v, w\rangle = \langle v, w\rangle_1 + \langle v, w\rangle_2$.

Solution. (1)–(3) are additive; for $v \neq 0$ both summands are positive reals. □

Exercise 10.1.11

$\langle f(t), g(t)\rangle = \int_0^1 f(t)g(t)\,dt$ is an inner product on $\mathbb{R}[t]$.

Solution. Bilinearity and symmetry are clear. If $f \neq 0$, $f^2 \geq 0$ is continuous and positive except at finitely many points, so $\int_0^1 f^2 > 0$. □

For $f = f_1 + i f_2 : \mathbb{R} \to \mathbb{C}$ (real and imaginary parts), $\int_a^b f = \int_a^b f_1 + i \int_a^b f_2$; f is continuous iff f_1, f_2 are.

Exercise 10.1.12

$\langle f(t), g(t)\rangle = \int_0^1 f(t)\overline{g(t)}\,dt$ is an inner product on $\mathbb{C}[t]$.

Solution. As in Exercise 10.1.11, with $\langle f, f\rangle = \int_0^1 |f|^2$ and $\overline{\int f\bar{g}} = \int \bar{f} g$. □

Exercise 10.1.13

$C^0[0, 2\pi] = \{f : [0, 2\pi] \to \mathbb{C} \text{ continuous}\}$ is a $\mathbb{C}$-vector space, and $\langle f, g\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(x)\overline{g(x)}\,dx$ is an inner product on it.

Solution. Sums and multiples of continuous functions are continuous. (1)–(3) as before; if $f \neq 0$, $|f|^2$ is continuous, nonnegative and positive on an interval, so $\langle f, f\rangle > 0$. □

Exercise 10.1.14

$\langle A, B\rangle = \operatorname{tr}(A^t \cdot \bar{B})$ is an inner product on $\mathfrak{M}_{n,n}(F)$ (the **trace form**).

Solution. $\operatorname{tr}(A^t\bar{B}) = \sum_{i,j} A_{ji}\bar{B}_{ji}$ is the dot product of A and B viewed as vectors in F^{n^2}, so Exercise 10.1.3 applies. □

10.2 Properties of inner product spaces

Motivation. Norm, Pythagoras, Cauchy–Schwarz and the triangle inequality hold in any inner product space.

Throughout, V is an inner product space over F.

Exercise 10.2.1

If $\langle w, v\rangle = 0$ for all $w \in V$, then $v = 0$.

Solution. Take $w = v$ and use positive definiteness. □

Definition 10.2.2 (Norm)

$\|v\| = \sqrt{\langle v, v\rangle}$.

Exercise 10.2.3

(a) $\|v\| \geq 0$, with equality iff $v = 0$.
(b) $\|cv\| = |c|\|v\|$.
(c) If $v \neq 0$, $\frac{1}{\|v\|}v$ is a **unit vector**.

Solution. (b) $\langle cv, cv\rangle = c\bar{c}\langle v, v\rangle = |c|^2\langle v, v\rangle$. The rest as in Exercise 9.1.5. □

Exercise 10.2.4 (Parallelogram law)

$\|v + w\|^2 + \|v - w\|^2 = 2(\|v\|^2 + \|w\|^2)$.

Solution. $\|v \pm w\|^2 = \|v\|^2 \pm (\langle v, w\rangle + \langle w, v\rangle) + \|w\|^2$; add. □

Definition 10.2.5 (Perpendicularity)

(a) $v \perp w$ if $\langle v, w\rangle = 0$ (equivalently $\langle w, v\rangle = 0$).
(b) For $S, T \subseteq V$, $S \perp T$ if $\langle v, w\rangle = 0$ for all $v \in S$, $w \in T$.

Exercise 10.2.6

If $v \perp w$: (a) $\|v+w\|^2 = \|v\|^2 + \|w\|^2$ (Pythagoras);
(b) $\|v+w\| = \|v-w\|$.

Solution. In the expansion of Exercise 10.2.4 the cross terms $\langle v,w\rangle + \langle w,v\rangle$ vanish. □

Exercise 10.2.7

If $w \neq 0$, then $\left(v - \frac{\langle v,w\rangle}{\langle w,w\rangle} w\right) \perp w$ (compare Example 9.4.1).

Solution. $\langle v,w\rangle - \frac{\langle v,w\rangle}{\langle w,w\rangle}\langle w,w\rangle = 0$. □

Observation 10.2.8

(a) (Cauchy–Schwarz) $|\langle v,w\rangle| \leq \|v\|\|w\|$, with equality iff $\{v,w\}$ is linearly dependent.
(b) (Triangle inequality) $\|v+w\| \leq \|v\| + \|w\|$.

Proof. (a) Trivial if $w = 0$. Otherwise, with $a = \frac{\langle v,w\rangle}{\langle w,w\rangle}$, Exercise 10.2.7 and Pythagoras give $\|v\|^2 = \|v - aw\|^2 + \|aw\|^2 \geq |a|^2\|w\|^2$, i.e. $|\langle v,w\rangle| \leq \|v\|\|w\|$; equality iff $v = aw$.

(b) Using $z + \bar{z} = 2\operatorname{Re} z \leq 2|z|$, $\|v+w\|^2 = \|v\|^2 + \langle v,w\rangle + \overline{\langle v,w\rangle} + \|w\|^2 = \|v\|^2 + 2\operatorname{Re}\langle v,w\rangle + \|w\|^2 \leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2$. ■

So $\|v-w\|$ is a distance on V:

Exercise 10.2.9

(a) $\|v-v\| = 0$.
(b) $\|v-w\| = \|w-v\|$.
(c) $\|v-w\| \leq \|v-u\| + \|u-w\|$.

Solution. As in Exercise 9.1.8, using Observation 10.2.8(b). □

Definition 10.2.10 (Orthogonal and orthonormal sets)

(a) Nonzero vectors $\{v_i\}_{i\in I}$ with $v_i \perp v_j$ for $i \neq j$ form an **orthogonal subset**; if all are unit vectors, an **orthonormal subset**.
(b) For $W \leq V$, a basis $\mathfrak{B}$ of W is an **orthogonal basis** if it is an orthogonal subset, an **orthonormal basis** if it is orthonormal.

Exercise 10.2.11

An orthogonal subset $\{v_i\}_{i\in I}$ is linearly independent.

Solution. If $\sum_{\text{finite}} c_i v_i = 0$, take the inner product with v_j: $c_j\|v_j\|^2 = 0$. □

Exercise 10.2.12

A basis $\{v_i\}$ of $W \leq V$ is orthonormal iff $\langle v_i, v_j\rangle = \delta_{ij}$.

Solution. This restates the definition. □

Exercise 10.2.13

(a) $\{(0,-i)^t,(i,0)^t\}$ is an orthonormal basis of $\mathbb{C}^2$ (dot product).
(b) $\{\frac{1}{\sqrt{3}}(1+i,i)^t,\frac{1}{\sqrt{6}}(1-i,-2)^t\}$ is an orthonormal basis of $\mathbb{C}^2$.
(c) $\{\frac{1}{\sqrt{2}}(1,0)^t,\frac{1}{\sqrt{6}}(1,2)^t\}$ is an orthonormal basis of $\mathbb{R}^2$ with the inner product of Exercise 10.1.9.
(d) $\{1,\sqrt{12}(t-\frac{1}{2})\}$ is an orthonormal basis of $\mathcal{P}_1(\mathbb{R})$ with the inner product of Exercise 10.1.11.
(e) For $f_n(x)=e^{inx}=\cos nx+i\sin nx$ $(n\in\mathbb{Z})$, $\{f_n\}_{n\in\mathbb{Z}}$ is an orthonormal subset of $C^0[0,2\pi]$ (Exercise 10.1.13).

Solution. (a) Norms 1, and $0\cdot\bar{i}+(-i)\cdot 0=0$. (b) $|1+i|^2+|i|^2=3$, $|1-i|^2+4=6$, and $(1+i)\overline{(1-i)}+i\cdot\overline{(-2)}=(1+i)^2-2i=0$. (c) $\langle(1,0),(1,0)\rangle=2$, $\langle(1,2),(1,2)\rangle=2-2-2+8=6$, $\langle(1,0),(1,2)\rangle=2-0-2+0=0$. (d) $\int_0^1 1=1$, $\int_0^1 12(t-\frac{1}{2})^2dt=1$, $\int_0^1(t-\frac{1}{2})dt=0$. (e) $\frac{1}{2\pi}\int_0^{2\pi}e^{i(m-n)x}dx=\delta_{mn}$, since for $m\neq n$ the antiderivative $\frac{e^{i(m-n)x}}{i(m-n)}$ is 2π-periodic. □

Question 10.2.14

If V is a (finite-dimensional) inner product space and $W\leq V$, does W always have an orthonormal basis?

Definition 10.2.15 (Orthogonal complement)

$S^\perp=\{v\in V: w\perp v \text{ for all } w\in S\}$; for $W\leq V$, $W^\perp$ is the **orthogonal complement**.

Exercise 10.2.16

For $S\subseteq V$, $W\leq V$ with basis $\mathfrak{C}$: (a) $S^\perp\leq V$;
(b) $S^\perp=\langle S\rangle^\perp$;
(c) $W^\perp=\mathfrak{C}^\perp$.

Solution. As in Observation 9.1.15; in (a) use conjugate linearity: $\langle w,cv\rangle=\bar{c}\langle w,v\rangle=0$. □

Exercise 10.2.17

For a basis $\mathfrak{B}$ of V and $W\leq V$: (a) $0^\perp=V$, $V^\perp=\mathfrak{B}^\perp=0$;
(b) $W\cap W^\perp=0$;
(c) $W\leq(W^\perp)^\perp$.

Solution. As in Exercise 9.1.16, using Exercise 10.2.1. □

Question 10.2.18

If V is finite-dimensional and $W\leq V$, is $\dim W^\perp=\dim V-\dim W$ and $V=W\oplus W^\perp$?

A **rigid motion** of V is $M:V\to V$ with $\|M(v)-M(w)\|=\|v-w\|$. For $F=\mathbb{C}$, Theorem 9.2.11 fails, since Observation 9.2.8(c) does.

Exercise 10.2.19

(a) On $\mathbb{C}=\mathbb{C}^1$ with the dot product, $M(\alpha)=\bar{\alpha}$ is a non-linear rigid motion fixing 0 (M is conjugate linear).

(b) On $\mathbb{C}^2$, $M((\alpha,\beta)^t) = (\bar\alpha,\beta)^t$ is a non-linear rigid motion fixing 0, not even conjugate linear.

Solution. (a) $|\bar\alpha-\bar\beta| = |\overline{\alpha-\beta}| = |\alpha-\beta|$, but $M(i\cdot 1) = -i \neq iM(1)$. (b) $\|M(v)-M(w)\|^2 = |\overline{\alpha-\alpha'}|^2 + |\beta-\beta'|^2 = \|v-w\|^2$; $M(i(1,0)^t) = (-i,0)^t \neq iM((1,0)^t)$, and $M(i(0,1)^t) = (0,i)^t \neq \bar iM((0,1)^t)$. □

Remark 10.2.20

The $\mathbb{R}$-isomorphism $\gamma(a+ib) = (a,b)^t$, $\mathbb{C}\to\mathbb{R}^2$, preserves distance, so rigid motions of $\mathbb{C}$ and of $\mathbb{R}^2$ coincide. Conjugation is a reflection of $\mathbb{R}^2$ ($\mathbb{R}$-linear, not $\mathbb{C}$-linear); a $\mathbb{C}$-linear rigid motion of $\mathbb{C}$ (multiplication by $e^{i\theta}$) is a rotation of $\mathbb{R}^2$.

10.3 Gram–Schmidt orthogonalization

Motivation. Subtracting projections onto earlier vectors orthogonalizes any independent list, answering Questions 9.1.19, 9.4.8, 10.2.14, 10.2.18.

Theorem 10.3.1 (Gram–Schmidt orthogonalization)

Let V be an inner product space and $\{v_1,\dots,v_r\}$ linearly independent. (a) Put $w_1 = v_1$ and, for $2\le i\le r$,

$$w_i = v_i - \frac{\langle v_i, w_{i-1}\rangle}{\langle w_{i-1}, w_{i-1}\rangle} w_{i-1} - \cdots - \frac{\langle v_i, w_1\rangle}{\langle w_1, w_1\rangle} w_1.$$

Then $\langle v_1,\dots,v_r\rangle = \langle w_1,\dots,w_r\rangle$ and $\{w_1,\dots,w_r\}$ is an orthogonal subset.
(b) Hence $\{\frac{1}{\|w_1\|}w_1,\dots,\frac{1}{\|w_r\|}w_r\}$ is an orthonormal subset.
(c) In particular, if $r=\dim V$, it is an orthonormal basis of V.

Proof. Induction on i: let $\langle v_1,\dots,v_i\rangle = \langle w_1,\dots,w_i\rangle$ with $\{w_1,\dots,w_i\}$ orthogonal. Then $w_{i+1}\in v_{i+1}+\langle w_1,\dots,w_i\rangle$ gives equal spans at $i+1$; $w_{i+1}\neq 0$ since $v_{i+1}\notin\langle v_1,\dots,v_i\rangle$; and for $j\le i$, $\langle w_{i+1},w_j\rangle = \langle v_{i+1},w_j\rangle - \frac{\langle v_{i+1},w_j\rangle}{\langle w_j,w_j\rangle}\langle w_j,w_j\rangle = 0$. Normalize for (b), (c) (Exercise 10.2.11). ■

Exercise 10.3.2

Apply Gram–Schmidt to the basis $\mathfrak{B}$: (a) $\mathfrak{B} = \{(1,1,0)^t,(-1,0,2)^t,(2,1,1)^t\}$ of $\mathbb{R}^3$ (dot product); (b) $\mathfrak{B} = \{(3,1,0,1)^t,(0,-1,2,0)^t,(1,1,-1,-1)^t\}$ of a subspace W of $\mathbb{R}^4$; (c) $\mathfrak{B} = \{(1,1)^t,(0,1)^t\}$ of $\mathbb{R}^2$ with the inner product of Exercise 10.1.9; (d) $\mathfrak{B} = \{1,t,t^2\}$ of $\mathcal{P}_2(\mathbb{R})$ with the inner product of Exercise 10.1.11.

Solution. (a) $w_1 = (1,1,0)$, $w_2 = (-\frac12,\frac12,2)$, $w_3 = (\frac23,-\frac23,\frac13)$; orthonormal basis $\frac{1}{\sqrt2}(1,1,0)$, $\frac{1}{3\sqrt2}(-1,1,4)$, $\frac13(2,-2,1)$.

(b) $w_1 = (3,1,0,1)$, $w_2 = \frac{1}{11}(3,-10,22,1)$, $w_3 = \frac19(3,2,1,-11)$; orthonormal basis $\frac{1}{\sqrt{11}}(3,1,0,1)$, $\frac{1}{\sqrt{594}}(3,-10,22,1)$, $\frac{1}{\sqrt{135}}(3,2,1,-11)$.

(c) $\langle(1,1),(1,1)\rangle = 2$, $\langle(0,1),(1,1)\rangle = 1$, so $w_2 = (0,1)-\frac12(1,1) = (-\frac12,\frac12)$ with $\langle w_2,w_2\rangle = \frac32$; orthonormal basis $\frac{1}{\sqrt2}(1,1)$, $\frac{1}{\sqrt6}(-1,1)$.

(d) 1, $t-\frac12$, $t^2-t+\frac16$; orthonormal basis 1, $\sqrt3(2t-1)$, $\sqrt5(6t^2-6t+1)$. □

Exercise 10.3.3

If $\{v_1, \dots, v_r\}$ in Theorem 10.3.1 is already orthogonal, then $w_i = v_i$ for all i.

Solution. All the subtracted terms have $\langle v_i, w_j\rangle = \langle v_i, v_j\rangle = 0$, by induction. □

Corollary 10.3.4 (Perp theorem, inner product case)

If V is a finite-dimensional inner product space and $W \le V$, then $\dim V = \dim W + \dim W^\perp$.

Proof. Extend an orthonormal basis $\{w_1, \dots, w_r\}$ of W to a basis of V and apply Gram–Schmidt: by Exercise 10.3.3 this gives an orthonormal basis $\{w_1, \dots, w_n\}$ of V. Then $W^\perp = \langle w_{r+1}, \dots, w_n\rangle$, since $v = \sum a_i w_i \in W^\perp$ iff $a_j = \langle v, w_j\rangle = 0$ for $j \le r$. ■

Corollary 10.3.5

If V is a finite-dimensional inner product space and $W \le V$, then $V = W \oplus W^\perp$.

Proof. $W \cap W^\perp = 0$ (Exercise 10.2.17) and the dimensions add up (Corollary 10.3.4). ■

Exercise 10.3.6

For a finite-dimensional inner product space V and $U, W \le V$: (a) $W = (W^\perp)^\perp$;
(b) $(U + W)^\perp = U^\perp \cap W^\perp$;
(c) $(U \cap W)^\perp = U^\perp + W^\perp$.

Solution. (a) $W \le (W^\perp)^\perp$ and both have dimension $\dim W$ by Corollary 10.3.4 (applied twice). (b) $v \perp U + W$ iff $v \perp U$ and $v \perp W$. (c) Apply (b) to $U^\perp, W^\perp$: $(U^\perp + W^\perp)^\perp = U^{\perp\perp} \cap W^{\perp\perp} = U \cap W$, and take $\perp$ using (a). □

Remark 10.3.7

Cancelling $W^\perp$ from $W \oplus W^\perp = V = W^\perp \oplus (W^\perp)^\perp$ to get (a) is nonsense: complements in a direct sum are not unique (Remark 7.6.10).

With the dot product on $\mathbb{R}^n$, $AX = 0$ reads $\langle [A]_i^t, X\rangle = 0$ for all rows i, so

$$\ker L_A = \langle [A]_1^t, \dots, [A]_m^t\rangle^\perp = [\text{row space of } A]^\perp.$$

Example 10.3.8

For $W = \langle X_1, \dots, X_m\rangle \le \mathbb{R}^n$, let A have rows X_i^t. Then $W^\perp = \ker L_A$ and $\dim W = \text{row rank} = \text{column rank} = \dim \operatorname{im} L_A$, so the dimension theorem gives $\dim W^\perp = n - \dim W$.

Proposition 10.3.9 (The three pillars, $F = \mathbb{R}$)

Of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (for the dot product on $\mathbb{R}^n$), any two imply the third.

Proof. With W the row space, $\dim \ker L_A = \dim W^\perp$, and the three are the linear relations (1) $\dim \ker L_A + \dim \operatorname{im} L_A = n$, (2) $\dim W = \dim \operatorname{im} L_A$, (3) $\dim W + \dim W^\perp = n$. ■

This gives a proof of the rank theorem without row reduction. Over $\mathbb{C}$, X^tY is not an inner product, so we need:

Exercise 10.3.10

For $W \leq \mathbb{C}^n$, $\overline{W} = \{\bar{X} : X \in W\}$ is a subspace of $\mathbb{C}^n$ with $\dim \overline{W} = \dim W$.

Solution. $\overline{W}$ is closed under sums and under $c\bar{X} = \bar{\bar{c}}\bar{X}$. If $\{X_i\}$ is a basis of W, then $\{\bar{X}_i\}$ is a basis of $\overline{W}$ (conjugate any relation). □

Proposition 10.3.11 (The three pillars)

For $F = \mathbb{R}$ or $\mathbb{C}$, of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (for the dot product on F^n), any two imply the third.

Proof. As in Proposition 10.3.9, using $A\bar{X} = 0 \iff \langle [A]_i^t, X\rangle = 0$: the solution space of $A\bar{X} = 0$ is $W^\perp$, and it equals $\overline{\ker L_A}$, of dimension $\dim \ker L_A$ (Exercise 10.3.10). ■

10.4 Standard basis versus orthonormal basis

Motivation. In any orthonormal basis, coordinates are inner products and the inner product is the dot product of coordinates.

For the dot product on F^n and the standard basis $\mathcal{E}$: (a) $X = \sum \langle X, e_i\rangle e_i$;

(b) $\langle X, Y\rangle = [X]_{\mathcal{E}}^t \overline{[Y]_{\mathcal{E}}}$.

Observation 10.4.1

If $\mathfrak{B} = \{v_1, \ldots, v_n\}$ is an orthogonal basis of V:
(a)′ $v = \sum_i \frac{\langle v, v_i\rangle}{\langle v_i, v_i\rangle} v_i$. If $\mathfrak{B}$ is an orthonormal basis:
(a) $v = \sum_i \langle v, v_i\rangle v_i$, i.e. the i-th coordinate of $[v]_{\mathfrak{B}}$ is $\langle v, v_i\rangle$;
(b) $\langle v, w\rangle = [v]_{\mathfrak{B}}^t \cdot \overline{[w]_{\mathfrak{B}}}$.

Proof. (a) If $v = \sum a_i v_i$, then $\langle v, v_j\rangle = a_j \langle v_j, v_j\rangle$; this gives (a)′ and (a).

(b) For $w = \sum b_i v_i$, $\langle v, w\rangle = \sum_{i,j} a_i \bar{b}_j \langle v_i, v_j\rangle = \sum a_i \bar{b}_i$. ■

E.g. the foot of the perpendicular from $X = (1, 2, 3)^t$ to the xy-plane is $(1, 2, 0)^t = \langle X, e_1\rangle e_1 + \langle X, e_2\rangle e_2$.

Observation 10.4.2

Let W be a finite-dimensional subspace of an inner product space V with orthonormal basis $\{v_1, \ldots, v_m\}$. (a) $V = W \oplus W^\perp$.
(b) (Closest vector problem) For $v \in V$, $w = \langle v, v_1\rangle v_1 + \cdots + \langle v, v_m\rangle v_m \in W$ satisfies $v - w \in W^\perp$, and w is the unique vector of W closest to v.

Proof. (a) With w as in (b), $\langle v - w, v_i\rangle = \langle v, v_i\rangle - \langle v, v_i\rangle = 0$ for all i, so $v - w \in W^\perp$ and $V = W + W^\perp$; also $W \cap W^\perp = 0$.

(b) For $u \in W$, $w - u \in W$ is perpendicular to $v - w$, so by Pythagoras $\|v - u\|^2 = \|v - w\|^2 + \|w - u\|^2 \geq \|v - w\|^2$, with equality iff $u = w$. ■

Exercise 10.4.3

(a) In Euclidean $\mathbb{R}^3$, find an orthonormal basis of $W = \langle (1,1,2)^t \rangle^\perp$ and the vector of W closest to $(1,0,-1)^t$.
(b) Find the foot of the perpendicular from $(1,0,0,0)^t$ to the subspace W of Exercise 10.3.2(b).

Solution. (a) $\frac{1}{\sqrt{2}}(1,-1,0)$, $\frac{1}{\sqrt{3}}(1,1,-1)$. The closest vector is $X - \frac{\langle X,Y\rangle}{\langle Y,Y\rangle}Y = (1,0,-1) + \frac{1}{6}(1,1,2) = (\frac{7}{6}, \frac{1}{6}, -\frac{2}{3})$ (with $Y = (1,1,2)$).
(b) Summing $\langle v, u_i\rangle u_i$ over the orthonormal basis found there: $(\frac{9}{10}, \frac{4}{15}, \frac{2}{15}, \frac{1}{30})$. □

Exercise 10.4.4

If $\{v_1, \ldots, v_n\}$ is an orthonormal basis and $v = \sum a_i v_i$, then $\|v\|^2 = \sum |a_i|^2 = \sum |\langle v, v_i\rangle|^2$.

Solution. Observation 10.4.1(b) with $w = v$, and (a). □

Observation 10.4.5 (Bessel's inequality)

If $\{v_1, \ldots, v_m\}$ is an orthonormal subset of an inner product space V (possibly infinite-dimensional), then $\sum_{i=1}^m |\langle v, v_i\rangle|^2 \le \|v\|^2$ for all $v \in V$.

Proof. With $W = \langle v_1, \ldots, v_m\rangle$, Observation 10.4.2 gives $v = w + w'$, $w = \sum \langle v, v_i\rangle v_i \in W$, $w' \in W^\perp$, and by Pythagoras $\|v\|^2 = \|w\|^2 + \|w'\|^2 \ge \|w\|^2 = \sum |\langle v, v_i\rangle|^2$. ■

For an orthonormal subset $\{v_i\}$, $\langle v, v_i\rangle$ is the i-th **Fourier coefficient** of v.

Exercise 10.4.6

With $f_n(x) = e^{inx}$ in $C^0[0, 2\pi]$ (Exercise 10.2.13(e)) and $f(x) = x$: (a) $\langle f, f_0\rangle = \pi$ and $\langle f, f_n\rangle = \frac{i}{n}$ for $n \neq 0$;
(b) $\|f\|^2 = \frac{4}{3}\pi^2$;
(c) Bessel's inequality for $\{f_n\}_{-k \le n \le k}$ gives $\sum_{n=1}^k \frac{1}{n^2} \le \frac{\pi^2}{6}$, hence $\sum_{n=1}^\infty \frac{1}{n^2} \le \frac{\pi^2}{6}$.

Solution. (a) $\frac{1}{2\pi}\int_0^{2\pi} x\,dx = \pi$; for $n \neq 0$, integrating by parts, $\frac{1}{2\pi}\int_0^{2\pi} xe^{-inx}dx = \frac{1}{2\pi}\left[\frac{xe^{-inx}}{-in}\right]_0^{2\pi} = \frac{1}{-in} = \frac{i}{n}$. (b) $\frac{1}{2\pi}\int_0^{2\pi} x^2 dx = \frac{4\pi^2}{3}$. (c) $\frac{4}{3}\pi^2 \ge \pi^2 + 2\sum_{n=1}^k \frac{1}{n^2}$. □

10.5 Isomorphisms of inner product spaces

Motivation. A finite-dimensional inner product space is F^n with the dot product under another name.

Definition 10.5.1 (Isomorphism of inner product spaces)

Inner product spaces V, W over F are **isomorphic** if there is a vector space isomorphism $\varphi : V \to W$ with $\langle \varphi(u), \varphi(v)\rangle = \langle u, v\rangle$ for all $u, v \in V$; φ is an **inner product space isomorphism**. Equivalently, with $(\varphi \times \varphi)(u, v) = (\varphi(u), \varphi(v))$, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{\langle\,,\rangle} & F \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle I} \\ W \times W & \xrightarrow[\langle\,,\rangle]{} & F \end{array}$$

commutes.

Exercise 10.5.2

Is $\varphi \times \varphi : V \times V \to W \times W$ a vector space isomorphism between the product spaces?

Solution. Yes: it is linear componentwise, with inverse $\varphi^{-1} \times \varphi^{-1}$. □

Exercise 10.5.3

Isomorphism of inner product spaces is an equivalence relation.

Solution. I_V works; inverses of isomorphisms preserve inner products ($\langle \varphi^{-1}u, \varphi^{-1}v\rangle = \langle \varphi\varphi^{-1}u, \varphi\varphi^{-1}v\rangle$); composites of isomorphisms preserve them. □

Example 10.5.4

For an orthonormal basis $\mathfrak{B} = \{v_1, \ldots, v_n\}$ of V and the coordinate isomorphism $\alpha_{\mathfrak{B}}(v_i) = e_i$, Observation 10.4.1(b) gives $\langle \alpha_{\mathfrak{B}}(v), \alpha_{\mathfrak{B}}(w)\rangle = \sum a_i \bar{b}_i = \langle v, w\rangle$ for $v = \sum a_i v_i$, $w = \sum b_i v_i$. So $\alpha_{\mathfrak{B}}$ is an inner product space isomorphism onto F^n with the dot product.

Theorem 10.5.5 (Classification of finite-dimensional inner product spaces)

For finite-dimensional inner product spaces V, W over F the following are equivalent: (1) V and W are isomorphic as inner product spaces; (2) $\dim V = \dim W$.

Proof. (1)⇒(2): clear. (2)⇒(1): for orthonormal bases $\mathfrak{B}, \mathfrak{C}$, $(\alpha_{\mathfrak{C}})^{-1} \circ \alpha_{\mathfrak{B}}$ is an inner product space isomorphism (Example 10.5.4). ■

Exercise 10.5.6

Write explicit inner product space isomorphisms (a) from $\mathbb{R}^2$ with the dot product to $\mathbb{R}^2$ with the inner product of Exercise 10.1.9; (b) from F^4 with the dot product to $\mathfrak{M}_{2,2}(F)$ with the trace form of Exercise 10.1.14.

Solution. (a) Send e_1, e_2 to the orthonormal basis of Exercise 10.2.13(c): $\varphi(a, b) = \frac{a}{\sqrt{2}}(1, 0) + \frac{b}{\sqrt{6}}(1, 2)$. (b) The matrix units $E_{11}, E_{12}, E_{21}, E_{22}$ are orthonormal for $\operatorname{tr}(A^t \bar{B})$; send $e_1, \ldots, e_4$ to them, i.e. $(a, b, c, d)^t \mapsto \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. □

For $F = \mathbb{R}$ the results of §9.2 on rigid motions carry over:

Theorem 10.5.7

Let V be a finite-dimensional inner product space over $\mathbb{R}$. (a) A rigid motion L of V with $L(0) = 0$ is linear. (b) Every rigid motion of V is a translation composed with a linear rigid motion. (c) Every rigid motion of V is a bijection.

First proof. V is isomorphic as an inner product space to $\mathbb{R}^n$ with the dot product (Theorem 10.5.5), and the statements are Theorem 9.2.11 and Corollary 9.2.12. ■

Second proof. The second proof of Theorem 9.2.11(a) works verbatim with an orthonormal basis in place of $\mathcal{E}$; (b), (c) follow as before. ■

Exercise 10.5.8

Let $\gamma : \mathbb{C}^n \to \mathbb{R}^{2n}$, $\gamma((a_1 + ib_1, \dots, a_n + ib_n)^t) = (a_1, b_1, \dots, a_n, b_n)^t$, with the dot products on both spaces. (a) γ is an isomorphism of $\mathbb{R}$-vector spaces.
(b) $\|\gamma(X) - \gamma(Y)\| = \|X - Y\|$.
(c) If M is a rigid motion of $\mathbb{C}^n$, then $\gamma \circ M \circ \gamma^{-1}$ is a rigid motion of $\mathbb{R}^{2n}$; hence M is a bijection.

Solution. (a) Clear.

(b) $|a + ib|^2 = a^2 + b^2$ coordinatewise.

(c) $\gamma M \gamma^{-1}$ preserves distances, so is a bijection (Corollary 9.2.12), and so is $M = \gamma^{-1}(\gamma M \gamma^{-1})\gamma$. □

10.6 Orthogonal groups and unitary groups

Motivation. Linear isometries of real and complex spaces form $\mathbf{O}(V)$ and $\mathbf{U}(V)$; their matrices satisfy $A^*A = I$.

Spaces are finite-dimensional.

Definition 10.6.1 (Orthogonal and unitary groups)

(a) For a real inner product space V: $\mathbf{O}(V) = \mathbf{O}(V, \langle\cdot,\cdot\rangle) = \{L \in \mathfrak{L}(V,V) : \|Lv - Lw\| = \|v - w\| \ \forall v, w\}$, the **orthogonal group**; its elements are **orthogonal operators**.
(b) For a complex inner product space V: $\mathbf{U}(V) = \mathbf{U}(V, \langle\cdot,\cdot\rangle)$ is defined the same way, the **unitary group**; its elements are **unitary operators**.

They depend on the inner product; their elements are injective, hence invertible.

Exercise 10.6.2

For $G = \mathbf{O}(V)$ or $\mathbf{U}(V)$: (a) $I_V \in G$;
(b) $L, M \in G \Rightarrow L \circ M \in G$;
(c) $L \in G \Rightarrow L^{-1} \in G$.

Solution. As in Exercise 9.2.7. □

Definition 10.6.3 (Unitary matrices)

With the Hermitian dot product on $\mathbb{C}^n$, $\mathbf{U}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{C}) : L_A \in \mathbf{U}(\mathbb{C}^n)\} = \{[L]_{\mathcal{E}}^{\mathcal{E}} : L \in \mathbf{U}(\mathbb{C}^n)\}$, the (complex) **unitary group**; its elements are **unitary matrices**. Likewise $\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : L_A \in \mathbf{O}(\mathbb{R}^n)\}$.

Remark 10.6.4

For an orthonormal basis $\mathfrak{B}$ of V, $\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{U}(V)\} = \mathbf{U}(n)$ (similarly for $\mathbf{O}$); cf. Exercises 10.6.11, 9.6.11 and §13.3.

Observation 10.6.5

For an n-dimensional inner product space V and $L \in \mathfrak{L}(V,V)$ the following are equivalent: (1) $\|Lv - Lw\| = \|v - w\|$ (i.e. $L \in \mathbf{O}(V)$ or $\mathbf{U}(V)$); (2) $\|Lv\| = \|v\|$; (3) $\langle Lv, Lv\rangle = \langle v, v\rangle$; (4) $\langle Lv, Lw\rangle = \langle v, w\rangle$; (5) if $\mathfrak{B}$ is an orthonormal basis, so is $L(\mathfrak{B})$.

Proof. As in Observation 9.3.4 (for (5)⇒(2) use Exercise 10.4.4), except (3)⇒(4) for $F = \mathbb{C}$: expanding (3) at $v + w$ and at $v + iw$ gives $\mathrm{Re}\langle Lv, Lw\rangle = \mathrm{Re}\langle v, w\rangle$ and $-i\langle Lv, Lw\rangle + i\langle Lw, Lv\rangle = -i\langle v, w\rangle + i\langle w, v\rangle$, i.e. $\mathrm{Im}\langle Lv, Lw\rangle = \mathrm{Im}\langle v, w\rangle$. ■

Exercise 10.6.6

One may add to Observation 10.6.5: (2)′ $\|v\| = 1 \Rightarrow \|Lv\| = 1$; (5)′ some orthonormal basis $\mathfrak{B}$ has $L\mathfrak{B}$ orthonormal.

Solution. As in Exercise 9.3.5. □

Definition 10.6.7 (Adjoint matrix)

For $A \in \mathfrak{M}_{m,n}(F)$, $A^* = \overline{A^t} = \bar{A}^t$ is the **adjoint matrix** (for real A, $A^* = A^t$; unrelated to the classical adjoint of §6.7).

Exercise 10.6.8

For $A, B \in \mathfrak{M}_{m,n}(F)$, $C \in \mathfrak{M}_{n,r}(F)$, $c \in F$: (a) $(A + B)^* = A^* + B^*$;
(b) $(cA)^* = \bar{c}A^*$;
(c) $(A^*)^* = A$;
(d) $(AC)^* = C^*A^*$.

Solution. Conjugation and transposition are additive, conjugation is multiplicative, $(AC)^t = C^tA^t$, and both operations are involutions. □

Exercise 10.6.9

For $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ (Hermitian dot product on $\mathbb{C}^n$) the following are equivalent: (1) $\|AX - AY\| = \|X - Y\|$ (i.e. $A \in \mathbf{U}(n)$); (2) $\|AX\| = \|X\|$; (3) $\langle AX, AX\rangle = \langle X, X\rangle$; (4) $\langle AX, AY\rangle = \langle X, Y\rangle$; (5) the columns form an orthonormal basis of $\mathbb{C}^n$; (6) $A^t\bar{A} = I = \bar{A}A^t$, i.e. $A^*A = I = AA^*$, $A^{-1} = A^*$; (7) $A^t \in \mathbf{U}(n)$; (7)′ $\bar{A} \in \mathbf{U}(n)$; (7)″ $A^* \in \mathbf{U}(n)$; (8) the rows form an orthonormal basis of $\mathfrak{M}_{1,n}(\mathbb{C})$.

Solution. (1)–(5) is Observation 10.6.5 with $\mathfrak{B} = \mathcal{E}$. The (i,j) entry of A^*A is $\overline{\langle [A]^i, [A]^j\rangle}$, which gives (5)⇔(6). (6) is preserved by $A \mapsto A^t, \bar{A}, A^*$ (take transposes and conjugates of $A^*A = I = AA^*$), giving (7), (7)′, (7)″. (8) is (5) for A^t. □

Exercise 10.6.10

For $G = \mathbf{O}(n)$ or $\mathbf{U}(n)$: (a) $I_n \in G$;
(b) $A, B \in G \Rightarrow AB \in G$;
(c) $A \in G \Rightarrow A^{-1} \in G$.

Solution. $(AB)^*AB = B^*A^*AB = I$; $(A^{-1})^*A^{-1} = (A^*)^*A^* = AA^* = I$. □

So $\mathbf{U}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{C}) : A^*A = I\}$; defining L^* for $\mathbf{U}(V)$ is Question 10.7.2.

Exercise 10.6.11

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal bases of an n-dimensional complex inner product space V, then $[I]_{\mathfrak{C}}^{\mathfrak{B}} \in \mathbf{U}(n)$.

Solution. The columns are $[w_j]_{\mathfrak{B}}$, $w_j \in \mathfrak{C}$, and $\langle [w_i]_{\mathfrak{B}}, [w_j]_{\mathfrak{B}} \rangle = \langle w_i, w_j \rangle = \delta_{ij}$ (Observation 10.4.1(b)). □

Exercise 10.6.12

If $A \in \mathbf{U}(n)$, then $|\det A| = 1$.

Solution. $1 = \det(A^*A) = \overline{\det A} \det A = |\det A|^2$. □

Definition 10.6.13 (Special unitary group)

$\mathbf{SU}(n) = \{A \in \mathbf{U}(n) : \det A = 1\}$.

Exercise 10.6.14

(a) $\mathbf{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$. What is $\mathbf{U}(2)$?
(b) Are the elements of $\mathbf{SU}(2)$ diagonalizable over $\mathbb{C}$? Those of $\mathbf{U}(2)$?

Solution. (a) For $A = \left(\begin{smallmatrix} \alpha & z \\ \beta & w \end{smallmatrix}\right) \in \mathbf{SU}(2)$, $|\alpha|^2 + |\beta|^2 = 1$ and $(z, w)^t$ is a unit vector in $\langle (\alpha, \beta)^t \rangle^{\perp} = \langle (-\bar{\beta}, \bar{\alpha})^t \rangle$: $(z, w) = \lambda(-\bar{\beta}, \bar{\alpha})$, $|\lambda| = 1$, and $\det A = \lambda = 1$. The converse is clear. Hence $\mathbf{U}(2) = \left\{ \left(\begin{smallmatrix} \alpha & -\lambda\bar{\beta} \\ \beta & \lambda\bar{\alpha} \end{smallmatrix}\right) : |\alpha|^2 + |\beta|^2 = 1,\ |\lambda| = 1 \right\}$.

(b) Yes in both cases (normal matrices, Chapter 15). Directly: take a unit eigenvector v_1 of unitary A and an orthonormal basis $\{v_1, v_2\}$; $Av_2 \perp Av_1 = \lambda v_1$ gives $Av_2 \in \langle v_2 \rangle$. □

10.7 The adjoint matrix and its applications

Motivation. The identity $\langle AX, Y \rangle = \langle X, A^*Y \rangle$ yields least squares and minimal solutions.

Observation 10.7.1

With dot products on F^n, F^m and $A \in \mathfrak{M}_{m,n}(F)$: $\langle AX, Y \rangle = \langle X, A^*Y \rangle$ and $\langle Y, AX \rangle = \langle A^*Y, X \rangle$ for $X \in F^n$, $Y \in F^m$.

Proof. $\langle AX, Y \rangle = (AX)^t\bar{Y} = X^tA^t\bar{Y}$, and since $\overline{A^*} = A^t$, $\langle X, A^*Y \rangle = X^t\overline{A^*Y} = X^tA^t\bar{Y}$. The second identity is the conjugate of the first. ■

Question 10.7.2

For finite-dimensional inner product spaces V, W and $L \in \mathfrak{L}(V,W)$, does there exist $L^* \in \mathfrak{L}(W,V)$ with $\langle Lv, w\rangle = \langle v, L^*w\rangle$ for all $v \in V$, $w \in W$? What does it mean?

For the rest of this section F^n carries the dot product.

Least squares approximation (LSA). An **LSA solution** of $AX = B$ ($A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$) minimizes $\|AX_0 - B\|$. E.g. fitting $g = \sum_{j\le n} c_j t^j$ to points $(a_i, b_i)_{i\le m}$, $m > n$, minimizes $\sum_i |g(a_i) - b_i|^2 = \|AX - B\|^2$ with $A = (a_i^j)$, $X = (c_j)$, $B = (b_i)$ ($m = n$: Lagrange interpolation).

Proposition 10.7.3

For $A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$: (a) an LSA solution X_0 ($\|AX_0 - B\| \le \|AX - B\|$ for all X) always exists; (b) if L_A is injective ($\operatorname{rk} A = n$), it is unique.

Proof. By Observation 10.4.2, $B = AX_0 + C$ with $C \in (\operatorname{im} A)^\perp$, and AX_0 is the unique closest vector of $\operatorname{im} A$ to B; if L_A is injective, it determines X_0. ■

Proposition 10.7.4

The solutions of $(A^*A)X = A^*B$ are exactly the LSA solutions.

Proof. X_0 is an LSA solution iff $AX_0 - B \in (\operatorname{im} A)^\perp$, i.e. $0 = \langle AX, AX_0 - B\rangle = \langle X, A^*(AX_0 - B)\rangle$ for all X (Observation 10.7.1), i.e. $A^*(AX_0 - B) = 0$. ■

Exercise 10.7.5

Find the least squares line for the points $(1,1)$, $(2,3)$, $(3,3)$, $(4,2)$ of $\mathbb{R}^2$.

Solution. For $y = ct + d$: $A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$, $B = (1,3,3,2)^t$, $A^*A = \left(\begin{smallmatrix} 30 & 10 \\ 10 & 4 \end{smallmatrix}\right)$, $A^*B = (24, 9)^t$. Solving, $(c,d) = (\frac{3}{10}, \frac{3}{2})$: $y = 0.3t + 1.5$. □

Exercise 10.7.6

Solve the LSA problem for $x + 3y + z = 3$, $2x - y - 2z = 2$, $5x + y - 3z = 6$.

Solution. The coefficient matrix has rank 2, so the normal equations $(A^*A)X = A^*B$ have a line of solutions: $X = (\frac{47}{42}, \frac{4}{7}, 0)^t + t(5, -4, 7)^t$, $t \in \mathbb{R}$. □

Observation 10.7.7

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{rk}(A^*A) = \operatorname{rk} A$;
(b) if $m \ge n$ and $\operatorname{rk} A = n$, then $A^*A \in \mathfrak{M}_{n,n}(F)$ is invertible.

Proof. (a) By the dimension theorem it suffices that $\ker(A^*A) = \ker A$. If $A^*AX = 0$, then $0 = \langle X, A^*AX\rangle = \langle (A^*)^*X, AX\rangle = \langle AX, AX\rangle$ (Observation 10.7.1, Exercise 10.6.8(c)), so $AX = 0$. (b) follows. ■

Exercise 10.7.8

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{rk} A = \operatorname{rk} A^t = \operatorname{rk} \bar{A} = \operatorname{rk} A^*$;
(b) $\operatorname{rk}(AA^*) = \operatorname{rk} A$.

Solution. (a) Rank theorem and Exercise 10.3.10.

(b) Observation 10.7.7(a) for A^* and (a). □

Observation 10.7.9

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{im}(A^*) = (\ker A)^\perp$, i.e. $\ker A = (\operatorname{im} A^*)^\perp$;
(b) $\operatorname{im} A = (\ker A^*)^\perp$, i.e. $\ker A^* = (\operatorname{im} A)^\perp$.

Proof. By Exercise 10.3.6(a) and $A \leftrightarrow A^*$ it suffices to show $\ker A = (\operatorname{im} A^*)^\perp$: $AX = 0$ gives $\langle X, A^*Y\rangle = \langle AX, Y\rangle = 0$; conversely $\langle Y, AX\rangle = \langle A^*Y, X\rangle = 0$ for all Y gives $AX = 0$. ■

Minimal solution problem. If $AX = B$ is solvable, find the solution of smallest norm (the foot of the perpendicular from 0 to the solution set).

Proposition 10.7.10

Let $AX = B$ ($A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$) have a solution. (a) There is a unique minimal solution X_0 ($AX_0 = B$ and $\|X_0\| \le \|X\|$ for every solution X).
(b) X_0 is the unique solution lying in $\operatorname{im}(A^*)$.
(c) If $(AA^*)Y_0 = B$, then $X_0 = A^*Y_0$.

Proof. (i) Let $AX_1 = B$. By Observation 10.7.9, $F^n = \operatorname{im}(A^*) \oplus \ker A$, so $X_1 = A^*Y_0 + Z$ with $Z \in \ker A$. Put $X_0 = A^*Y_0$; then $AX_0 = AX_1 = B$, and X_0 is a solution in $\operatorname{im} A^*$.

(ii) Every solution is $X = X_0 + Z'$ with $Z' \in \ker A = (\operatorname{im} A^*)^\perp$, so $X_0 \perp Z'$ and $\|X\|^2 = \|X_0\|^2 + \|Z'\|^2 \ge \|X_0\|^2$, with equality iff $Z' = 0$.

(iii) If $X \in \operatorname{im} A^*$ is a solution, then $X_0 - X \in \operatorname{im} A^* \cap \ker A = 0$. (c) follows: $A^*Y_0 \in \operatorname{im} A^*$ is a solution. ■

Exercise 10.7.11

Find the minimal solutions of (a) $x + 3y + z = 3$, $2x - y - 2z = 2$;
(b) $x + 3y + z = 3$, $2x - y - 2z = 2$, $5x + y - 3z = 7$.

Solution. (a) $AA^* = \begin{pmatrix} 11 & -3 \\ -3 & 9 \end{pmatrix}$, and $Y_0 = (AA^*)^{-1}(3,2)^t$ gives $X_0 = A^*Y_0 = (\frac{19}{18}, \frac{34}{45}, -\frac{29}{90})^t$. (b) Equation 3 = equation 1 $+2\times$ equation 2: same solutions, same X_0 as (a). □

Chapter 11. Groups

11.1 Binary operations and groups

Motivation. Vector addition, matrix multiplication and composition of rigid motions share associativity, identity and inverses: a group.

$X \times Y = \{(x, y) : x \in X, y \in Y\}$ is the **Cartesian product**; (x, y) is an **ordered pair**.

Definition 11.1.1 (Binary operation)

A function $* : X \times X \to X$ is a **binary operation** on X; we write $x * y$, or $x \cdot y$, xy.

Definition 11.1.2 (Associativity)

A binary operation is **associative** if $(xy)z = x(yz)$ for all $x, y, z \in X$; then xyz is meaningful.

Example 11.1.3

On $\mathfrak{M}_{n,n}(F)$, the bracket $[A, B] = AB - BA$ is not associative: for $n = 2$, $[[E_{12}, E_{21}], E_{21}] = [E_{11} - E_{22}, E_{21}] = -2E_{21}$, while $[E_{12}, [E_{21}, E_{21}]] = 0$.

From now on every binary operation is assumed associative.

Definition 11.1.4 (Group)

A set G with a binary operation is a **group** if (G1) $(xy)z = x(yz)$ for all $x, y, z \in G$; (G2) there is $e \in G$ with $xe = x = ex$ for all $x \in G$; (G3) for each $x \in G$ there is $\tilde{x} \in G$ with $x\tilde{x} = e = \tilde{x}x$.

Example 11.1.5

(a) $(\{-1, 1\}, +)$ is not a group ($1 + 1 \notin \{-1, 1\}$), but $(\{-1, 1\}, \times)$ is.
(b) $\{e\}$ with $ee = e$ is the **trivial group**.
(c) $(\mathbb{Z}, +)$ is a group; $(\mathbb{Z}, \times)$ is not.
(d) $(F, +)$ and $(F - \{0\}, \times)$ are groups; $(F, \times)$ is not.
(e) For a vector space V, $(V, +)$ is a group (axioms (V1)–(V4)).
(f) The symmetric group $(S_n, \circ)$ of Chapter 6 is a group.

Observation 11.1.6

A group is not empty.

Proof. By (G2) it contains e. ■

Observation 11.1.7

The element e of (G2) is unique; it is the **identity** of G, written e_G.

Proof. If e' is another, $e = ee' = e'$. ■

Observation 11.1.8

For each $x \in G$, the $\tilde{x}$ of (G3) is unique; it is the **inverse** x^{-1}.

Proof. If $xx' = e = x'x$ as well, then $\tilde{x} = \tilde{x}e = \tilde{x}(xx') = (\tilde{x}x)x' = ex' = x'$. ■

Exercise 11.1.9

For $g, h \in G$: (a) $(gh)^{-1} = h^{-1}g^{-1}$;
(b) $(g^{-1})^{-1} = g$.

Solution. (a) $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = e$ and likewise on the other side; use uniqueness of inverses.

(b) g satisfies the defining property of the inverse of g^{-1}. □

Definition 11.1.10 (Order)

G is a **finite group** if it is a finite set; then $|G|$ is its **order**. Otherwise G is an **infinite group**.

Definition 11.1.11 (Commutative group)

G is **commutative** (**abelian**) if $xy = yx$ for all $x, y \in G$.

Exercise 11.1.12

$\mu_n = \{\alpha \in \mathbb{C} : \alpha^n = 1\}$, under multiplication of complex numbers, is a finite commutative group of order n.

Solution. μ_n contains 1 and is closed under products and inverses; associativity and commutativity come from $\mathbb{C}$. Its elements are the n distinct $e^{2\pi ik/n}$, $0 \le k < n$ (at most n roots of $t^n - 1$). □

Exercise 11.1.13

For $n \ge 3$, S_n is a finite non-commutative group of order $n!$.

Solution. $|S_n| = n!$. $(1\,2)(1\,3)$ sends $1 \mapsto 3 \mapsto 3$ but $(1\,3)(1\,2)$ sends $1 \mapsto 2$; so they differ. □

Exercise 11.1.14

On F^3 define $(a_1, a_2, a_3)^t \bullet (b_1, b_2, b_3)^t = (a_1 + b_1,\ a_2 + b_2 + a_1b_3,\ a_3 + b_3)^t$. Then $(F^3, \bullet)$ is an infinite non-commutative group.

Solution. Both $(a \bullet b) \bullet c$ and $a \bullet (b \bullet c)$ equal $(a_1{+}b_1{+}c_1,\ a_2{+}b_2{+}c_2{+}a_1b_3{+}a_1c_3{+}b_1c_3,\ a_3{+}b_3{+}c_3)$. The identity is 0, and the inverse of a is $(-a_1, -a_2{+}a_1a_3, -a_3)$. Non-commutative: $e_1 \bullet e_3 = (1, 1, 1)$ but $e_3 \bullet e_1 = (1, 0, 1)$. (This is the Heisenberg group of upper unitriangular $3{\times}3$ matrices in disguise.) □

11.2 Elementary properties of groups

Motivation. Powers, cancellation, and multiplication tables classify the smallest groups.

Notation 11.2.1

(a) (Multiplicative notation) The identity is 1; $g^1 = g$, $g^2 = gg$, …, $g^0 = 1$, $g^{-2} = g^{-1}g^{-1}$, ….
(b) (Additive notation) For an abelian group A written with $+$: identity 0, inverse $-a$, and $\ldots, (-2)a = (-a) + (-a)$, $0a = 0$, $1a = a$, $2a = a + a, \ldots$; $a - b = a + (-b)$.
(c) Commutative groups are usually written multiplicatively and abelian groups additively.
(d) Unless stated otherwise we use multiplicative notation.

Exercise 11.2.2

(a) $(g^n)^{-1} = (g^{-1})^n = g^{-n}$ for $n \in \mathbb{Z}$.
(b) Additively: $-(na) = n(-a) = (-n)a$.

Solution. $g^n(g^{-1})^n = 1$ by cancelling adjacent gg^{-1}; and $g^{-n} = (g^{-1})^n$ by definition (for $n > 0$; the other cases follow). □

Exercise 11.2.3

$(h^{-1}gh)^n = h^{-1}g^nh$ for $n \in \mathbb{Z}$.

Solution. For $n > 0$, the inner hh^{-1} cancel; $n = 0$ is clear; for $n < 0$ apply this to g^{-1}, using $(h^{-1}gh)^{-1} = h^{-1}g^{-1}h$. □

Exercise 11.2.4

Elements $ghg^{-1}h^{-1}$ are **commutators**. G is commutative iff all commutators are 1.

Solution. $ghg^{-1}h^{-1} = 1 \iff gh = hg$. □

Exercise 11.2.5

If $g^2 = 1$ for all $g \in G$, then G is commutative.

Solution. Then $g = g^{-1}$ for all g, so $ghg^{-1}h^{-1} = ghgh = (gh)^2 = 1$. □

Exercise 11.2.6 (Exponent laws, multiplicative)

For $g, h \in G$, $m, n \in \mathbb{Z}$: (a) $g^m g^n = g^{m+n}$;
(b) $(g^m)^n = g^{mn}$;
(c) if G is commutative, $g^n h^n = (gh)^n$.

Solution. For $m, n \geq 0$ count factors; negative exponents reduce to this via $g^{-k} = (g^{-1})^k$. (c): regroup $(gh)\cdots(gh)$ by commutativity. □

Exercise 11.2.7 (Exponent laws, additive)

For $a, b \in A$, $m, n \in \mathbb{Z}$: (a) $ma + na = (m + n)a$;
(b) $m(na) = (mn)a$;
(c) $na + nb = n(a + b)$.

Solution. Exercise 11.2.6 in additive notation. □

Observation 11.2.8 (Cancellation law)

If $xy = xz$, then $y = z$; if $xz = yz$, then $x = y$.

Proof. $y = (x^{-1}x)y = x^{-1}(xy) = x^{-1}(xz) = z$; the other side is symmetric. ■

Exercise 11.2.9

State Observation 11.2.8 additively.

Solution. If $a + b = a + c$ then $b = c$; if $a + c = b + c$ then $a = b$. □

Example 11.2.10

$xy = x$ implies $y = 1$, since $xy = x1$.

Exercise 11.2.11

(a) If $y = u^{-1}xu$, then $x = uyu^{-1}$.
(b) If $u^{-1}xu = 1$, then $x = 1$.

Solution. (a) Multiply by u on the left and u^{-1} on the right.

(b) By (a), $x = u1u^{-1} = 1$. □

A finite group $G = \{1, x_2, x_3, \dots\}$ can be described by its **multiplication table**, whose (x_i, x_j) entry is $x_i x_j$.

Example 11.2.12

All groups of order 1 are "the same": $\{e\}$ and $(\{0\}, +)$ differ only in names.

Example 11.2.13

If $G = \{1, x\}$, then $x^2 \neq x$ (cancellation), so $x^2 = 1$: the table is forced, and G is μ_2 up to renaming.

Observation 11.2.14

In a multiplication table every row contains each element exactly once, and so does every column.

Proof. $xy = xz$ forces $y = z$ (cancellation), so a row has no repetitions; G is finite, so each element appears. Similarly for columns. ■

Observation 11.2.15

For fixed $x \in G$, $\lambda_x : G \to G$, $\lambda_x(y) = xy$, is a bijection (it permutes G).

Proof. Injective by cancellation; surjective since $z = \lambda_x(x^{-1}z)$. ■

Example 11.2.16

If $G = \{1, x, y\}$, then $x^2 \neq x$, and $x^2 = 1$ would force $xy = y$, contradicting cancellation; so $x^2 = y$, and by Observation 11.2.14 the table is

	1	x	y
1	1	x	y
x	x	y	1
y	y	1	x

So there is at most one group of order 3 up to renaming.

Question 11.2.17

How can we check that a given table is associative?

If a known group realises the table, associativity is automatic: for $\zeta = e^{2\pi i/3}$, the renaming $x \mapsto \zeta$, $y \mapsto \zeta^2$ carries the table of Example 11.2.16 to that of μ_3.

Example 11.2.18

For $G = \{1, x, y, z\}$, cancellation and Observation 11.2.14 leave four possible tables (up to symmetry): one in which every element squares to 1, and three in which one non-identity element squares to 1 (the second, third and fourth tables), which are $\mu_4 = \{1, -1, i, -i\}$ up to renaming.

Exercise 11.2.19

The second, third and fourth tables of Example 11.2.18 are associative.

Solution. Each is the table of μ_4 after a renaming (e.g. $x \mapsto -1$, $y \mapsto i$, $z \mapsto -i$). □

Example 11.2.20

For groups G_1, G_2, the Cartesian product $G_1 \times G_2$ with $(g_1, g_2)(h_1, h_2) = (g_1h_1, g_2h_2)$ is a group, the **direct product**.

Example 11.2.21

$\mu_2 \times \mu_2$ realises the first table of Example 11.2.18; e.g. $(1, -1)(1, -1) = (1, 1)$. It is the **Klein 4-group**, not μ_4 (all its elements square to 1).

Exercise 11.2.22

(If you can,) show with tables that a group of order 5 is μ_5 up to renaming.

Solution [Supplement] (via Lagrange's theorem, Observation 12.1.10). Let $x \neq 1$. The powers of x form a subgroup $\langle x \rangle$ (§11.3) whose order divides 5, hence equals 5. So $G = \{1, x, x^2, x^3, x^4\}$ with $x^5 = 1$, and $x^k \mapsto e^{2\pi ik/5}$ is a renaming onto μ_5 preserving the table. □

Exercise 11.2.23

If $|G| \leq 5$, then G is commutative; so S_3 is the smallest non-commutative group.

Solution. By Examples 11.2.12, 11.2.13, 11.2.16, 11.2.18 and Exercise 11.2.22, every such group is μ_n or $\mu_2 \times \mu_2$, whose tables are symmetric. S_3 has order 6 and is non-commutative. □

Remark 11.2.24

Semigroups, monoids, and one-sided identities and inverses appear in other texts; they are not needed here, nor is the classification of finite groups.

11.3 Subgroups

Motivation. Subgroups play the role of subspaces; the subgroup generated by S consists of words in $S \cup S^{-1}$.

Definition 11.3.1 (Subgroup)

A subset $H \subseteq G$ that is a group under the operation of G is a **subgroup**, $H \leq G$.

Observation 11.3.2

Let $H \leq G$. (a) $e_H = e_G$.
(b) The inverse of $h \in H$ in H equals its inverse h^{-1} in G.

Proof. (a) In G, $e_H e_H = e_H = e_H e_G$; cancel e_H.

(b) In G, $hh' = e_H = e_G = hh^{-1}$; cancel h. ■

Exercise 11.3.3

Why was $0_W = 0_V$ for subspaces $W \leq V$ never an issue?

Solution. $0_W = 0 \cdot w = 0_V$ for any $w \in W$. □

Observation 11.3.4

$H \subseteq G$ is a subgroup iff (S1) $h_1, h_2 \in H \Rightarrow h_1 h_2 \in H$; (S2) $e \in H$; (S3) $h \in H \Rightarrow h^{-1} \in H$.

Proof. Necessity: the operation must restrict to H (S1), and Observation 11.3.2 gives (S2), (S3). Sufficiency: associativity is inherited from G. ■

Exercise 11.3.5

A nonempty $H \subseteq G$ is a subgroup iff $h_1, h_2 \in H \Rightarrow h_1 h_2^{-1} \in H$.

Solution. Necessity is clear. For sufficiency, take $h \in H$: then $e = hh^{-1} \in H$, $h^{-1} = eh^{-1} \in H$, and $h_1 h_2 = h_1 (h_2^{-1})^{-1} \in H$. □

Example 11.3.6

(a) $\{e\} \leq G \leq G$. (b) Under addition, $\mathbb{Z} < \mathbb{Q} < \mathbb{R} < \mathbb{C}$. (c) Under multiplication, $\mu_2 < \mathbb{Q}-\{0\} < \mathbb{R}-\{0\} < \mathbb{C}-\{0\}$. (d) The alternating group A_n is a subgroup of S_n. (e) If $m \mid n$, then $\mu_m \leq \mu_n$. (f) If W is a subspace of V, then $(W,+) \leq (V,+)$.

Observation 11.3.7

If $K \leq H$ and $H \leq G$, then $K \leq G$.

Proof. K is a group under the operation of H, which is that of G. ■

Exercise 11.3.8

If $I \neq \emptyset$ and $H_i \leq G$ for all $i \in I$, then $\bigcap_i H_i \leq G$.

Solution. Each H_i contains e and is closed under products and inverses; so is the intersection. □

Notation 11.3.9

For $H, K \subseteq G$, $HK = \{hk : h \in H, k \in K\}$.

Exercise 11.3.10

(a) If G is commutative and $H, K \leq G$, then $HK \leq G$.
(b) In $S_3 = \{id, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}$, $H = \{id, (1\,2)\}$ and $K = \{id, (2\,3)\}$ are subgroups but $HK \not\leq S_3$.

Solution. (a) $(h_1k_1)(h_2k_2)^{-1} = h_1h_2^{-1}k_1k_2^{-1} \in HK$ by commutativity; apply Exercise 11.3.5. (b) $HK = \{id, (2\,3), (1\,2), (1\,2)(2\,3)\}$ does not contain $((1\,2)(2\,3))^{-1} = (2\,3)(1\,2)$. □

Definition 11.3.11 (Generated subgroup)

For $S \subseteq G$, the smallest subgroup of G containing S is the **subgroup generated by** S, $\langle S \rangle$: $S \subseteq \langle S \rangle \leq G$, and $S \subseteq H \leq G$ implies $\langle S \rangle \subseteq H$.

Observation 11.3.12

$\langle S \rangle = \bigcap_{S \subseteq H \leq G} H$; in particular $\langle S \rangle$ exists and is unique.

Proof. By Exercise 11.3.8 the intersection is a subgroup containing S, and it lies in every subgroup containing S. ■

Example 11.3.13

$\langle \emptyset \rangle = \{e\}$, and $\langle x \rangle = \{x^i : i \in \mathbb{Z}\}$ (a subgroup by the exponent laws); possibly $x^i = x^j$ for $i \neq j$.

Definition 11.3.14 (Cyclic group)

(a) $\langle x\rangle$ is the **cyclic subgroup** generated by x.
(b) If $G = \langle x\rangle$ for some x, G is a **cyclic group** with generator x. Cyclic groups are commutative (§11.7).

Definition 11.3.15 (Alphabet and words)

(a) For $S \subseteq G$ let $S^{-1} = \{s^{-1} : s \in S\}$; $S \cup S^{-1}$ is the **alphabet**.
(b) Finite products of letters are **words**; e is the empty word. (Different words may represent the same element.)

Observation 11.3.16

The set of all words in the alphabet $S \cup S^{-1}$ is a subgroup of G.

Proof. It contains the empty word, products of words are words, and $(s_1 \cdots s_k)^{-1} = s_k^{-1} \cdots s_1^{-1}$ is a word (Exercise 11.1.9); apply Observation 11.3.4. ■

Proposition 11.3.17

$\langle S\rangle$ is the set K of all words in the alphabet $S \cup S^{-1}$.

Proof. $\langle S\rangle$ is a subgroup containing S, so it contains all such words: $K \subseteq \langle S\rangle$. Conversely K is a subgroup (Observation 11.3.16) containing S, so $\langle S\rangle \subseteq K$. ■

Exercise 11.3.18

(a) $(\mathbb{Z}, +)$ is cyclic; how many generators does it have?
(b) $3\mathbb{Z} = \{3a : a \in \mathbb{Z}\}$ is a cyclic subgroup of $(\mathbb{Z}, +)$.
(c) $\langle 2, 3\rangle = \mathbb{Z} = \langle 4, 15\rangle$, but $\langle 6, 9\rangle \neq \mathbb{Z}$.

Solution. (a) $\mathbb{Z} = \langle 1\rangle = \langle -1\rangle$, and these are the only generators: $\langle n\rangle = n\mathbb{Z} \neq \mathbb{Z}$ for $|n| \neq 1$.

(b) $3\mathbb{Z} = \langle 3\rangle$.

(c) $1 = 3 - 2 = 4 \cdot 4 - 15$, so 1 lies in both; but every element of $\langle 6, 9\rangle$ is a multiple of 3, so $\langle 6, 9\rangle = 3\mathbb{Z}$. □

Exercise 11.3.19

(a) μ_5 is cyclic; how many generators?
(b) μ_6 is cyclic; how many generators?

Solution. $\mu_n = \langle \zeta\rangle$ with $\zeta = e^{2\pi i/n}$, and ζ^k generates iff $\gcd(k, n) = 1$. (a) 4 generators ($k = 1, 2, 3, 4$).

(b) 2 generators ($k = 1, 5$). □

Exercise 11.3.20

(a) Find all subgroups of S_3.
(b) Is S_3 cyclic?
(c) $\langle (1\,2), (2\,3)\rangle = S_3 = \langle (1\,2), (1\,2\,3)\rangle$.

Solution. (a) $\{id\}$, $\langle(1\,2)\rangle$, $\langle(1\,3)\rangle$, $\langle(2\,3)\rangle$, $A_3 = \langle(1\,2\,3)\rangle$, S_3 (two transpositions, or a transposition and a 3-cycle, generate S_3 by (c)).

(b) No: element orders are $1, 2, 3$.

(c) $(1\,2)(2\,3)$, $(2\,3)(1\,2)$ are the two 3-cycles and $(1\,2)(2\,3)(1\,2) = (1\,3)$. Likewise $(1\,2\,3)^2$, $(1\,2)(1\,2\,3)$, $(1\,2\,3)(1\,2)$ give the rest. □

11.4 The heart of undergraduate algebra

Motivation. Every subgroup of $(\mathbb{Z}, +)$ is $n\mathbb{Z}$, by division by its least positive element.

Example 11.4.1

For $n \in \mathbb{Z}$, $n\mathbb{Z} = \{na : a \in \mathbb{Z}\} = \langle n\rangle$ is a subgroup of $(\mathbb{Z}, +)$.

Theorem 11.4.2 (The heart of undergraduate algebra)

The subgroups of $(\mathbb{Z}, +)$ are exactly the $n\mathbb{Z}$, $0 \le n \in \mathbb{Z}$.

Proof. Let $0 \ne H \le \mathbb{Z}$ and n its least positive element. For $a \in H$, $a = qn + r$ with $0 \le r < n$ gives $r = a - qn \in H$, so $r = 0$. Hence $H \le n\mathbb{Z} = \langle n\rangle \le H$. ($H = 0$ is $0\mathbb{Z}$.) ■

11.5 Group isomorphisms

Motivation. Isomorphic groups differ only in the names of their elements.

Definition 11.5.1 (Isomorphism)

Groups G, G' are **isomorphic**, $G \approx G'$, if there is a bijection $\varphi : G \to G'$ with $\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2)$; φ is a (**group**) **isomorphism**. Equivalently, with $(\varphi \times \varphi)(g_1, g_2) = (\varphi g_1, \varphi g_2)$, the square

$$\begin{array}{ccc} G \times G & \xrightarrow{*} & G \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ G' \times G' & \xrightarrow[*']{} & G' \end{array}$$

commutes.

Example 11.5.2

Let $\varphi : G \to H$ be an isomorphism, $g \in G$, $S \subseteq G$. Then (a) $\varphi(e_G) = e_H$;
(b) $\varphi(g^{-1}) = \varphi(g)^{-1}$;
(c) if $\langle S\rangle = G$, then $\langle\varphi(S)\rangle = H$;
(d) if $g^2 = e_G$, then $\varphi(g)^2 = e_H$;
(e) if G is commutative, so is H;
(f) if G is cyclic, so is H;
(g) if G has six subgroups, so does H.

Proof [Supplement]. (a) $\varphi(e)\varphi(e) = \varphi(e) = \varphi(e)e_H$; cancel.

(b) $\varphi(g)\varphi(g^{-1}) = \varphi(e) = e_H$.

(c) φ maps words in $S \cup S^{-1}$ onto words in $\varphi S \cup (\varphi S)^{-1}$ (Proposition 11.3.17) and is onto.

(d) Apply φ.

(e) $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$, and φ is onto.

(f) Case of (c).

(g) $K \mapsto \varphi(K)$ is a bijection between the sets of subgroups (with inverse $K' \mapsto \varphi^{-1}(K')$). ■

Exercise 11.5.3

Is $\mathbb{Z} \approx \mathbb{Z} \times \mathbb{Z}$ (additive groups)?

Solution. No (Example 11.5.2(f)): $\mathbb{Z} \times \mathbb{Z}$ is not cyclic, as $\langle (a,b) \rangle = \{(na, nb)\}$ never contains both $(1,0)$ and $(0,1)$. □

Example 11.5.4

Properties *not* preserved by isomorphisms: "the operation is called multiplication", "G has an element named x", "the elements are complex numbers", "$|g| = 7$ as a real number", "g integrates to h".

Observation 11.5.5

$\approx$ is an equivalence relation.

Proof. Exercise 11.5.6. ■

Exercise 11.5.6

(a) id_G is an isomorphism.
(b) If $\varphi : G \to H$ is an isomorphism, so is φ^{-1}.
(c) If $\psi : K \to G$, $\varphi : G \to H$ are isomorphisms, so is $\varphi \circ \psi$.

Solution. (b) With $h_i = \varphi(g_i)$: $\varphi^{-1}(h_1 h_2) = \varphi^{-1}(\varphi(g_1 g_2)) = g_1 g_2 = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$. (a), (c) are direct. □

Example 11.5.7

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a group isomorphism, with inverse log: $e^{a+b} = e^a e^b$.

Remark 11.5.8

Unlike vector spaces (dimension), deciding whether two groups are isomorphic is in general hard.

11.6 Group homomorphisms

Motivation. Homomorphisms are the analogue of linear maps, with kernels and images.

Definition 11.6.1 (Homomorphism)

$\varphi : G \to H$ is a (**group**) **homomorphism** if $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in G$.

Observation 11.6.2

For a homomorphism φ: (a) $\varphi(e) = e$;
(b) $\varphi(x^{-1}) = \varphi(x)^{-1}$.

Proof. As in Example 11.5.2(a), (b). ■

Definition 11.6.3 (Kinds of homomorphisms)

A homomorphism $\varphi : G \to H$ is a **monomorphism** if injective, an **epimorphism** if surjective, an **isomorphism** if bijective, an **endomorphism** if $G = H$, and an **automorphism** if it is a bijective endomorphism.

Observation 11.6.4

A homomorphism $\varphi : G \to H$ is an isomorphism iff there is a homomorphism $\psi : H \to G$ with $\psi\varphi = id_G$ and $\varphi\psi = id_H$.

Proof. If φ is an isomorphism, take $\psi = \varphi^{-1}$ (Exercise 11.5.6). Conversely such a ψ is an inverse function, so φ is bijective. ■

Definition 11.6.5 (Kernel and image)

$\ker \varphi = \varphi^{-1}(e) = \{x \in G : \varphi(x) = e\}$ and $\operatorname{im} \varphi = \varphi(G)$.

Observation 11.6.6

For a homomorphism $\varphi : G \to H$: (a) $\ker \varphi \leq G$ and $\operatorname{im} \varphi \leq H$;
(b) if φ is a monomorphism, $G \approx \operatorname{im} \varphi$;
(c) if $K \leq G$, then $\varphi(K) \leq H$.

Proof [Supplement]. (a) If $\varphi x = \varphi y = e$, then $\varphi(xy^{-1}) = e$, and $e \in \ker \varphi$ (Exercise 11.3.5).

(c) $\varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}) \in \varphi(K)$; the case $K = G$ gives $\operatorname{im} \varphi \leq H$.

(b) $\varphi : G \to \operatorname{im} \varphi$ is a bijective homomorphism. ■

Observation 11.6.7

For a homomorphism φ the following are equivalent: (1) φ is a monomorphism; (2) $\varphi x = \varphi y \Rightarrow x = y$; (3) $\varphi x = \varphi e = e \Rightarrow x = e$; (4) $\ker \varphi = \{e\}$.

Proof. All but (3)⇒(2) are immediate. If (3) holds and $\varphi x = \varphi y$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = e$, so $xy^{-1} = e$, $x = y$. ■

Observation 11.6.8

For a homomorphism $\varphi : G \to H$ and $S \subseteq G$: (a) $\varphi\langle S\rangle = \langle \varphi S\rangle$;
(b) if $G = \langle S\rangle$, then $\operatorname{im} \varphi = \langle \varphi S\rangle$;

(c) φ is an epimorphism iff it maps a generating set of G to a generating set of H.

Proof. (a) Images of words in $S \cup S^{-1}$ are words in $\varphi S \cup (\varphi S)^{-1}$ (Proposition 11.3.17), so $\varphi\langle S\rangle \leq \langle \varphi S\rangle$; conversely $\varphi\langle S\rangle \leq H$ contains φS (Observation 11.6.6(c)).

(b) The case $\langle S\rangle = G$.

(c) By (b), $\operatorname{im}\varphi = H$ iff $\langle \varphi S\rangle = H$. ■

Exercise 11.6.9

If $\varphi : G \to H$ is a homomorphism and G is cyclic, then $\operatorname{im}\varphi$ is cyclic.

Solution. If $G = \langle x\rangle$, then $\operatorname{im}\varphi = \langle \varphi x\rangle$ by Observation 11.6.8(b). □

Exercise 11.6.10

Let $\varphi(g) = g^{-1}$ and $\psi(g) = g^2$ on G. (a) φ is a bijection; is it a homomorphism?
(b) Is ψ a homomorphism?

Solution. (a) $\varphi = \varphi^{-1}$; it is a homomorphism iff $y^{-1}x^{-1} = x^{-1}y^{-1}$ for all x, y, iff G is commutative.

(b) $xyxy = xxyy$ iff $yx = xy$: again iff G is commutative. □

Exercise 11.6.11

(a) For $n \geq 2$, $\operatorname{sgn} : S_n \to \mu_2$ is an epimorphism.
(b) Describe $\ker(\operatorname{sgn})$.

Solution. (a) $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}\sigma \operatorname{sgn}\tau$, and $\operatorname{sgn}(1\,2) = -1$.

(b) $\ker(\operatorname{sgn}) = A_n$, the even permutations. □

11.7 Cyclic groups

Motivation. Up to isomorphism a cyclic group is $\mathbb{Z}$ or μ_n.

Example 11.7.1

(a) $\mathbb{Z} = \langle 1\rangle$ is an infinite cyclic group, and so is $n\mathbb{Z} = \langle n\rangle$ for $n > 0$; $a \mapsto na$ is an isomorphism $\mathbb{Z} \approx n\mathbb{Z}$ onto a proper subgroup.
(b) For $\zeta = e^{2\pi i/n}$, $\mu_n = \{1, \zeta, \ldots, \zeta^{n-1}\} = \langle \zeta\rangle$ is cyclic of order n.

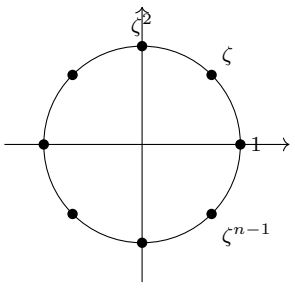


Figure. *μ_n ($n = 8$): the vertices of a regular n-gon on the unit circle.*

Definition 11.7.2 (Order of an element)

The **order** of $x \in G$ is $|x| = |\langle x \rangle|$.

Observation 11.7.3

If x has finite order, then $|x| = n$ iff (1) $x^n = 1$ and (2) $x^m \neq 1$ for $0 < m < n$; i.e. n is the least positive integer with $x^n = 1$.

Proof. $\langle x \rangle$ is finite, so $x^i = x^j$ for some $i < j$: $\{a \geq 1 : x^a = 1\} \neq \emptyset$; let n be its least element (the n of (1), (2)). For $b = qn + r$, $0 \leq r < n$, $x^b = (x^n)^q x^r = x^r$, so $\langle x \rangle = \{1, x, \ldots, x^{n-1}\}$; these are distinct, since $x^k = x^\ell$, $0 \leq k < \ell < n$, gives $x^{\ell-k} = 1$. So $|x| = n$. ■

Definition 11.7.4 (Order, second definition)

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, is a homomorphism (exponent laws), so $\ker \gamma_x = n\mathbb{Z}$ for a unique $n \geq 0$ (Theorem 11.4.2). Set $|x| = \infty$ if $n = 0$ and $|x| = n$ otherwise.

Exercise 11.7.5

Definitions 11.7.2 and 11.7.4 agree.

Solution. If $\ker \gamma_x = 0$, γ_x is injective and $\langle x \rangle = \operatorname{im} \gamma_x$ is infinite. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, then n is the least positive integer with $x^n = 1$, and $|\langle x \rangle| = n$ by Observation 11.7.3. □

Exercise 11.7.6

(a) $|x| = 1$ iff $x = 1$.
(b) $|x| = 2$ iff $1 \neq x = x^{-1}$.
(c) Every nonzero element of $(\mathbb{R}, +)$ has order ∞.
(d) A k-cycle in S_n has order k.

Solution. (a),

(b) by Observation 11.7.3.

(c) $na \neq 0$ for $n \neq 0$, $a \neq 0$.

(d) $(i_1 \ldots i_k)^m$ sends $i_1 \mapsto i_{1+m}$ (indices mod k), which is i_1 iff $k \mid m$; the k-th power is id. □

Exercise 11.7.7

If $|x| = n$ and $a, b \in \mathbb{Z}$, then $x^a = x^b$ iff $n \mid (a - b)$.

Solution. $x^a = x^b \iff x^{a-b} = 1 \iff a - b \in \ker \gamma_x = n\mathbb{Z}$. □

Exercise 11.7.8

Every subgroup of a cyclic group is cyclic.

Solution. Let $G = \langle x \rangle$, $H \leq G$. Then $\gamma_x : \mathbb{Z} \to G$ is onto, $\gamma_x^{-1}(H) \leq \mathbb{Z}$ equals $m\mathbb{Z}$ (Theorem 11.4.2), and $H = \gamma_x(m\mathbb{Z}) = \langle x^m \rangle$. □

Exercise 11.7.9

Up to isomorphism the cyclic groups are $\mathbb{Z}$ and the μ_n.

Solution. Let $G = \langle x \rangle$. If $\ker \gamma_x = 0$, $\gamma_x : \mathbb{Z} \to G$ is an isomorphism. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, $\varphi(x^a) = \zeta^a$ is well defined and injective since $x^a = x^b \iff n \mid a-b \iff \zeta^a = \zeta^b$ (Exercise 11.7.7); it is onto and a homomorphism. □

11.8 Examples of groups and homomorphisms

Motivation. Symmetric groups, Cayley's theorem, conjugacy, automorphisms and products.

Exercise 11.8.1

(a) Composites of homomorphisms are homomorphisms.
(b) If $L \in \mathfrak{L}(V, W)$, then $L : (V, +) \to (W, +)$ is a homomorphism.

Solution. (a) $\varphi\psi(xy) = \varphi(\psi x \psi y) = \varphi\psi x \, \varphi\psi y$.

(b) $L(v + w) = Lv + Lw$. □

Exercise 11.8.2

(a) The constant map $\mathbf{1}(g) = 1$ is a homomorphism, the **trivial map** (written 0 for abelian groups). (b) For $H \leq G$, the inclusion $\iota_H : H \to G$ is a monomorphism, the **natural embedding**. (c) For $H \leq G$ and a homomorphism $\varphi : G \to G'$, the restriction $\varphi|_H$ is a homomorphism.

Solution. All follow directly from the definitions ($1 \cdot 1 = 1$; the operation of H is that of G). □

Example 11.8.3

S_n is generated by transpositions, and $A_n = \ker(\mathrm{sgn}) \leq S_n$ (Observation 11.6.6).

Example 11.8.4

(a) In S_3, put $\sigma = (1\,2)$, $\tau = (1\,2\,3)$. Then $\sigma^2 = id = \tau^3$, $\tau^2 = (1\,3\,2)$, $\sigma\tau = (2\,3)$, $\sigma\tau^2 = (1\,3)$, so

$$S_3 = \{id, \tau, \tau^2, \sigma, \sigma\tau, \sigma\tau^2\} = \{\sigma^i\tau^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\} = \langle \sigma, \tau \rangle.$$

(b) $\sigma^{-1}\tau\sigma = \tau^2 = \tau^{-1}$, i.e. $\tau\sigma = \sigma\tau^2$, moves σ to the left, so the relations

$$\sigma^2 = 1, \qquad \tau^3 = 1, \qquad \tau\sigma = \sigma\tau^2$$

determine the whole multiplication table of S_3; e.g. $(\sigma\tau)(\sigma\tau^2) = \sigma(\tau\sigma)\tau^2 = \sigma\sigma\tau^2\tau^2 = \tau$.
(c) This is the plane rule of Observation 9.5.9; see Example 11.9.23.

Definition 11.8.5 (Symmetric group on a set)

For a nonempty set X, $S_X = \{\sigma : X \to X \text{ bijection}\}$ is a group under composition, the **symmetric group** on X.

Exercise 11.8.6

(a) How should "X and Y are the same as sets" be defined?
(b) If X, Y are isomorphic as sets, then $S_X \approx S_Y$.

Solution. (a) There is a bijection $f : X \to Y$ (a renaming).

(b) $\sigma \mapsto f \circ \sigma \circ f^{-1}$ is a bijection $S_X \to S_Y$ with inverse $\tau \mapsto f^{-1}\tau f$, and $f\sigma\sigma' f^{-1} = (f\sigma f^{-1})(f\sigma' f^{-1})$. □

Exercise 11.8.7

(a) The **automorphism group** $\mathrm{Aut}(G) = \{\varphi : G \to G \text{ automorphism}\}$ is a subgroup of S_G.
(b) For $g \in G$, $\mathrm{Int}(g)(x) = gxg^{-1}$ defines $\mathrm{Int}(g) \in \mathrm{Aut}(G)$, an **inner automorphism**; $\mathrm{Int}(G) = \{\mathrm{Int}(g) : g \in G\} \leq \mathrm{Aut}(G)$.
(c) (Remark) Automorphisms that are not inner are **outer**.

Solution. (a) Composites and inverses of automorphisms are automorphisms (Exercise 11.5.6), and $id \in \mathrm{Aut}(G)$. (b) $gxyg^{-1} = gxg^{-1}gyg^{-1}$; inverse $\mathrm{Int}(g^{-1})$; $\mathrm{Int}(g) \circ \mathrm{Int}(h) = \mathrm{Int}(gh)$. □

Theorem 11.8.8 (Cayley)

$\lambda : G \to S_G$, $\lambda(x) = \lambda_x$ with $\lambda_x(g) = xg$, is a monomorphism. Hence every group is isomorphic to a subgroup of a symmetric group.

Proof. $\lambda_x \in S_G$ (Observation 11.2.15); $\lambda_{xy}(g) = xyg = \lambda_x\lambda_y(g)$; and $\lambda_x = id$ gives $x = \lambda_x(1) = 1$. Apply Observations 11.6.7, 11.6.6(b). ■

(Right multiplication gives $\rho_{xy} = \rho_y \circ \rho_x$, an anti-homomorphism.)

Definition 11.8.9 (Conjugacy)

$x \sim y$ iff $y = g^{-1}xg$ for some $g \in G$; this is an equivalence relation, the **conjugacy relation**, and its classes are the **conjugacy classes** (x is **conjugate** to y).

Exercise 11.8.10

(a) The conjugacy class of 1 is $\{1\}$.
(b) If G is commutative, conjugacy is equality.
(c) How many conjugacy classes does S_3 have?

Solution. (a) $g^{-1}1g = 1$.

(b) $g^{-1}xg = x$.

(c) Three: $\{id\}$, the transpositions, the 3-cycles (cycle type is preserved; $\sigma^{-1}\tau\sigma = \tau^2$, $\tau^{-1}\sigma\tau = \sigma\tau^2$, ...). □

Exercise 11.8.11

(a) For $H \leq G$, $g \in G$: $g^{-1}Hg = \{g^{-1}hg : h \in H\} \leq G$.
(b) On the set $\mathcal{S}$ of subgroups of G, $H \sim K$ iff $K = g^{-1}Hg$ for some g is an equivalence relation (K is a **conjugate subgroup** of H).

Solution. (a) $g^{-1}Hg = \mathrm{Int}(g^{-1})(H)$, the image of a subgroup under a homomorphism.

(b) Reflexive with $g = 1$; symmetric since $H = gKg^{-1}$; transitive since $h^{-1}(g^{-1}Hg)h = (gh)^{-1}H(gh)$. □

Exercise 11.8.12

(a) Give the Cartesian product $G = G_1 \times \cdots \times G_n$ of groups a natural group structure. (b) The i-th **projection** $\pi_i : G \to G_i$ is an epimorphism and the j-th **embedding** $\iota_j : G_j \to G$ is a monomorphism.

Solution. (a) Componentwise multiplication $(g_i)(h_i) = (g_i h_i)$; identity $(1, \ldots, 1)$, inverse (g_i^{-1}).
(b) $\pi_i(g) = g_i$; $\iota_j(g) = (1, \ldots, g, \ldots, 1)$ (g in position j). □

Exercise 11.8.13

(a) $G \times H \approx H \times G$.
(b) $(G \times H) \times K \approx G \times (H \times K) \approx G \times H \times K$.

Solution. $(g, h) \mapsto (h, g)$ and $((g, h), k) \mapsto (g, h, k)$ and $(g, (h, k)) \mapsto (g, h, k)$ are bijective homomorphisms. □

Exercise 11.8.14

$G_1 \times \cdots \times G_n$ is commutative iff all G_i are commutative.

Solution. Componentwise; conversely $G_i \approx \iota_i(G_i)$ is a subgroup. □

Exercise 11.8.15

(a) $\mu_3 \times \mu_3 \not\approx \mu_9$.
(b) $\mu_2 \times \mu_3 \approx \mu_6$.

Solution. (a) Elements of $\mu_3 \times \mu_3$ have order dividing 3; μ_9 has one of order 9.
(b) $(-1, \omega)^k = 1$ ($\omega = e^{2\pi i/3}$) iff $2 \mid k$ and $3 \mid k$, so it has order 6 and $\mu_2 \times \mu_3 \approx \mu_6$ (Exercise 11.7.9). □

Exercise 11.8.16

In Exercise 11.8.12(b): (a) find $\ker \pi_i$ and $\operatorname{im} \iota_j$;
(b) $\pi_i \circ \iota_i = id$;
(c) $\pi_i \circ \iota_j = \mathbf{1}$ for $i \neq j$.

Solution. (a) $\ker \pi_i = \{g : g_i = 1\}$ and $\operatorname{im} \iota_j = \{g : g_k = 1 \text{ for } k \neq j\}$. (b), (c) are direct. □

Remark 11.8.17

The pointwise product $(\varphi\psi)(g) = \varphi(g)\psi(g)$ of homomorphisms need not be one: $\varphi(x)\varphi(y)\psi(x)\psi(y) \neq \varphi(x)\psi(x)\varphi(y)\psi(y)$ unless H is commutative.

11.9 Linear groups

Motivation. A basis identifies $\mathrm{GL}(V)$ with $\mathrm{GL}_n(F)$, and every finite group is a matrix group.

Example 11.9.1

(a) $\mathrm{GL}_n(F) = \{A \in \mathfrak{M}_{n,n}(F) : A \text{ invertible}\}$, the **general linear group**, is a group under matrix multiplication; $F^\times = \mathrm{GL}_1(F) = F - \{0\}$ is the **unit** (multiplicative) **group** of F.
(b) Subgroups of $\mathrm{GL}_n(F)$ are **linear** (**matrix**) **groups**.
(c) The **special linear group** $\mathrm{SL}_n(F) = \{A \in \mathrm{GL}_n(F) : \det A = 1\} = \ker(\det : \mathrm{GL}_n(F) \to F^\times)$ is a subgroup.

Example 11.9.2

For a finite-dimensional F-vector space V: (a) $\mathrm{GL}(V) = \{L \in \mathfrak{L}(V,V) : L \text{ invertible}\}$ is a group under composition; (b) its subgroups are **linear groups**;
(c) $\mathrm{SL}(V) = \{L \in \mathrm{GL}(V) : \det L = 1\} \leq \mathrm{GL}(V)$.

Example 11.9.3

(a) For $\dim V = n$ and a basis $\mathfrak{B}$, the vector space isomorphism $\Psi_{\mathfrak{B}}^{\mathfrak{B}} : \mathfrak{L}(V,V) \to \mathfrak{M}_{n,n}(F)$, $L \mapsto [L]_{\mathfrak{B}}^{\mathfrak{B}}$, restricts to a group isomorphism $\mathrm{GL}(V) \to \mathrm{GL}_n(F)$ ($[LM] = [L][M]$).

$$\begin{array}{ccc} \mathrm{GL}(V) & \hookrightarrow & \mathfrak{L}(V,V) \\ \Psi_{\mathfrak{B}}^{\mathfrak{B}}\downarrow & & \downarrow\Psi_{\mathfrak{B}}^{\mathfrak{B}} \\ \mathrm{GL}_n(F) & \hookrightarrow & \mathfrak{M}_{n,n}(F) \end{array}$$

(c) Since $\det L = \det[L]_{\mathfrak{B}}^{\mathfrak{B}}$, it also restricts to $\mathrm{SL}(V) \approx \mathrm{SL}_n(F)$.

Example 11.9.4

If $\dim V = \dim W = n$, then $\mathrm{GL}(V) \approx \mathrm{GL}(W)$. (a) Via bases: $(\Psi_{\mathfrak{C}}^{\mathfrak{C}})^{-1} \circ \Psi_{\mathfrak{B}}^{\mathfrak{B}}$. (b) Via a vector space isomorphism $\varphi : V \to W$ (e.g. with $\varphi(\mathfrak{B}) = \mathfrak{C}$): $\Gamma(L) = \varphi \circ L \circ \varphi^{-1}$, with $\Gamma(LL') = \Gamma(L)\Gamma(L')$ and $\Gamma^{-1}(M) = \varphi^{-1} M \varphi$. (c) If $\varphi(\mathfrak{B}) = \mathfrak{C}$ they coincide: $[\varphi L \varphi^{-1}]_{\mathfrak{C}}^{\mathfrak{C}} = [\varphi]_{\mathfrak{B}}^{\mathfrak{C}}[L]_{\mathfrak{B}}^{\mathfrak{B}}[\varphi^{-1}]_{\mathfrak{C}}^{\mathfrak{B}} = [L]_{\mathfrak{B}}^{\mathfrak{B}}$, as $[\varphi]_{\mathfrak{B}}^{\mathfrak{C}} = I$.

Example 11.9.5

$\mu_2 < \mathbb{R}^\times < \mathbb{C}^\times$ and $\mu_n < \mathbb{C}^\times$.

Example 11.9.6

Theorem 1.3.6(3) (every invertible matrix is a product of elementary matrices) says: $\mathrm{GL}_n(F)$ is generated by elementary matrices.

Example 11.9.7

$\mathbf{T}_n(F) = \{A \in \mathrm{GL}_n(F) : A \text{ upper triangular}\}$ and $\mathbf{V}_n(F) = \{A \in \mathbf{T}_n(F) : A \text{ unipotent}\}$ (diagonal entries 1) are matrix groups.

Proof [Supplement]. Products and inverses (back substitution) of invertible upper triangular matrices are upper triangular, with diagonal entries multiplied resp. inverted. ■

Exercise 11.9.8

$(F,+)$ is a matrix group: $(F,+) \approx \mathbf{V}_2(F)$.

Solution. $a \mapsto \left(\begin{smallmatrix}1 & a\\ 0 & 1\end{smallmatrix}\right)$ is a bijection, and $\left(\begin{smallmatrix}1 & a\\ 0 & 1\end{smallmatrix}\right)\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right) = \left(\begin{smallmatrix}1 & a+b\\ 0 & 1\end{smallmatrix}\right)$. □

Exercise 11.9.9

The group $(F^3,\bullet)$ of Exercise 11.1.14 is isomorphic to $\mathbf{V}_3(F)$.

Solution. $(a_1,a_2,a_3)^t \mapsto \begin{pmatrix}1 & a_1 & a_2\\ 0 & 1 & a_3\\ 0 & 0 & 1\end{pmatrix}$; the product has entries a_1+b_1, $a_2+b_2+a_1b_3$, a_3+b_3, the formula for $\bullet$. □

Example 11.9.10

Conjugacy in $\mathrm{GL}_n(F)$ is the similarity relation of §5.5.

Exercise 11.9.11

If $L \in \mathrm{GL}(V)$ is unipotent and $L \neq I$, then $|L| = \infty$ in $\mathrm{GL}(V)$ (over $F = \mathbb{R}$ or $\mathbb{C}$).

Solution. $L = I + N$, $0 \neq N$ nilpotent, $k \geq 1$ largest with $N^k \neq 0$. Then $L^m - I = \sum_{1\leq j\leq k} \binom{m}{j} N^j$ with coefficient $m \neq 0$ at N, and $I, N, \ldots, N^k$ are independent (for $\sum c_j N^j = 0$ with first nonzero c_r, multiply by N^{k-r}: $c_r N^k = 0$). So $L^m \neq I$ for $m \geq 1$. □

Exercise 11.9.12

For fixed $C \in \mathrm{GL}_n(\mathbb{R})$, $\varphi(A) = CAC^{-1}$ and $\psi(A) = (A^t)^{-1}$ belong to $\mathrm{Aut}(\mathrm{SL}_n(\mathbb{R}))$. If $C \notin \mathrm{SL}_n(\mathbb{R})$, are φ, ψ outer automorphisms?

Solution. Both preserve $\mathrm{SL}_n(\mathbb{R})$ (det is unchanged) and products ($((AB)^t)^{-1} = (A^t)^{-1}(B^t)^{-1}$), and are bijective ($\psi^2 = id$, $\varphi^{-1} = \mathrm{Int}(C^{-1})$).

[Supplement] φ. $C' \in \mathrm{SL}_n(\mathbb{R})$ induces φ iff $C^{-1}C'$ commutes with $\mathrm{SL}_n(\mathbb{R})$ (all type-3 elementary matrices), i.e. is λI. So φ is inner iff $\lambda^n \det C = 1$ for some real λ: always for odd n; never for even n with $\det C < 0$ (outer).

ψ. For $n = 2$, $(A^t)^{-1} = JAJ^{-1}$ with $J = \left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$: inner. For $n \geq 3$, outer, since inner automorphisms preserve trace: $A = \mathrm{diag}(2, 3, \frac{1}{6}, 1, \ldots, 1)$ has trace $5\frac{1}{6} + (n-3)$, $\psi(A) = \mathrm{diag}(\frac{1}{2}, \frac{1}{3}, 6, 1, \ldots, 1)$ has trace $6\frac{5}{6} + (n-3)$. □

Exercise 11.9.13

For the invertible diagonal matrices $\mathbf{D}_n(F)$: (a) $\mathbf{D}_n(F) \approx F^\times \times \cdots \times F^\times$ (n factors);
(b) $\mathbf{D}_n(F) \cap \mathrm{SL}_n(F) \approx (F^\times)^{n-1}$ for $n \geq 2$.

Solution. (a) $\mathrm{diag}(a_1, \ldots, a_n) \mapsto (a_1, \ldots, a_n)$.

(b) The last entry is determined by $a_n = (a_1 \cdots a_{n-1})^{-1}$; project to the first $n-1$ entries. □

Exercise 11.9.14

(a) If $m \leq n$, $\mathrm{GL}_m(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.
(b) If $n_1 + \cdots + n_r = n$, $\mathrm{GL}_{n_1}(F) \times \cdots \times \mathrm{GL}_{n_r}(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.

Solution. (b) $(A_1, \ldots, A_r) \mapsto A_1 \oplus \cdots \oplus A_r$ (block diagonal) is a monomorphism. (a) is the case $A \mapsto A \oplus I_{n-m}$. □

Exercise 11.9.15

(a) $\mathbf{SO}(n) < \mathbf{O}(n) < \mathrm{GL}_n(\mathbb{R})$;
(b) $\mathbf{SU}(n) < \mathbf{U}(n) < \mathrm{GL}_n(\mathbb{C})$;
(c) $\mathbf{SO}(n) = \mathrm{SL}_n(\mathbb{R}) \cap \mathbf{O}(n)$;
(d) $\mathbf{SU}(n) = \mathrm{SL}_n(\mathbb{C}) \cap \mathbf{U}(n)$.

Solution. By Exercises 9.3.3, 9.3.10, 10.6.10 and the definitions. □

Example 11.9.16

$\Psi_{\mathcal{E}}^{\mathcal{E}}$ restricts to $\mathbf{O}(\mathbb{R}^n) \approx \mathbf{O}(n)$ (dot product), and so does $\Psi_{\mathfrak{B}}^{\mathfrak{B}}$ for any orthonormal $\mathfrak{B}$ (Remark 10.6.4, Exercise 9.6.11).

Exercise 11.9.17

(a) $\mathbf{U}(\mathbb{C}^n) \approx \mathbf{U}(n)$.
(b) $\mathbf{SO}(\mathbb{R}^n) \approx \mathbf{SO}(n)$.
(c) Define $\mathbf{SU}(\mathbb{C}^n)$ and show $\mathbf{SU}(\mathbb{C}^n) \approx \mathbf{SU}(n)$.

Solution. As in Example 11.9.16, restricting $\Psi_{\mathcal{E}}^{\mathcal{E}}$; for (b), (c) also $\det L = \det[L]_{\mathcal{E}}^{\mathcal{E}}$, with $\mathbf{SU}(\mathbb{C}^n) = \{L \in \mathbf{U}(\mathbb{C}^n) : \det L = 1\}$. □

Example 11.9.18

If $\varphi : V \to W$ is an inner product space isomorphism, $\Gamma(L) = \varphi L \varphi^{-1}$ (Example 11.9.4) restricts to $\mathbf{O}(V) \approx \mathbf{O}(W)$ (resp. $\mathbf{U}$), since L preserves inner products iff $\varphi L \varphi^{-1}$ does.

Example 11.9.19

The **circle group** $S^1 = \{X \in \mathbb{R}^2 : \|X\| = 1\} = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times$, and $\varphi : \mathbf{SO}(2) \to S^1$, $\varphi(R_\theta) = e^{i\theta}$, is a group isomorphism.

Exercise 11.9.20

$\mathbf{SO}(2) \approx S^1 \approx \mathbf{U}(1)$.

Solution. $\varphi(R_\theta R_\eta) = \varphi(R_{\theta+\eta}) = e^{i(\theta+\eta)} = e^{i\theta}e^{i\eta}$, and φ is bijective (Theorem 9.5.2). $\mathbf{U}(1) = \{z \in \mathbb{C} : \bar{z}z = 1\} = S^1$. □

Exercise 11.9.21

$S_n \approx \{I_\sigma \in \mathfrak{M}_{n,n}(F) : \sigma \in S_n\} \approx \{P_\sigma \in \mathfrak{L}(V,V) : \sigma \in S_n\}$ ($\dim V = n$), via permutation matrices and permutation operators.

Solution. The permutation matrix I_σ (with $I_\sigma e_j = e_{\sigma(j)}$) satisfies $I_\sigma I_\tau = I_{\sigma\tau}$, and $\sigma \mapsto I_\sigma$ is injective; similarly for P_σ on a basis. □

Corollary 11.9.22

Every finite group is isomorphic to a linear (matrix) group.

Proof. By Cayley's theorem G embeds in $S_G \approx S_n$ ($n = |G|$), and S_n embeds in $\mathrm{GL}_n(F)$ by Exercise 11.9.21. ■

Example 11.9.23

In $\mathbf{O}(2)$ let $R = R_{2\pi/3}$, $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $D_3 = \langle R, S\rangle \leq \mathbf{O}(2)$.
(a) The plane rule gives $RS = SR^2$. D_3 consists of words in R, R^2, S; moving each S left gives $D_3 = \{S^i R^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\}$, six distinct matrices (determinants, first columns). $S^2 = I$, $R^3 = I$, $RS = SR^2$ determine the table; e.g. $(SR)(SR^2) = S(RS)R^2 = S^2R^4 = R$.
(b) These are the relations $\sigma^2 = 1$, $\tau^3 = 1$, $\tau\sigma = \sigma\tau^2$ of S_3 (Example 11.8.4).
(c) So $\sigma^i\tau^j \mapsto S^iR^j$ is a table-preserving bijection: $S_3 \approx D_3$.

Exercise 11.9.24

For $n \geq 3$, the **dihedral group** $D_n = \langle R_{2\pi/n}, S\rangle \leq \mathbf{O}(2)$ has $|D_n| = 2n$.

Solution. $R^n = I$, $S^2 = I$, $RS = SR^{-1}$ give $D_n = \{S^iR^j : 0 \leq i \leq 1, 0 \leq j < n\}$ as in Example 11.9.23; distinct since $\det(S^iR^j) = (-1)^i$ and the R^j are distinct. □

Exercise 11.9.25

(a) For $\sigma = (2\,5)(3\,4)$, $\tau = (1\,2\,3\,4\,5) \in S_5$, $G_5 = \langle\sigma,\tau\rangle \approx D_5$. (b) For $\sigma = (2\,6)(3\,5)$, $\tau = (1\,2\,3\,4\,5\,6) \in S_6$, $G_6 = \langle\sigma,\tau\rangle \approx D_6$. (c) Interpret (a),
(b) as permutations of the vertices of a regular pentagon/hexagon, and generalize to the regular n-gon.

Solution. $\sigma^2 = 1$, $\tau^n = 1$, $\sigma\tau\sigma = \tau^{-1}$ (e.g. $\sigma\tau\sigma(1) = \sigma(2) = n = \tau^{-1}(1)$), so $\sigma^i\tau^j \mapsto S^iR^j$ is an isomorphism onto D_n (Example 11.9.23). With vertices of the regular n-gon numbered counterclockwise, τ is rotation by $2\pi/n$ and σ the reflection through vertex 1: D_n is the symmetry group of the n-gon. □

Chapter 12. Quotients

12.1 Cosets

Motivation. Cosets of H partition G into pieces of size $|H|$: Lagrange's theorem.

Definition 12.1.1 (Cosets)

For $H \leq G$, $x \in G$: $xH = \{xh : h \in H\}$ (or $x + H$ additively) is the **left coset** of H with **representative** x, and Hx the **right coset**. "Coset" means left coset from now on.

Observation 12.1.2

For $H \leq G$, $x, y \in G$: (a) $x \in xH$, so $G = \bigcup_{x \in G} xH$;
(b) $xH = yH$ iff $y^{-1}x \in H$;
(c) $xH \cap yH = \emptyset$ if $xH \neq yH$.

Proof. (a) $x = x1$. (b) If $xH = yH$, then $x = yh'$ by (a), so $y^{-1}x = h' \in H$. Conversely, if $y^{-1}x = h_0 \in H$, then $xh = yh_0h$ gives $xH \subseteq yH$, and $x^{-1}y = h_0^{-1}$ gives $yH \subseteq xH$. (c) If $xh_1 = yh_2$, then $y^{-1}x = h_2h_1^{-1} \in H$, so $xH = yH$ by (b). ■

So the cosets partition G by the equivalence relation

$$x \equiv y \pmod{H} \iff xH = yH \iff y^{-1}x \in H,$$

with classes $\bar{x} = xH$. The set of cosets $G/H = \frac{G}{H} = \{xH : x \in G\} = \{\bar{x} : x \in G\}$ is the **coset space** of G modulo H. For a **complete set of representatives** $\{x_i\}_{i \in I}$, $G/H = \{\bar{x}_i\}$ and $G = \coprod_i x_i H$ (**coset decomposition**).

Example 12.1.3

$1H = H$ is always a coset.

Exercise 12.1.4

$|G/G| = 1$ and $|G/\{1\}| = |G|$.

Solution. $xG = G$ for all x; $x\{1\} = \{x\}$ are all distinct. □

Example 12.1.5

The solution set $X_0 + \ker L_A$ of $AX = B$ is a coset of the subgroup $\ker L_A$ of the abelian group $(F^n, +)$.

Example 12.1.6

For $n\mathbb{Z} \leq \mathbb{Z}$ $(n \neq 0)$: $\bar{a} = a + n\mathbb{Z} = \bar{b} \iff a - b \in n\mathbb{Z} \iff n \mid (a - b)$, i.e. equal remainders mod n. So $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \ldots, \overline{n-1}\}$; $\bar{a}$ is the **residue class** of a, and a complete set of representatives a **complete residue system**.

Example 12.1.7

In S_3 (notation of Example 11.8.4) with $H = \{1, \sigma\}$: $\tau H = \{\tau, \sigma\tau^2\}$, $\tau^2 H = \{\tau^2, \sigma\tau\}$, so $S_3/H = \{\bar{1}, \bar{\tau}, \overline{\tau^2}\}$. Since $\sigma\tau \neq \tau\sigma$, $\tau H \neq H\tau$: left and right cosets need not coincide.

Exercise 12.1.8

For $K = \{1, \tau, \tau^2\} \leq S_3$: $|S_3/K| = 2$, and left and right cosets coincide (e.g. $\sigma K = K\sigma$).

Solution. $S_3 = K \amalg \sigma K$ with $\sigma K = \{\sigma, \sigma\tau, \sigma\tau^2\}$, the three transpositions. Since $K\sigma$ is a coset of size 3 disjoint from K, it equals σK. □

Observation 12.1.9

For $H \leq G$, $x, y \in G$: $|xH| = |yH| = |H|$.

Proof. $\lambda_x : H \to xH$, $h \mapsto xh$, is a bijection (injective by cancellation, onto by definition). ■

Observation 12.1.10 (Lagrange's theorem)

If G is finite and $H \leq G$, then $|H|$ divides $|G|$ and $|G/H| = \frac{|G|}{|H|}$.

Proof. The distinct cosets partition G and each has $|H|$ elements (Observation 12.1.9). ■

Notation 12.1.11

If $|G/H| < \infty$ (G may be infinite), $[G : H] = |G/H|$ is the **index** of H in G.

Observation 12.1.12

If $K \leq H \leq G$ with $|G/H|, |H/K| < \infty$, then $[G : K] = [G : H][H : K]$.

Proof. For representatives $\{g_i\}$ of G/H and $\{h_j\}$ of H/K, $G = \coprod_i g_i H = \bigcup_{i,j} g_i h_j K$, disjointly: $g_i h_j K = g_{i'} h_{j'} K$ forces $g_i H = g_{i'} H$, so $i = i'$, then $h_j K = h_{j'} K$, so $j = j'$. ■

Example 12.1.13

By Example 11.3.6(e) and Lagrange, $\mu_m \leq \mu_n$ iff $m \mid n$.

Observation 12.1.14

If G is finite and $x \in G$, then $|x|$ divides $|G|$.

Proof. $|x| = |\langle x \rangle|$; apply Lagrange. ■

Proposition 12.1.15

If $|G| = p$ is prime, then G is cyclic, $G \approx \mu_p$.

Proof. Take $x \neq 1$. Then $|x| \neq 1$ divides p, so $|x| = p$, $\langle x \rangle = G$, and $\langle x \rangle \approx \mu_p$ (Exercise 11.7.9). ■

Example 12.1.16

Proper nontrivial subgroups of S_3 have order 2 or 3, hence are cyclic (Proposition 12.1.15): $\{id, (1\,2)\}$, $\{id, (2\,3)\}$, $\{id, (1\,3)\}$ and $\{id, (1\,2\,3), (1\,3\,2)\}$.

Exercise 12.1.17

If $|G|$ is even, G has an element of order 2.

Solution. Pairing x with x^{-1} shows $\{x : x = x^{-1}\}$ has even size; it contains 1, hence some $x \neq 1$ with $|x| = 2$. □

Remark 12.1.18

But if $d \mid |G|$, G need not have an element, or even a subgroup, of order d.

Exercise 12.1.19

(a) S_4 and A_4 have no elements of order 6, 8 or 12. (b) S_4 has exactly four subgroups of order 6. (c) A_4 has no subgroup of order 6. (d) How many subgroups of order 8 does S_4 have? (e) Of order 12?

Solution. (a) Elements of S_4 have cycle types $1, (2), (3), (4), (2,2)$, of orders $1, 2, 3, 4, 2$.

(b) The four stabilizers of a point ($\approx S_3$). Conversely, $|H| = 6$ gives a 3-cycle fixing some i (Cauchy, Theorem 12.1.20); the order-2 elements of H normalize its subgroup, so also fix i.

(c) Such H would have index 2, hence be normal (Observation 12.2.9) and contain all squares, so all eight 3-cycles ($c = (c^2)^2$): too many.

(d) Three (the dihedral groups D_4, Sylow 2-subgroups).

(e) One, namely A_4 (a subgroup of index 2 contains all squares, which generate A_4). □

Theorem 12.1.20 (Cauchy)

If a prime p divides $|G|$, then G has an element of order p.

Proof [Supplement] (McKay; the book defers it to abstract algebra). Let $X = \{(x_1, \ldots, x_p) \in G^p : x_1 \cdots x_p = 1\}$; $|X| = |G|^{p-1} \equiv 0 \pmod p$. Since $x_2 \cdots x_p x_1 = x_1^{-1}(x_1 \cdots x_p)x_1 = 1$, cyclic shifts preserve X; their orbits have size p or 1, the latter being $(x, \ldots, x)$ with $x^p = 1$. So p divides $|\{x : x^p = 1\}| \geq 1$, which is thus $\geq p \geq 2$: some $x \neq 1$ has $|x| = p$. ■

12.2 Normal subgroups and quotient groups

Motivation. $\bar{x}\,\bar{y} = \overline{xy}$ is well defined exactly when the subgroup is normal.

For $\bar{x}\,\bar{y} = \overline{xy}$ on G/H to be *well defined* we need: $x'^{-1}x, y'^{-1}y \in H \Rightarrow (x'y')^{-1}xy = y'^{-1}x'^{-1}xy \in H$.

Example 12.2.1

For $H = \{1, \sigma\} \leq S_3$ (Example 12.1.7): $\bar{1} = \bar{\sigma}$ and $\bar{\tau} = \overline{\sigma\tau^2}$, but $1 \cdot \tau = \tau$ while $\sigma \cdot \sigma\tau^2 = \tau^2$, and $\bar{\tau} \neq \overline{\tau^2}$. So the operation is not well defined. (For $K = \{1, \tau, \tau^2\}$ it is.)

Definition 12.2.2 (Normal subgroup)

$H \leq G$ is **normal**, $H \trianglelefteq G$, if $g^{-1}hg \in H$ for all $g \in G$, $h \in H$.

Observation 12.2.3

For $H \leq G$, $\bar{x}\,\bar{y} = \overline{xy}$ is well defined on G/H iff $H \trianglelefteq G$.

Proof. If $H \trianglelefteq G$ and $x'^{-1}x, y'^{-1}y \in H$, then $y'^{-1}x'^{-1}xy = (y'^{-1}y)(y^{-1}(x'^{-1}x)y) \in H$. Conversely, if the operation is well defined, take $x = h$, $x' = 1$, $y = y' = g$: then $\bar{h} = \bar{1}$, so $\overline{hg} = \bar{g}$, i.e. $g^{-1}hg \in H$. ■

Proposition 12.2.4

If $N \trianglelefteq G$, then G/N with $\bar{x}\,\bar{y} = \overline{xy}$ is a group, the **quotient group** of G modulo N.

Proof. $(\bar{x}\bar{y})\bar{z} = \overline{xyz} = \bar{x}(\bar{y}\bar{z})$; $\bar{1}$ is the identity; $\bar{x}\,\overline{x^{-1}} = \bar{1} = \overline{x^{-1}}\,\bar{x}$, so $\bar{x}^{-1} = \overline{x^{-1}}$. ■

For $N \trianglelefteq G$: $ng = gn'$ with $n' = g^{-1}ng \in N$.

Observation 12.2.5

For $H \leq G$ the following are equivalent: (1) $H \trianglelefteq G$, i.e. $g^{-1}Hg \leq H$ for all g; (2) $gHg^{-1} = H$ for all g; (3) $g^{-1}Hg = H$ for all g.

Proof. (3)⇔(2): replace g by g^{-1}. (3)⇒(1) is clear. (1)⇒(3): (1) for g^{-1} gives $gHg^{-1} \leq H$, i.e. $H \leq g^{-1}Hg$. ■

Exercise 12.2.6

For $H \leq G$ the following are equivalent: (1) $H \trianglelefteq G$; (2) $\varphi(H) = H$ for every inner automorphism φ; (3) $gH = Hg$ for all g.

Solution. (1)⇔(2) is Observation 12.2.5 with $\varphi = \mathrm{Int}(g)$. (3)⇔ $gHg^{-1} = H$ (multiply by g^{-1} on the right). □

Example 12.2.7

$\{1\} \trianglelefteq G$ and $G \trianglelefteq G$; $G/\{1\} \approx G$ and $G/G \approx \{1\}$.

Example 12.2.8

(a) In a commutative group every subgroup is normal, and G/N is commutative.
(b) If $G = \langle x \rangle$ is cyclic, so is $G/N = \langle \bar{x} \rangle$.

Observation 12.2.9

If $[G : H] = 2$, then $H \trianglelefteq G$.

Proof. $G = H \amalg xH$ for $x \notin H$. For $h \in H$ and $g \notin H$, $g = xh_1$ and $ghg^{-1} = xh_2x^{-1}$, $h_2 = h_1hh_1^{-1} \in H$. If $xh_2x^{-1} \notin H$, then $xh_2x^{-1} = xh_3$, $h_3 \in H$, so $x = h_3^{-1}h_2 \in H$: contradiction. ■

(For index $3, 4, \ldots$ this fails, e.g. Example 12.2.1.)

Example 12.2.10

Groups of order 6. By Cauchy take $|x| = 2$, $|y| = 3$; $N = \langle y \rangle$ has index 2, so $N \trianglelefteq G = \{1, y, y^2, x, xy, xy^2\}$ and $x^{-1}yx \in \{y, y^2\}$. If $x^{-1}yx = y$, $|xy| = 6$ and $G \approx \mu_6$; if $x^{-1}yx = y^2$, $G \approx S_3$ (Examples 11.8.4, 11.9.23).

Observation 12.2.11

If $\varphi : G \to H$ is a homomorphism, then $\ker \varphi \trianglelefteq G$.

Proof [Supplement]. For $k \in \ker \varphi$, $\varphi(g^{-1}kg) = \varphi(g)^{-1}\varphi(g) = e$. ■

(But $\operatorname{im} \varphi$ need not be normal in H.)

Definition 12.2.12 (Natural projection)

For $N \trianglelefteq G$, $\pi : G \to G/N$, $\pi(x) = \bar{x}$, is the **natural projection**.

Exercise 12.2.13

(a) π is an epimorphism.
(b) $\ker \pi = N$.
(c) For the natural embedding $\iota : N \to G$, $\pi \circ \iota = \mathbf{1}$.

Solution. (a) $\pi(xy) = \overline{xy} = \bar{x}\bar{y}$, onto by definition.

(b) $\bar{x} = \bar{1} \iff x \in N$.

(c) is (b). □

The quotient operation is the unique one making π a homomorphism.

Notation 12.2.14

For $1 \leq n \in \mathbb{Z}$, $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, with $\bar{a} + \bar{b} = \overline{a + b}$.

Observation 12.2.15

$\mathbb{Z}_n$ is cyclic, so $\mathbb{Z}_n \approx \mu_n$.

Proof. $\mathbb{Z}_n = \langle \bar{1} \rangle$ has order n; apply Exercise 11.7.9. ■

Exercise 12.2.16

(a) In A_4, $N = \{id, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}$ satisfies $N \approx \mathbb{Z}_2 \times \mathbb{Z}_2$ and $N \trianglelefteq A_4$.
(b) $H = \{id, (1\,2)(3\,4)\}$ satisfies $H \trianglelefteq N$ and $N \trianglelefteq A_4$, but H is not normal in A_4.

Solution. (a) The non-identity elements have order 2 and any two multiply to the third: the Klein 4-group. N is the identity plus all elements of type $(2,2)$, and conjugation preserves cycle type, so N is normal (even in S_4).

(b) $[N : H] = 2$, so $H \trianglelefteq N$; but $(1\,2\,3)^{-1}(1\,2)(3\,4)(1\,2\,3) = (1\,4)(2\,3) \notin H$. □

Remark 12.2.17

This N underlies solving the quartic by radicals; A_5 has no such normal subgroup, hence no quintic formula (Galois theory).

Definition 12.2.18 (Center)

$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$ is the **center** of G.

Exercise 12.2.19

(a) $Z(G) \trianglelefteq G$.
(b) If $\varphi \in \mathrm{Aut}(G)$, then $\varphi(Z(G)) = Z(G)$.

Solution. (a) $Z(G)$ is a subgroup, and $g^{-1}zg = z$ for $z \in Z(G)$.

(b) If z commutes with all g, $\varphi(z)$ commutes with all $\varphi(g)$, i.e. with G; likewise for φ^{-1}. □

Exercise 12.2.20

(a) $Z(\mathrm{GL}_n(F)) = \{cI : c \in F^\times\}$.
(b) $Z(\mathrm{SL}_n(F)) = \{cI : c \in F,\ c^n = 1\}$.

Solution. (a) Scalars are central. If A commutes with $I + E_{ij}$ $(i \neq j)$, it commutes with E_{ij}; $AE_{ij} = E_{ij}A$ forces $A_{ki} = 0$ for $k \neq i$, $A_{jk} = 0$ for $k \neq j$, and $A_{ii} = A_{jj}$. So A is scalar.

(b) The same argument works, since $I + E_{ij} \in \mathrm{SL}_n(F)$; and $cI \in \mathrm{SL}_n(F)$ iff $c^n = 1$. □

Definition 12.2.21 (Projective linear groups)

$\mathbf{PGL}_n(F) = \mathrm{GL}_n(F)/Z(\mathrm{GL}_n(F))$ and $\mathbf{PSL}_n(F) = \mathrm{SL}_n(F)/Z(\mathrm{SL}_n(F))$; similarly **PSO**, **PSU**, ...

GL, **PGL**, SL, **PSL**, **O**, **SO**, **PSO**, **U**, **SU**, **PSU** (and **Sp**, **PSp** of Chapter 13) are Weyl's **classical groups**.

12.3 Quotient spaces

Motivation. V/W is a vector space of dimension $\dim V - \dim W$, the tool for induction on dimension.

For $W \leq V$, V/W is an abelian group with $\bar{u} + \bar{v} = \overline{u+v}$, $\bar{v} = v + W$, and $c\bar{v} = \overline{cv}$ is well defined: $v - u \in W$ gives $c(v-u) \in W$.

Proposition 12.3.1

For $W \leq V$, V/W with $\bar{u} + \bar{v} = \overline{u+v}$ and $c\bar{v} = \overline{cv}$ is a vector space, the **quotient space** of V modulo W.

Proof. Each axiom follows from the one in V by putting bars on; zero $\bar{0}$, $-\bar{v} = \overline{-v}$. ■

Example 12.3.2

$V/0 \approx V$ and $V/V \approx 0$.

Exercise 12.3.3

For $W \leq V$: (a) $\pi : V \to V/W$ is a linear epimorphism;
(b) $\ker \pi = W$;
(c) for the inclusion $\iota : W \to V$, $\pi \circ \iota = 0$.

Solution. $\pi(cv + u) = \overline{cv+u} = c\bar{v} + \bar{u}$; onto by definition; $\bar{v} = \bar{0} \iff v \in W$. □

Remark 12.3.4

Heuristic: for $|F| = q$, Lagrange gives $|V/W| = q^n/q^m$, suggesting $\dim V/W = n - m$.

Theorem 12.3.5

If V is finite-dimensional and $W \leq V$, then $\dim V = \dim W + \dim(V/W)$.

Proof. Apply the dimension theorem to $\pi : V \to V/W$ ($\ker \pi = W$, $\operatorname{im} \pi = V/W$). ■

Exercise 12.3.6

If a basis $\{v_1, \ldots, v_m\}$ of W is extended to a basis $\{v_1, \ldots, v_n\}$ of V, then $\{\overline{v_{m+1}}, \ldots, \overline{v_n}\}$ is a basis of V/W.

Solution. Spanning: $\bar{v} = \overline{\sum a_i v_i} = \sum_{i>m} a_i \bar{v}_i$. Independence: $\sum_{i>m} a_i \bar{v}_i = \bar{0}$ gives $\sum_{i>m} a_i v_i \in \langle v_1, \ldots, v_m \rangle$, so all $a_i = 0$. □

Observation 12.3.7

Every subspace of F^n is the solution space of $AX = 0$ for some $A \in \mathfrak{M}_{m,n}(F)$.

Proof. Take A the matrix of $\pi : F^n \to F^n/W$ in some bases; $\ker \pi = W$. ■

12.4 Isomorphism theorems

Motivation. Dividing out its kernel turns a homomorphism into an isomorphism onto its image.

Observation 12.4.1

If $\varphi : G \to H$ is a homomorphism and $\varphi(g) = h$, then $\varphi^{-1}(h) = g \ker \varphi$.

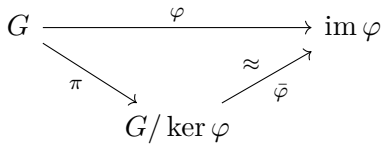
Proof. $x \in \varphi^{-1}(h) \iff \varphi(x) = \varphi(g) \iff g^{-1}x \in \ker\varphi \iff x \in g\ker\varphi$. ■

So the fibres of φ are the cosets of $\ker\varphi$.

Theorem 12.4.2 (First isomorphism theorem for groups)

For a homomorphism $\varphi : G \to H$, $\bar\varphi : G/\ker\varphi \to \operatorname{im}\varphi$, $\bar\varphi(\bar g) = \varphi(g)$, is a well-defined isomorphism: $G/\ker\varphi \approx \operatorname{im}\varphi$.

Proof. Well defined: $g'^{-1}g \in \ker\varphi$ gives $\varphi(g) = \varphi(g')$. Homomorphism: $\bar\varphi(\bar g\bar g') = \varphi(gg') = \varphi(g)\varphi(g')$. Onto: clear. Injective: $\bar\varphi(\bar g) = 1$ gives $g \in \ker\varphi$, $\bar g = \bar 1$ (Observation 11.6.7). ■



Remark 12.4.3

(a) Writing $\bar G = G/\ker\varphi$, $\bar\varphi\bar g = \varphi g$. (b) Equivalently, $\psi : \operatorname{im}\varphi \to G/\ker\varphi$, $\psi(\varphi(g)) = \bar g$, is an isomorphism; its well-definedness ($\varphi(g) = \varphi(g') \Rightarrow \bar g = \bar g'$) is the injectivity of $\bar\varphi$.

Example 12.4.4

$\operatorname{sgn} : S_n \to \mu_2$ ($n \geq 2$) is onto with kernel A_n; so $A_n \trianglelefteq S_n$, $S_n/A_n \approx \mu_2$, $[S_n : A_n] = 2$, $S_n = A_n \amalg \tau A_n$ ($\tau \notin A_n$), and $|A_n| = n!/2$.

Example 12.4.5

$\det : \mathrm{GL}_n(F) \to F^\times$ is onto with kernel $\mathrm{SL}_n(F)$; so $\mathrm{SL}_n(F) \trianglelefteq \mathrm{GL}_n(F)$ and $\mathrm{GL}_n(F)/\mathrm{SL}_n(F) \approx F^\times$.

Example 12.4.6

(a) $\det : \mathbf{O}(n) \to \mu_2$ is onto with kernel $\mathbf{SO}(n)$; so $\mathbf{SO}(n) \trianglelefteq \mathbf{O}(n)$, $\mathbf{O}(n)/\mathbf{SO}(n) \approx \mu_2$, and $\mathbf{O}(n) = \mathbf{SO}(n) \amalg S \cdot \mathbf{SO}(n)$ ($S \notin \mathbf{SO}(n)$; Exercise 9.3.11).
(b) $\det : \mathbf{U}(n) \to S^1$ is onto (e.g. $\operatorname{diag}(z, 1, \dots, 1)$) with kernel $\mathbf{SU}(n)$; so $\mathbf{U}(n)/\mathbf{SU}(n) \approx S^1 \approx \mathbf{SO}(2)$.

Exercise 12.4.7

(a) For $A, B \in \mathrm{GL}_n(F)$, $\bar A = \bar B$ in $\mathrm{GL}_n(F)/\mathrm{SL}_n(F)$ iff $\det A = \det B$. (b) Describe the coset decomposition of $\mathrm{GL}_n(F)$ modulo $\mathrm{SL}_n(F)$. (c) Describe that of $\mathbf{U}(n)$ modulo $\mathbf{SU}(n)$.

Solution. (a) $\bar A = \bar B \iff B^{-1}A \in \mathrm{SL}_n(F) \iff \det A = \det B$. (b) $\mathrm{GL}_n(F) = \coprod_{c\in F^\times} \operatorname{diag}(c, 1, \dots, 1)\,\mathrm{SL}_n(F)$: one coset for each value of the determinant. (c) $\mathbf{U}(n) = \coprod_{z\in S^1} \operatorname{diag}(z, 1, \dots, 1)\mathbf{SU}(n)$. □

Exercise 12.4.8

$\rho : (\mathbb{R}, +) \to \mathbf{SO}(2)$, $\rho(\theta) = R_\theta$, is a homomorphism, and $\mathbb{R}/2\pi\mathbb{Z} \approx \mathbf{SO}(2) \approx S^1$.

Solution. $R_\theta R_\eta = R_{\theta+\eta}$; ρ is onto (Theorem 9.5.2) with kernel $2\pi\mathbb{Z}$; apply Theorem 12.4.2 and Example 11.9.19. □

Example 12.4.9

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, has $\ker \gamma_x = n\mathbb{Z}$ for some $n \geq 0$ (Theorem 11.4.2) and image $\langle x \rangle$, so $\langle x \rangle \approx \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ (with $\mathbb{Z}_0 = \mathbb{Z}$, $\mathbb{Z}_1 = 0$).

Theorem 12.4.10

Up to isomorphism the cyclic groups are exactly the $\mathbb{Z}_n$, $n \geq 0$.

Proof. Example 12.4.9 with $G = \langle x \rangle$; and $\mathbb{Z}_n$ is cyclic. ■

Exercise 12.4.11

For $\mathrm{Int} : G \to \mathrm{Aut}(G)$, $g \mapsto \mathrm{Int}(g)$: (a) Int is a homomorphism;
(b) $\mathrm{Int}(G) \trianglelefteq \mathrm{Aut}(G)$;
(c) $\ker(\mathrm{Int}) = Z(G)$, so $\mathrm{Int}(G) \approx G/Z(G)$.

Solution. (a) $\mathrm{Int}(gh)(x) = ghxh^{-1}g^{-1} = \mathrm{Int}(g)\,\mathrm{Int}(h)(x)$ (this is why $\mathrm{Int}(g)(x) = gxg^{-1}$ rather than $g^{-1}xg$). (b) For $\varphi \in \mathrm{Aut}(G)$: $\varphi \circ \mathrm{Int}(g) \circ \varphi^{-1}(x) = \varphi(g\varphi^{-1}(x)g^{-1}) = \varphi(g)x\varphi(g)^{-1}$, i.e. $= \mathrm{Int}(\varphi(g))$. (c) $\mathrm{Int}(g) = id \iff gx = xg$ for all x; apply Theorem 12.4.2. □

Exercise 12.4.12

Define $\psi : \mathrm{SL}_2(F) \to \mathrm{GL}_3(F)$ by

$$\psi\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2 & -2ac & -c^2 \\ -ab & ad+bc & cd \\ -b^2 & 2bd & d^2 \end{pmatrix}.$$

(a) $\mathrm{im}\,\psi \leq \mathrm{SL}_3(F)$.
(b) ψ is a homomorphism with $\ker \psi = \{\pm I\} = Z(\mathrm{SL}_2(F))$.
(c) Hence $\mathbf{PSL}_2(F) \approx \mathrm{im}\,\psi$ is a matrix group.

Solution. [Supplement] $\psi(A)$ is the action of A on binary quadratic forms; multiplying out gives $\psi(AB) = \psi(A)\psi(B)$ and $\det \psi(A) = (ad - bc)^3$. (a) For $A \in \mathrm{SL}_2(F)$, $\det \psi(A) = 1^3 = 1$. (b) $\psi(A) = I$ forces $c = b = 0$ (off-diagonal entries $-c^2 = -b^2 = 0$) and $a^2 = d^2 = ad = 1$, so $a = d = \pm 1$; conversely $\psi(\pm I) = I$. And $Z(\mathrm{SL}_2(F)) = \{\pm I\}$ (Exercise 12.2.20, $c^2 = 1$). (c) First isomorphism theorem. □

Exercise 12.4.13

Is there an epimorphism $S_3 \to \mathbb{Z}_3$?

Solution. No: its kernel would be a normal subgroup of order 2, and there is none (Example 12.2.1). □

Theorem 12.4.14 (First isomorphism theorem for vector spaces)

For a linear map $L : V \to W$, $\bar{L} : V/\ker L \to \operatorname{im} L$, $\bar{L}\bar{v} = Lv$, is a well-defined vector space isomorphism.

Proof. By Theorem 12.4.2 for $L : (V,+) \to (W,+)$, $\bar{L}$ is a well-defined group isomorphism; and $\bar{L}(c\bar{v}) = \bar{L}(\overline{cv}) = L(cv) = cL(v) = c\bar{L}(\bar{v})$. ■

Remark 12.4.15

By Theorem 12.4.14, the dimension theorem and Theorem 12.3.5 are equivalent statements.

Exercise 12.4.16

For $\operatorname{tr} : \mathfrak{M}_{n,n}(F) \to F$: (a) $\ker \operatorname{tr} = \mathfrak{sl}_n(F)$;
(b) $\mathfrak{M}_{n,n}(F)/\mathfrak{sl}_n(F) \approx F$, so $\dim \mathfrak{sl}_n(F) = n^2 - 1$.

Solution. tr is linear and onto ($\operatorname{tr}(cE_{11}) = c$); $\mathfrak{sl}_n(F)$ is by definition the trace-zero matrices; apply Theorem 12.4.14. □

Exercise 12.4.17

For $T \in \mathfrak{L}(V,V)$ and a T-cyclic space $V = F[t]v$, $\varphi : F[t] \to V$, $\varphi(f) = f(T)v$: (a) φ is linear with $\ker \varphi = \mathcal{I}_T$ (the annihilator ideal of v);
(b) $V \approx F[t]/\mathcal{I}_T$.

Solution. φ is linear and onto by definition of $F[t]v$; its kernel is $\{f : f(T)v = 0\}$; apply Theorem 12.4.14. □

Observation 12.4.18

If $H \trianglelefteq G$ and $K \leq G$, then $HK \leq G$.

Proof. $1 \in HK$. For $h, h' \in H$, $k, k' \in K$: $(hk)(h'k') = h(kh'k^{-1})kk' \in HK$ and $(hk)^{-1} = k^{-1}h^{-1} = (k^{-1}h^{-1}k)k^{-1} \in HK$. ■

Theorem 12.4.19 (Second isomorphism theorem for groups)

If $H \trianglelefteq G$ and $K \leq G$, then $\bar{\jmath} : K/(H \cap K) \to HK/H$, $\bar{\jmath}(k(H \cap K)) = kH$, is a well-defined isomorphism.

Proof. By Observation 12.4.18 $HK \leq G$, and clearly $H \trianglelefteq HK$, $H \cap K \trianglelefteq K$.

(i) Well defined: if $k'^{-1}k \in H \cap K$, then $k'^{-1}k \in H$.

(ii) Injective: if $k'^{-1}k \in H$, then (as $k, k' \in K$) $k'^{-1}k \in H \cap K$.

(iii) Homomorphism: $\bar{\jmath}(\bar{k}\bar{k}') = \overline{kk'} = \bar{k}\,\bar{k}'$.

(iv) Onto: for $h \in H$, $k \in K$, $\overline{hk} = \bar{h}\,\bar{k} = \bar{1}\bar{k} = \bar{k}$ in HK/H (H normal). ■

Second proof. $j = \pi' \circ \iota : K \to HK \to HK/H$ is an epimorphism by (iv), with $\ker j = H \cap K$; apply Theorem 12.4.2. ■

Corollary 12.4.20

If G is finite, $H \trianglelefteq G$, $K \leq G$, then $|HK| \cdot |H \cap K| = |H| \cdot |K|$.

Proof. Lagrange and Theorem 12.4.19: $\frac{|K|}{|H \cap K|} = \frac{|HK|}{|H|}$. ■

Theorem 12.4.21 (Second isomorphism theorem for vector spaces)

For $U, W \leq V$, $\bar{\jmath} : W/(U \cap W) \to (U + W)/U$, $\bar{\jmath}(\bar{w}) = \bar{w}$, is a well-defined vector space isomorphism.

Proof. Theorem 12.4.19 for $(V, +)$ gives a group isomorphism; it is F-linear since $\bar{\jmath}(c\bar{w}) = \overline{cw} = c\bar{w}$. ■

Corollary 12.4.22

For finite-dimensional V and $U, W \leq V$: $\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$.

Proof. By Theorems 12.4.21 and 12.3.5, $\dim W - \dim(U \cap W) = \dim(U + W) - \dim U$. ■

12.5 Triangularization II

Motivation. An invariant W splits T into $T|_W$ and $\bar{T}$ on V/W; induction on $\dim V$ triangularizes.

Spaces are finite-dimensional. For T-invariant W with basis $\mathfrak{C} = \{v_1, \ldots, v_m\}$ extended to $\mathfrak{B} = \{v_1, \ldots, v_n\}$, $[T]^{\mathfrak{B}}_{\mathfrak{B}} = \begin{pmatrix} [T|_W]^{\mathfrak{C}}_{\mathfrak{C}} & * \\ 0 & ? \end{pmatrix}$; the block "?" is $\bar{T}\bar{v} = \overline{Tv}$ on $\bar{V} = V/W$ with basis $\bar{\mathfrak{B}} = \{\bar{v}_{m+1}, \ldots, \bar{v}_n\}$ (Exercise 12.3.6).

Observation 12.5.1

If $W \leq V$, $W' \leq V'$ and $L : V \to V'$ is linear with $LW \leq W'$, then $\bar{L} : V/W \to V'/W'$, $\bar{L}\bar{v} = \overline{Lv}$, is a well-defined linear map (L **induces** $\bar{L}$).

Proof. If $v - u \in W$, then $Lv - Lu = L(v - u) \in W'$; linearity is inherited. (Conversely, well-definedness forces $LW \leq W'$: take $u = 0$.) ■

$$\begin{array}{ccc} V & \xrightarrow{L} & V' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ V/W & \xrightarrow[\bar{L}]{} & V'/W' \end{array}$$

Exercise 12.5.2

State and prove Observation 12.5.1 for groups.

Solution. If $N \trianglelefteq G$, $N' \trianglelefteq G'$ and $\varphi : G \to G'$ is a homomorphism with $\varphi(N) \leq N'$, then $\bar{\varphi}(\bar{g}) = \overline{\varphi g}$ is a well-defined homomorphism $G/N \to G'/N'$: if $g'^{-1}g \in N$, then $\varphi(g')^{-1}\varphi(g) = \varphi(g'^{-1}g) \in N'$. □

Exercise 12.5.3

The well-definedness of $\bar{L}$ in Theorem 12.4.14 and of $\bar{\jmath}$ in Theorem 12.4.21 are special cases of Observation 12.5.1.

Solution. $\bar{L}$ is induced by $L : V \to \operatorname{im} L$ with $L(\ker L) \leq 0$; $\bar{\jmath}$ is induced by the inclusion $W \to U+W$ with $U \cap W \leq U$. □

Lemma 12.5.4

Let W be T-invariant, $\bar{T} : \bar{V} \to \bar{V}$ as above, $\mathfrak{C} = \{v_1, \ldots, v_m\}$ a basis of W, $\mathfrak{B} = \{v_1, \ldots, v_n\}$ a basis of V, $\bar{\mathfrak{B}} = \{\bar{v}_{m+1}, \ldots, \bar{v}_n\}$. Then

$$[T]_{\mathfrak{B}}^{\mathfrak{B}} = \begin{pmatrix} [T|_W]_{\mathfrak{C}}^{\mathfrak{C}} & * \\ 0 & [\bar{T}]_{\bar{\mathfrak{B}}}^{\bar{\mathfrak{B}}} \end{pmatrix}.$$

Proof. The first m columns are as stated since $TW \leq W$. From $Tv_{m+j} = \sum_i a_{i,m+j} v_i$ and $\bar{v}_1 = \cdots = \bar{v}_m = \bar{0}$, $\bar{T}\bar{v}_{m+j} = \sum_{i>m} a_{i,m+j}\bar{v}_i$: column j of $[\bar{T}]$ is the lower part of column $m+j$. ■

Exercise 12.5.5

In Lemma 12.5.4: (a) $\operatorname{tr} T = \operatorname{tr}(T|_W) + \operatorname{tr}\bar{T}$;
(b) $\det T = \det(T|_W) \det \bar{T}$;
(c) $\phi_T(t) = \phi_{T|_W}(t)\phi_{\bar{T}}(t)$.

Solution. Read off the block-triangular matrix (for (c), apply the block determinant to $tI - [T]$). □

Theorem 12.5.6 (Triangularization)

If $F = \mathbb{C}$ and $T \in \mathfrak{L}(V, V)$, then V has a basis $\mathfrak{B}$ with $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ upper triangular.

Second proof. Induction on $n = \dim V$. Take an eigenvector $Tv = \lambda v$, $W = \langle v \rangle$, $\bar{V} = V/W$ of dimension $n-1$. By induction $\bar{V}$ has a basis $\bar{\mathfrak{D}} = \{\bar{u}_2, \ldots, \bar{u}_n\}$ with $[\bar{T}]_{\bar{\mathfrak{D}}}^{\bar{\mathfrak{D}}}$ upper triangular. $\mathfrak{D} = \{v, u_2, \ldots, u_n\}$ is a basis ($cv + \sum c_i u_i = 0$ gives $\sum c_i \bar{u}_i = 0$ under π, so $c_i = 0$, $c = 0$), and $[T]_{\mathfrak{D}}^{\mathfrak{D}} = \left(\begin{smallmatrix} \lambda & * \\ 0 & [\bar{T}]_{\bar{\mathfrak{D}}}^{\bar{\mathfrak{D}}} \end{smallmatrix}\right)$ (Lemma 12.5.4). ■

Example 12.5.7

Let G be finite and $\varphi : G \to \mathrm{GL}_n(\mathbb{C})$ a homomorphism, $\chi = \operatorname{tr} \circ \varphi$ (the **character** of φ). Triangularize $\varphi(g)$ with diagonal $\lambda_1, \ldots, \lambda_n$; $\varphi(g)^{|g|} = I$ gives $\lambda_i^{|g|} = 1$, $|\lambda_i| = 1$, and $\varphi(g^{-1}) = \varphi(g)^{-1}$ has diagonal $\lambda_i^{-1} = \bar{\lambda}_i$, so

$$\chi(g^{-1}) = \sum \lambda_i^{-1} = \sum \bar{\lambda}_i = \overline{\chi(g)}.$$

Exercise 12.5.8

In Example 12.5.7, if g is conjugate to h, then $\chi(g) = \chi(h)$.

Solution. $h = x^{-1}gx$ gives $\varphi(h) = \varphi(x)^{-1}\varphi(g)\varphi(x)$, and similar matrices have equal traces. □

Definition 12.5.9 (Simultaneous triangularization)

$T, S \in \mathfrak{L}(V, V)$ are **simultaneously triangularizable** if $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ and $[S]_{\mathfrak{B}}^{\mathfrak{B}}$ are upper triangular for one basis $\mathfrak{B}$ (likewise for any family). Then the first vector of $\mathfrak{B}$ is a common eigenvector.

Proposition 12.5.10

If $F = \mathbb{C}$ and $TS = ST$, then T and S have a common eigenvector.

Proof. An eigenspace E_λ^T is S-invariant ($T(Sv) = S(Tv) = \lambda Sv$); take an eigenvector of $S|_{E_\lambda^T}$. ■

Proposition 12.5.11

If $F = \mathbb{C}$ and $TS = ST$, then T, S are simultaneously triangularizable.

Proof. For a common eigenvector v (Proposition 12.5.10), $\bar{T}, \bar{S}$ on $V/\langle v \rangle$ commute; by induction they are simultaneously triangularizable, and lift as in Theorem 12.5.6. ■

Exercise 12.5.12

If $\{T_i\}_{i \in I} \subseteq \mathfrak{L}(V, V)$ ($F = \mathbb{C}$) pairwise commute, they are simultaneously triangularizable.

Solution.

Step 1 (a common eigenvector). Take a nonzero U invariant under all T_i, of minimal dimension. If some $T_j|_U$ is not scalar, an eigenspace $\{u \in U : T_j u = \lambda u\}$ is a smaller such subspace (commutativity). So all T_i are scalar on U, and any $0 \neq v \in U$ works.

Step 2 (induction). Pass to $V/\langle v \rangle$ as in Proposition 12.5.11. □

Chapter 13. Bilinear Forms

From now on all vector spaces are finite-dimensional.

13.1 Bilinear forms

Motivation. Given a basis, a bilinear form is a matrix, just as a linear map is.

For a bilinear form $B : V \times V \to F$, a basis $\mathfrak{B} = \{v_i\}$ of V, $v = \sum a_i v_i$, $w = \sum b_i v_i$:

$$B(v,w) = B\Big(\sum_i a_i v_i, \sum_j b_j v_j\Big) = \sum_{i,j} a_i b_j B(v_i, v_j). \qquad (*)$$

Observation 13.1.1

With $[B]_{\mathfrak{B}} = (B(v_i,v_j))_{n\times n}$, $B(v,w) = [v]_{\mathfrak{B}}^t\,[B]_{\mathfrak{B}}\,[w]_{\mathfrak{B}}$ for all $v, w \in V$. $[B]_{\mathfrak{B}}$ is the **matrix of** B with respect to $\mathfrak{B}$.

Proof. $(*)$ reads $\sum_{i,j} a_i B(v_i,v_j) b_j = [v]_{\mathfrak{B}}^t [B]_{\mathfrak{B}} [w]_{\mathfrak{B}}$. ■

Notation 13.1.2

For $J \in \mathfrak{M}_{n,n}(F)$, $B_{\mathcal{E}}^J(X,Y) = X^t J Y$ is a bilinear form on F^n with $[B_{\mathcal{E}}^J]_{\mathcal{E}} = J$.

Example 13.1.3

(a) $B_{\mathcal{E}}^I$ is the **ordinary inner product** X^tY of F^n (not Hermitian on $\mathbb{C}^n$). (b) The inner product $\sum j a_j b_j$ of Exercise 10.1.8 is $B_{\mathcal{E}}^J$ with $J = \operatorname{diag}(1,2,\dots,n)$. (c) The inner product of Exercise 10.1.9 is $B_{\mathcal{E}}^J$ with $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$.

Notation 13.1.4

For a basis $\mathfrak{B}$ of V and $J \in \mathfrak{M}_{n,n}(F)$, $B_{\mathfrak{B}}^J(v,w) = [v]_{\mathfrak{B}}^t J [w]_{\mathfrak{B}}$ is a bilinear form on V with $[B_{\mathfrak{B}}^J]_{\mathfrak{B}} = J$.

So $B \leftrightarrow [B]_{\mathfrak{B}}$ is a bijection between bilinear forms on V and $\mathfrak{M}_{n,n}(F)$ (for fixed $\mathfrak{B}$).

Exercise 13.1.5

$\mathfrak{Bil}(V) = \{B : V\times V \to F \text{ bilinear}\}$, with $(B+C)(v,w) = B(v,w)+C(v,w)$ and $(aB)(v,w) = aB(v,w)$, is an F-vector space.

Solution. $B + C$ and aB are bilinear; the axioms hold pointwise. □

Theorem 13.1.6 (Fundamental theorem for bilinear forms)

For a fixed basis $\mathfrak{B}$ of n-dimensional V, $\Omega_{\mathfrak{B}} : \mathfrak{Bil}(V) \to \mathfrak{M}_{n,n}(F)$, $\Omega_{\mathfrak{B}}(B) = [B]_{\mathfrak{B}}$, is a vector space isomorphism.

Proof. Injective by Observation 13.1.1, surjective by Notation 13.1.4, and linear since $[B+C]_{\mathfrak{B}} = [B]_{\mathfrak{B}} + [C]_{\mathfrak{B}}$ and $[aB]_{\mathfrak{B}} = a[B]_{\mathfrak{B}}$ (entrywise). ■

Remark 13.1.7

With the coordinate isomorphism $\alpha_{\mathfrak{B}} : V \to F^n$, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{B} & F \\ {\scriptstyle \alpha_{\mathfrak{B}} \times \alpha_{\mathfrak{B}}}\downarrow & & \downarrow{\scriptstyle I} \\ F^n \times F^n & \xrightarrow[B^J_{\mathcal{E}}]{} & F \end{array}$$

($J = [B]_{\mathfrak{B}}$) commutes (Observation 13.1.1).

Observation 13.1.8

If J is invertible, then $B^J_{\mathcal{E}}(AX, Y) = B^J_{\mathcal{E}}(X, (J^{-1}A^tJ)Y)$ for $A \in \mathfrak{M}_{n,n}(F)$, $X, Y \in F^n$.

Proof. $(AX)^tJY = X^tA^tJY$ and $X^tJ(J^{-1}A^tJ)Y = X^tA^tJY$. ■

(This generalizes Observation 10.7.1; compare Question 10.7.2.)

Definition 13.1.9 (Symmetric, alternating)

A bilinear form B is (a) **symmetric** if $B(v, w) = B(w, v)$;
(b) **alternating** if $B(v, v) = 0$ for all v.

Observation 13.1.10

B is symmetric iff $[B]_{\mathfrak{B}}$ is a symmetric matrix.

Proof. If B is symmetric, $B(v_i, v_j) = B(v_j, v_i)$. Conversely, $(*)$ shows $B(v, w) = B(w, v)$ when $B(v_i, v_j) = B(v_j, v_i)$. ■

Exercise 13.1.11

B is alternating iff $B(v, w) = -B(w, v)$ for all v, w (when $1 + 1 \neq 0$ in F).

Solution. If alternating, $0 = B(v + w, v + w) = B(v, w) + B(w, v)$. Conversely, $B(v, v) = -B(v, v)$ gives $2B(v, v) = 0$. (In characteristic 2 only "alternating ⇒ skew" holds.) □

Exercise 13.1.12

Define an **alternating matrix** ($J^t = -J$ with zero diagonal) and show: B is alternating iff $[B]_{\mathfrak{B}}$ is alternating. (For $1 + 1 \neq 0$, alternating = skew-symmetric.)

Solution. If B is alternating, $B(v_i, v_i) = 0$ and $B(v_i, v_j) = -B(v_j, v_i)$. Conversely, by $(*)$, $B(v, v) = \sum_i a_i^2 J_{ii} + \sum_{i<j} a_i a_j (J_{ij} + J_{ji}) = 0$. □

Exercise 13.1.13

Let $\mathfrak{Sym}(V)$ and $\mathfrak{Alt}(V)$ be the symmetric and the alternating bilinear forms on V ($1+1 \neq 0$). (a) They are subspaces of $\mathfrak{Bil}(V)$. (b) $\dim \mathfrak{Bil}(V) = n^2$, $\dim \mathfrak{Sym}(V) = \frac{n(n+1)}{2}$, $\dim \mathfrak{Alt}(V) = \frac{n(n-1)}{2}$. (c) $\mathfrak{Bil}(V) = \mathfrak{Sym}(V) \oplus \mathfrak{Alt}(V)$.

Solution. (a) The defining conditions are linear.

(b) Via $\Omega_{\mathfrak{B}}$: symmetric matrices are determined by the entries on and above the diagonal, alternating ones by those strictly above.

(c) $B = \frac{1}{2}(B+B^t)+\frac{1}{2}(B-B^t)$ with $B^t(v,w) = B(w,v)$, and a form both symmetric and alternating is 0. □

Question 13.1.14

For $F = \mathbb{R}$, how can we tell from $[B]_{\mathfrak{B}}$ whether a symmetric bilinear form is an inner product (positive definite)? (Answered in §15.3.)

13.2 Quadratic forms

Motivation. A symmetric bilinear form is the same as a quadratic form $Q(v) = \frac{1}{2}B(v,v)$.

Assume $1+1 \neq 0$ in F.

Example 13.2.1

For $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, $B^J_{\mathcal{E}}((x,y)^t,(x,y)^t) = 2(x^2 - xy + y^2)$, and $Q(x,y) = x^2 - xy + y^2$ is the familiar quadratic form.

Definition 13.2.2 (Quadratic form)

$Q : V \to F$ is a **quadratic form** if (Q1) $B_Q(v,w) = Q(v+w) - Q(v) - Q(w)$ is a symmetric bilinear form, and (Q2) $Q(cv) = c^2Q(v)$.

Exercise 13.2.3

For a symmetric bilinear form B, $Q(v) = \frac{1}{2}B(v,v)$ is a quadratic form with $B_Q = B$.

Solution. $Q(v+w) - Q(v) - Q(w) = \frac{1}{2}(B(v,w) + B(w,v)) = B(v,w)$, and $Q(cv) = c^2Q(v)$. □

Observation 13.2.4

$Q \mapsto B_Q$ is a bijection between quadratic forms and symmetric bilinear forms on V.

Proof. $B_Q(v,v) = Q(2v) - 2Q(v) = 4Q(v) - 2Q(v) = 2Q(v)$, so Q is recovered from B_Q: injective. Surjective by Exercise 13.2.3. ■

Exercise 13.2.5

The quadratic forms on V form a vector space, and $Q \mapsto B_Q$ is a vector space isomorphism.

Solution. Pointwise operations preserve (Q1), (Q2), and $B_{aQ+Q'} = aB_Q + B_{Q'}$. □

Why "quadratic". With $[B_Q]_{\mathfrak{B}} = J = (a_{ij})$ (symmetric) and $[v]_{\mathfrak{B}} = (x_1,\dots,x_n)^t$,

$$2Q(v) = B_Q(v,v) = \sum_{i,j} a_{ij}x_ix_j = \sum_i a_{ii}x_i^2 + 2\sum_{i<j} a_{ij}x_ix_j,$$

a homogeneous quadratic polynomial in the coordinates; conversely such a polynomial gives, with $a_{ji} = a_{ij}$, $B_Q(v,w) = [v]^tJ[w]$.

Definition 13.2.6 (Quadratic space)

A vector space with a symmetric bilinear form B (hence a quadratic form Q) is a **quadratic space** $(V, B) = (V, B, Q)$.

13.3 Orthogonal groups and symplectic groups

Motivation. Operators preserving B are, in a basis, $\{A : A^tJA = J\}$: orthogonal or symplectic groups.

Definition 13.3.1 (Orthogonal group of a quadratic space)

$\mathbf{O}(V, B) = \{L \in \mathrm{GL}(V) : B(Lv, Lw) = B(v, w)\ \forall v, w\}$, the **orthogonal group**; its elements are **orthogonal operators**. Studying (V, B) and $\mathbf{O}(V, B)$ is **orthogonal geometry**.

Exercise 13.3.2

(a) $\mathbf{O}(V, B) \leq \mathrm{GL}(V)$.
(b) $\mathbf{SO}(V, B) = \{L \in \mathbf{O}(V, B) : \det L = 1\} \trianglelefteq \mathbf{O}(V, B)$.

Solution. (a) I preserves B; if L, M do, so do LM and L^{-1} ($B(L^{-1}v, L^{-1}w) = B(LL^{-1}v, LL^{-1}w)$).
(b) It is the kernel of $\det : \mathbf{O}(V, B) \to F^\times$. □

Exercise 13.3.3

(a) $\mathbf{O}(V, B) = \{L \in \mathrm{GL}(V) : Q(Lv) = Q(v)\ \forall v\}$.
(b) If $\mathfrak{B} = \{v_i\}$ is a basis, $\mathbf{O}(V, B) = \{L \in \mathrm{GL}(V) : B(Lv_i, Lv_j) = B(v_i, v_j)\ \forall i, j\}$.

Solution. (a) As in Observation 9.3.4: $B(v, w) = Q(v + w) - Q(v) - Q(w)$ is determined by Q.
(b) By bilinearity ($*$). □

Exercise 13.3.4

Can $\mathbf{O}(V, B) = \mathrm{GL}(V)$?

Solution. Yes, iff $B = 0$ (or $V = 0$). If $B \neq 0$, take v with $Q(v) \neq 0$ ($1 + 1 \neq 0$, Exercise 13.2.3); then $L = cI$ with $c^2 \neq 1$ (e.g. $c = 2$ if $3 \neq 0$; in characteristic 3 this needs $|F| > 3$) gives $Q(Lv) = c^2Q(v) \neq Q(v)$. □

Proposition 13.3.5

Let $\mathfrak{B}$ be a basis of n-dimensional (V, B) and $J = [B]_{\mathfrak{B}}$. Then

$$\{[L]^{\mathfrak{B}}_{\mathfrak{B}} : L \in \mathbf{O}(V, B)\} = \{A \in \mathrm{GL}_n(F) : A^tJA = J\}.$$

Proof. By Observation 13.1.1 and $[Lv]_{\mathfrak{B}} = [L]^{\mathfrak{B}}_{\mathfrak{B}}[v]_{\mathfrak{B}}$, $B(Lv, Lw) = B(v, w)$ for all v, w reads $[v]^t([L]^tJ[L])[w] = [v]^tJ[w]$, i.e. $[L]^tJ[L] = J$ (Exercise 9.3.7); and conversely. (Symmetry of J is not used.) ■

With $J = I$ this settles Remark 10.6.4.

Definition 13.3.6 (Matrix orthogonal groups)

For a symmetric $J \in \mathfrak{M}_{n,n}(F)$, $\mathbf{O}_n^J(F) = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}$, the **orthogonal group** of J. (Thus $\mathbf{O}(n) = \mathbf{O}_n^I(\mathbb{R})$.)

$\mathbf{O}_n^J(F) \leq \mathrm{GL}_n(F)$, $\mathbf{SO}_n^J(F) \trianglelefteq \mathbf{O}_n^J(F)$, and with $J = [B]_{\mathfrak{B}}$, $\Psi_{\mathfrak{B}}^{\mathfrak{B}}$ restricts to $\mathbf{O}(V,B) \approx \mathbf{O}_n^J(F)$ (compare §11.9).

Exercise 13.3.7

If J is invertible symmetric and $A \in \mathbf{O}_n^J(F)$, then $\det A = \pm 1$.

Solution. $\det(A)^2 \det J = \det J \neq 0$. □

Definition 13.3.8 (Symplectic group)

For an alternating form B on V, $\mathbf{Sp}(V,B) = \{L \in \mathrm{GL}(V) : B(Lv, Lw) = B(v,w)\ \forall v, w\}$ is the **symplectic group**; this is **symplectic geometry**.

Proposition 13.3.9

If B is alternating, $\mathfrak{B}$ a basis and $J = [B]_{\mathfrak{B}}$ (skew-symmetric), then $\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{Sp}(V,B)\} = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}$.

Proof. Word for word the proof of Proposition 13.3.5, which did not use symmetry of J. ■

Definition 13.3.10 (Matrix symplectic groups)

For skew-symmetric J, $\mathbf{Sp}_n^J(F) = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}$.

Exercise 13.3.11

For $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $\mathbf{Sp}_2^J(F) = \mathrm{SL}_2(F)$.

Solution. $A^t J A$ is alternating, hence a multiple of J, with $(1,2)$ entry $ad - bc$: $A^t J A = (\det A) J$. □

In general $\mathbf{Sp}_n^J(F) \leq \mathrm{SL}_n(F)$ for invertible skew-symmetric J (proof postponed).

Question 13.3.12

For invertible symmetric J, is there always $A \in \mathbf{O}_n^J(F)$ with $\det A = -1$? How should "reflections" of a quadratic space be defined? (See §13.5.)

13.4 $\mathbf{O}(1,1)$ and $\mathbf{O}(3,1)$

Motivation. For $x^2 - y^2$, cosh and sinh replace cos and sin; adding space coordinates gives the Lorentz group.

For $J = \mathrm{diag}(1,-1)$, $B_{\mathcal{E}}^J((a,b)^t,(c,d)^t) = ac - bd$ (quadratic form $2Q = x^2 - y^2$), and

$$\mathbf{O}(1,1) = \mathbf{O}_2^J(\mathbb{R}) = \{A \in \mathrm{GL}_2(\mathbb{R}) : A^t J A = J\}.$$

$A = \left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \in \mathbf{O}(1,1)$ iff $x^2 - y^2 = 1 = w^2 - z^2$ and $xz - yw = 0$: the columns lie on the hyperbolas $x^2 - y^2 = 1$, $w^2 - z^2 = 1$. Recall $\cosh t = \frac{e^t + e^{-t}}{2}$, $\sinh t = \frac{e^t - e^{-t}}{2}$.

Exercise 13.4.1

For $t, s \in \mathbb{R}$: (a) $\cosh^2 t - \sinh^2 t = 1$;
(b) $\cosh(-t) = \cosh t$, $\sinh(-t) = -\sinh t$;
(c) $\cosh(t+s) = \cosh t \cosh s + \sinh t \sinh s$;
(d) $\sinh(t+s) = \cosh t \sinh s + \sinh t \cosh s$.

Solution. Expand the exponentials; e.g. (a): $\frac{(e^t+e^{-t})^2 - (e^t - e^{-t})^2}{4} = 1$. □

$(\pm \cosh t, \sinh t)^t$ parametrize the two branches of $x^2 - y^2 = 1$. If $x, w > 0$, write $(x, y) = (\cosh t, \sinh t)$, $(z, w) = (\sinh s, \cosh s)$; then $0 = xz - yw = \sinh(s - t)$, so $s = t$ and

$$A = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}.$$

Similarly: if $x > 0$, $w < 0$: $A = \left(\begin{smallmatrix} \cosh u & \sinh u \\ -\sinh u & -\cosh u \end{smallmatrix}\right)$; if $x < 0$, $w > 0$: $A = \left(\begin{smallmatrix} -\cosh t & -\sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)$; if $x, w < 0$: $A = \left(\begin{smallmatrix} -\cosh u & -\sinh u \\ -\sinh u & -\cosh u \end{smallmatrix}\right)$.

Exercise 13.4.2

With $\mathbf{SO}(1,1) = \mathrm{SL}_2(\mathbb{R}) \cap \mathbf{O}(1,1)$: $\mathbf{O}(1,1)/\mathbf{SO}(1,1) \approx \mu_2$, $[\mathbf{O}(1,1) : \mathbf{SO}(1,1)] = 2$, and $\mathbf{O}(1,1) = \mathbf{SO}(1,1) \amalg \mathrm{diag}(-1,1)\mathbf{SO}(1,1)$.

Solution. $\det : \mathbf{O}(1,1) \to \mu_2$ is onto ($\det \mathrm{diag}(-1,1) = -1$) with kernel $\mathbf{SO}(1,1)$; apply the First Isomorphism Theorem. □

Exercise 13.4.3

Let $\mathbf{SO}^\circ(1,1) = \left\{ \left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right) \right\} = \left\{ \left(\begin{smallmatrix} x & y \\ y & x \end{smallmatrix}\right) : x \geq 1,\ x^2 - y^2 = 1 \right\}$. Then $\mathbf{SO}^\circ(1,1) < \mathbf{SO}(1,1)$, $\mathbf{SO}(1,1) = \mathbf{SO}^\circ(1,1) \amalg (-I \cdot \mathbf{SO}^\circ(1,1))$, $[\mathbf{SO}(1,1) : \mathbf{SO}^\circ(1,1)] = 2$, and $\mathbf{SO}^\circ(1,1) \trianglelefteq \mathbf{SO}(1,1)$.

Solution. Products and inverses correspond to $t + s$ and $-t$ (Exercise 13.4.1). Of the four cases, $\det A = 1$ exactly for $x, w > 0$ ($\mathbf{SO}^\circ$) and $x, w < 0$ ($-\mathbf{SO}^\circ$). Index 2 gives normality (Observation 12.2.9). □

Hence $[\mathbf{O}(1,1) : \mathbf{SO}^\circ] = 4$ (Observation 12.1.12) and, with $\mathbf{SO}^\circ = \mathbf{SO}^\circ(1,1)$,

$$\mathbf{O}(1,1) = \mathbf{SO}^\circ \amalg (-I \cdot \mathbf{SO}^\circ) \amalg (\mathrm{diag}(-1,1)\mathbf{SO}^\circ) \amalg (\mathrm{diag}(1,-1)\mathbf{SO}^\circ),$$

matching the four cases.

Exercise 13.4.4

Is $\mathbf{SO}^\circ(1,1)$ normal in $\mathbf{O}(1,1)$?

Solution. Yes: $S = \mathrm{diag}(1,-1)$ gives $S\left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)S^{-1} = \left(\begin{smallmatrix} \cosh t & -\sinh t \\ -\sinh t & \cosh t \end{smallmatrix}\right) \in \mathbf{SO}^\circ$, and $\mathbf{O}(1,1)$ is generated by $\mathbf{SO}(1,1)$ (in which $\mathbf{SO}^\circ$ is normal) and S. □

Exercise 13.4.5

$\varphi : (\mathbb{R}, +) \to \mathbf{SO}^\circ(1,1)$, $\varphi(t) = \left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)$, is a group isomorphism.

Solution. A homomorphism by Exercise 13.4.1(c), (d); onto by the parametrization; injective since $\sinh t = 0$ only for $t = 0$. □

So $\mathbf{O}(1,1)$ is four "real lines", whereas $\mathbf{O}(2)$ is two circles.

Exercise 13.4.6

$\mathbf{O}(2) \not\approx \mathbf{O}(1,1)$.

Solution. [Supplement] $R_{2\pi/3} \in \mathbf{O}(2)$ has order 3. For $A \in \mathbf{O}(1,1)$, $A^2 \in \mathbf{SO}^\circ$ (quotient $\mu_2 \times \mu_2$), and $\mathbf{SO}^\circ \approx (\mathbb{R}, +)$ is torsion-free, so $A^{2k} = 1$ forces $A^2 = 1$: finite orders in $\mathbf{O}(1,1)$ are 1 or 2. □

Definition 13.4.7 (Lorentz group)

For $J = \operatorname{diag}(1,1,1,-1)$, $\mathbf{O}(3,1) = \mathbf{O}_4^J(\mathbb{R})$ is the **Lorentz group**, the symmetry group of space-time $\mathbb{R}^4$.

Exercise 13.4.8

$\mathbf{O}(3)$ and $\mathbf{O}(1,1)$ are isomorphic to subgroups of $\mathbf{O}(3,1)$.

Solution. $A \mapsto \operatorname{diag}(A, 1)$ and $A \mapsto \operatorname{diag}(1, 1, A)$; $A^t J A = J$ holds blockwise. □

13.5 Non-degenerate bilinear forms

Motivation. Without positivity, $W \oplus W^\perp = V$ may fail, but non-degenerate forms keep $\dim W^\perp = \dim V - \dim W$.

V is n-dimensional with a bilinear form B, and $1 + 1 \neq 0$.

Definition 13.5.1 (Orthogonality)

$v \perp w$ if $B(v,w) = 0$; $S \perp T$ if $B(v,w) = 0$ for all $v \in S$, $w \in T$.

Remark 13.5.2

$\perp$ is symmetric iff B is symmetric or alternating.

Definition 13.5.3 (Orthogonal basis)

A basis $\mathfrak{B} = \{v_i\}$ of mutually perpendicular vectors is an **orthogonal basis**; if moreover $B(v_i, v_i) = 1$ for all i, an **orthonormal basis**.

Exercise 13.5.4

(a) $\mathfrak{B}$ is orthogonal iff $[B]_\mathfrak{B}$ is diagonal.
(b) $\mathfrak{B}$ is orthonormal iff $[B]_\mathfrak{B} = I$.

Solution. The entries of $[B]_{\mathfrak{B}}$ are $B(v_i, v_j)$. □

Since $B(v, v) = 0$ is possible for $v \neq 0$, Gram–Schmidt is unavailable.

Notation 13.5.5

For $S \subseteq V$, $S^\perp = \{v \in V : B(w, v) = 0 \text{ for all } w \in S\}$.

Exercise 13.5.6

(a) $S^\perp = \langle S \rangle^\perp \leq V$.
(b) If B is symmetric or alternating, $W \leq (W^\perp)^\perp$.

Solution. (a) Linearity in the second variable, and Exercise 10.2.16.

(b) For $w \in W$, $v \in W^\perp$: $B(v, w) = \pm B(w, v) = 0$. □

($B = 0$ gives $W^\perp = V$.)

Exercise 13.5.7

If $B(w, w) \neq 0$, then $\dim \langle w \rangle^\perp = \dim V - 1$.

Solution. $v \mapsto B(w, v)$ is a nonzero linear functional (it is $\neq 0$ at w); its kernel $\langle w \rangle^\perp$ has dimension $n - 1$. □

Question 13.5.8

When is $\dim W^\perp = \dim V - \dim W$ for $W \leq V$?

Definition 13.5.9 (Non-degenerate)

B is **non-degenerate** if $V^\perp = 0$.

(Inner products and $B^I_{\mathcal{E}}$ are non-degenerate; Exercises 9.1.3, 10.2.1.)

Proposition 13.5.10

For a basis $\mathfrak{B}$ of V: B is non-degenerate iff $\det [B]_{\mathfrak{B}} \neq 0$.

First proof (elementary). With $J = [B]_{\mathfrak{B}}$: B is degenerate iff some $0 \neq Y$ has $X^t J Y = 0$ for all $X \in F^n$ (Observation 13.1.1), iff (taking $X = e_i$) some $0 \neq Y$ has $JY = 0$, iff $\det J = 0$. ■

(Second proof after Observation 13.8.3.)

Theorem 13.5.11 (Perp theorem, non-degenerate case)

If B is non-degenerate and $W \leq V$, then $\dim W^\perp = \dim V - \dim W$. (This does *not* say $V = W \oplus W^\perp$.)

Proof. Extend a basis $\{v_1, \ldots, v_m\}$ of W to a basis $\mathfrak{B}$ of V, $J = [B]_{\mathfrak{B}}$. Then $v \in W^\perp$ iff $0 = e_i^t J [v] = [J]_i [v]$ for $i \leq m$, i.e. $[v]_{\mathfrak{B}} \in \ker L_K$ with K the first m rows of J. J is invertible, so $\operatorname{rk} K = m$; the rank and dimension theorems give $\dim \ker L_K = n - m$. ■

Proposition 13.5.12 (The three pillars)

Of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (non-degenerate case), any two imply the third.

Proof. As in Proposition 10.3.11 with $B_{\mathcal{E}}^I$ on F^n: the row space W of A has $W^\perp = \ker L_A$. ■

This works over any F (no trick as in §10.3 for $\mathbb{C}$).

From now on in this section B is symmetric.

Exercise 13.5.13

If (V, B) is a non-degenerate quadratic space and $U, W \leq V$: (a) $(W^\perp)^\perp = W$;
(b) $(U + W)^\perp = U^\perp \cap W^\perp$;
(c) $U^\perp + W^\perp = (U \cap W)^\perp$.

Solution. As in Exercise 10.3.6, using Theorem 13.5.11 and Exercise 13.5.6(b). □

Lemma 13.5.14

If B is symmetric, $w \in V$ and $B(w, w) \neq 0$, then $V = \langle w \rangle \oplus \langle w \rangle^\perp$.

Proof. For $v \in V$, $v = \frac{B(v,w)}{B(w,w)} w + \left(v - \frac{B(v,w)}{B(w,w)} w \right)$ with the second term in $\langle w \rangle^\perp$. And $cw \in \langle w \rangle^\perp$ means $cB(w, w) = 0$, so $c = 0$. ■

Observation 13.5.15

If B is symmetric and $B(v, v) = 0$ for all v, then $B = 0$.

Proof. $2B(v, w) = B(v + w, v + w) - B(v, v) - B(w, w) = 0$. ■

Theorem 13.5.16

Every quadratic space (V, B) $(V \neq 0)$ has an orthogonal basis (B need not be non-degenerate).

Proof. Trivial if $B = 0$. Otherwise take $B(w, w) \neq 0$ (Observation 13.5.15); $V = \langle w \rangle \oplus \langle w \rangle^\perp$ (Lemma 13.5.14), and by induction $\langle w \rangle^\perp$ has an orthogonal basis; add w. ■

Definition 13.5.17 (Reflection)

For symmetric B and w with $B(w, w) \neq 0$, $S_w(v) = v - \frac{2B(v,w)}{B(w,w)} w$ is the **reflection** (**symmetry**) with respect to w.

Exercise 13.5.18

For symmetric B and $B(w, w) \neq 0$: (a) S_w is linear and $S_w \in \mathbf{O}(V, B)$;
(b) $S_w = S_{cw}$ for $c \neq 0$;
(c) $S_w(w) = -w$;
(d) $S_w(v) = v$ for $v \in \langle w \rangle^\perp$.

Solution. (a) Linearity is clear. $B(S_w v, S_w u) = B(v,u) - \frac{2B(v,w)B(w,u)}{B(w,w)} - \frac{2B(u,w)B(v,w)}{B(w,w)} + \frac{4B(v,w)B(u,w)}{B(w,w)^2}B(w,w) = B(v,u)$, using symmetry ($B(w,u) = B(u,w)$). (b)–(d) as in Exercise 9.4.4. □

By Lemma 13.5.14, (c), (d) determine S_w. For an orthogonal basis $\mathfrak{B} = \{w, v_2, \dots, v_n\}$, $[S_w]^{\mathfrak{B}}_{\mathfrak{B}} = \mathrm{diag}(-1, 1, \dots, 1) \in \mathbf{O}^J_n(F)$ ($J = [B]_{\mathfrak{B}}$), so $\det S_w = -1$, $S_w^2 = I$.

Exercise 13.5.19

For a quadratic space (V, B): (a) some $L \in \mathbf{O}(V,B)$ has $\det L = -1$;
(b) if B is non-degenerate, $\mathbf{O}(V,B)/\mathbf{SO}(V,B) \approx \mu_2$.

Solution. (a) If $B \neq 0$, a reflection S_w. If $B = 0$, $\mathbf{O}(V,B) = \mathrm{GL}(V)$ contains $\mathrm{diag}(-1, 1, \dots)$.

(b) $\det : \mathbf{O}(V,B) \to F^\times$ takes values ± 1 (Exercise 13.3.7) and is onto μ_2 by (a); apply the First Isomorphism Theorem. □

Exercise 13.5.20

(a) $-1 + 2\cosh^2 \frac{t}{2} = \cosh t = 1 + 2\sinh^2 \frac{t}{2}$ and $2\cosh \frac{t}{2} \sinh \frac{t}{2} = \sinh t$. (b) For $J = \mathrm{diag}(1, -1)$ and $Y = (\cosh \frac{t}{2}, -\sinh \frac{t}{2})^t$, $B^J_{\mathcal{E}}(Y,Y) = 1$ and $[S_Y]_{\mathcal{E}} = \begin{pmatrix} -\cosh t & -\sinh t \\ \sinh t & \cosh t \end{pmatrix}$, so this matrix is a reflection of $\mathbf{O}(1,1)$.

Solution. (a) Exercise 13.4.1(c),

(d) with $s = t$, replaced by $t/2$. (b) $B(Y,Y) = \cosh^2 \frac{t}{2} - \sinh^2 \frac{t}{2} = 1$. $S_Y e_1 = e_1 - 2\cosh \frac{t}{2} Y = (1 - 2\cosh^2 \frac{t}{2},\ 2\cosh \frac{t}{2} \sinh \frac{t}{2})^t = (-\cosh t, \sinh t)^t$, and $S_Y e_2 = e_2 - 2B(e_2, Y)Y = e_2 - 2\sinh \frac{t}{2} Y = (-2\sinh \frac{t}{2} \cosh \frac{t}{2},\ 1 + 2\sinh^2 \frac{t}{2})^t = (-\sinh t, \cosh t)^t$. □

13.6 Dual spaces and dual maps

Motivation. The matrix of the dual map $L^*(g) = g \circ L$ is the transpose.

$V^* = \mathfrak{L}(V, F)$ is the **dual space**, its elements **linear functionals**; $V^{**} = (V^*)^*$ is the **double dual**. For a basis $\mathfrak{B} = \{v_1, \dots, v_n\}$, $v_i^*(v_j) = \delta_{ij}$ defines the **dual basis** $\mathfrak{B}^* = \{v_1^*, \dots, v_n^*\}$ of V^* (§5.1), so $\dim V = \dim V^* = \dim V^{**}$.

Proposition 13.6.1

$\psi^V : V \to V^{**}$, $\psi^V(v)(f) = f(v)$, is a vector space isomorphism, independent of any choice of basis (a **natural isomorphism**); we use it to identify V and V^{**}.

Proof. $\psi^V(v) \in V^{**}$ and ψ^V is linear. $\psi^V(v) = 0$ means $f(v) = 0$ for all f, so $v = 0$ (Observation 13.6.2); $\dim V = \dim V^{**}$ gives surjectivity. ■

Observation 13.6.2

If $f(v) = 0$ for all $f \in V^*$, then $v = 0$.

Proof. With $v = \sum a_i v_i$, $0 = v_i^*(v) = a_i$ for all i. ■

Observation 13.6.3

For a basis $\{v_i\}$ of V, $\psi^V(v_i) = v_i^{**}$ (the dual basis of $\{v_i^*\}$).

Proof. $\psi^V(v_i)(v_j^*) = v_j^*(v_i) = \delta_{ij} = v_i^{**}(v_j^*)$; two linear maps agreeing on a basis are equal. ■

Notation 13.6.4

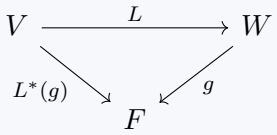
When there is a natural isomorphism, we identify the two spaces and abuse notation: $V = V^{**}$, $v_i = v_i^{**}$.

Remark 13.6.5

The isomorphisms of the first and second isomorphism theorems (Theorems 12.4.14, 12.4.21) are also natural.

Definition 13.6.6 (Dual map)

For linear $L : V \to W$, $L^* : W^* \to V^*$, $L^*(g) = g \circ L$, is the **dual map** of L.



Exercise 13.6.7

(a) For $g \in W^*$, $g \circ L \in V^*$.
(b) L^* is linear.

Solution. (a) A composite of linear maps.

(b) $(ag + h) \circ L = a(g \circ L) + h \circ L$. □

Observation 13.6.8

For bases $\mathfrak{B} = \{v_j\}$ of V, $\mathfrak{C} = \{w_i\}$ of W and $L \in \mathfrak{L}(V, W)$: $[L^*]_{\mathfrak{C}^*}^{\mathfrak{B}^*} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^t$.

Proof. With $Lv_j = \sum_i a_{ij} w_i$: $(L^* w_i^*)(v_\ell) = w_i^*(Lv_\ell) = w_i^*(\sum_k a_{k\ell} w_k) = a_{i\ell}$, i.e. $L^* w_i^* = \sum_j a_{ij} v_j^*$. ■

This answers Question 5.4.22.

Exercise 13.6.9

For $L \in \mathfrak{L}(V, W)$, $M \in \mathfrak{L}(W, U)$: (a) $(I_V)^* = I_{V^*}$;
(b) $(M \circ L)^* = L^* \circ M^*$;
(c) if L is invertible, so is L^*, with $(L^*)^{-1} = (L^{-1})^*$.

Solution. (a) $g \circ I = g$.

(b) $(ML)^*(g) = g \circ M \circ L = L^*(M^*(g))$.

(c) Apply (a), (b) to $LL^{-1} = I$ and $L^{-1}L = I$. □

Exercise 13.6.10

(a) $(L_1 + L_2)^* = L_1^* + L_2^*$;
(b) $(cL)^* = cL^*$;
(c) $* : \mathfrak{L}(V, W) \to \mathfrak{L}(W^*, V^*)$, $L \mapsto L^*$, is an isomorphism.

Solution. (a), (b): $g \circ (L_1 + L_2) = g \circ L_1 + g \circ L_2$, $g \circ (cL) = c(g \circ L)$.

(c) $*$ is linear by (a), (b), and via Observation 13.6.8 it corresponds to the transpose $\mathfrak{M}_{m,n}(F) \to \mathfrak{M}_{n,m}(F)$, which is bijective. □

Exercise 13.6.11

Translate Exercises 13.6.9 and 13.6.10 into matrix language.

Solution. $I^t = I$, $(AB)^t = B^t A^t$, $(A^{-1})^t = (A^t)^{-1}$, $(A + B)^t = A^t + B^t$, $(cA)^t = cA^t$, and $A \mapsto A^t$ is an isomorphism $\mathfrak{M}_{m,n}(F) \to \mathfrak{M}_{n,m}(F)$. □

Exercise 13.6.12

For $L \in \mathfrak{L}(V, W)$, the square

$$\begin{array}{ccc} V & \xrightarrow{L} & W \\ {\scriptstyle \psi^V}\downarrow & & \downarrow{\scriptstyle \psi^W} \\ V^{**} & \xrightarrow[L^{**}]{} & W^{**} \end{array}$$

commutes; so with $V = V^{**}$, $W = W^{**}$, $L = L^{**}$ (in matrices, $A = (A^t)^t$).

Solution. For $v \in V$, $g \in W^*$: $L^{**}(\psi^V v)(g) = \psi^V v(L^* g) = (g \circ L)(v) = g(Lv) = \psi^W(Lv)(g)$. □

13.7 Duality

Motivation. The pairing $\varepsilon(v, f) = f(v)$ behaves like a non-degenerate form: $\dim W^{\text{perp}} = \dim V - \dim W$.

V comes with the bilinear form $\varepsilon : V \times V^* \to F$, $\varepsilon(v, f) = f(v)$ (a **bilinear pairing**); we write $v \perp f$ if $\varepsilon(v, f) = 0$.

Notation 13.7.1

For $S \subseteq V$, $T \subseteq V^*$: $S^{\text{perp}} = \{f \in V^* : \varepsilon(v, f) = 0 \ \forall v \in S\}$ and $T^{\text{perp}} = \{v \in V : \varepsilon(v, f) = 0 \ \forall f \in T\}$ (the **annihilators**).

Exercise 13.7.2

(a) $S^{\text{perp}} = \langle S \rangle^{\text{perp}} \leq V^*$.
(b) $T^{\text{perp}} = \langle T \rangle^{\text{perp}} \leq V$.

Solution. Linearity of ε in each variable. □

Theorem 13.7.3 (Duality)

For $W \leq V$, $Y \leq V^*$: (a) $\dim W^{\text{perp}} = \dim V - \dim W$;
(b) $\dim Y^{\text{perp}} = \dim V - \dim Y$.

Proof. (a) Extend a basis $\{v_1, \dots, v_r\}$ of W to a basis $\{v_1, \dots, v_n\}$ of V. Then $\{v_{r+1}^*, \dots, v_n^*\}$ is a basis of W^{perp}: these vanish on W, and if $f = \sum a_i v_i^* \in W^{\text{perp}}$, then $a_i = f(v_i) = 0$ for $i \leq r$.

(b) (First proof.) Extend a basis $\{f_1, \dots, f_s\}$ of Y to a basis $\{f_i\}$ of V^* with dual basis $\{f_i^*\}$; we show $\psi^V(Y^{\text{perp}}) = \langle f_{s+1}^*, \dots, f_n^* \rangle$. If $v \in Y^{\text{perp}}$, $\psi^V(v) = \sum a_i f_i^*$, then $a_i = f_i(v) = 0$ for $i \leq s$. Conversely, $f_j^* = \psi^V(w_j)$ and, for $i \leq s < j$, $f_i(w_j) = f_j^*(f_i) = 0$, so $w_j \in Y^{\text{perp}}$. ■

(b) Second proof. By (a) for $\delta(\alpha, f) = \alpha(f)$ on $V^{**} \times V^*$, $Y^{\text{d-perp}} = \{\alpha \in V^{**} : \alpha(f) = 0 \; \forall f \in Y\}$ has dimension $\dim V - \dim Y$, and $\delta(\psi^V v, f) = \varepsilon(v, f)$ (square below) gives $\psi^V(Y^{\text{perp}}) = Y^{\text{d-perp}}$.

$$\begin{array}{ccc} V \times V^* & \xrightarrow{\varepsilon} & F \\ {\scriptstyle \psi^V \times I}\downarrow & & \downarrow{\scriptstyle I} \\ V^{**} \times V^* & \xrightarrow[\delta]{} & F \end{array}$$

■

The dual of "a proper W has a nonzero functional vanishing on it" needs the theorem:

Corollary 13.7.4

If $Y \leq V^*$, $Y \neq V^*$, then some $0 \neq v \in V$ has $f(v) = 0$ for all $f \in Y$.

Proof. By duality, $\dim Y^{\text{perp}} = \dim V - \dim Y > 0$. ■

Exercise 13.7.5

For $W \leq V$: (a) $(W^{\text{perp}})^{\text{perp}} = W$;
(b) with $V = V^{**}$, $(W^{\text{perp}})^{\text{d-perp}} = W$.

Solution. $W \leq (W^{\text{perp}})^{\text{perp}}$ is clear, and the dimensions agree by Theorem 13.7.3 applied twice.

(b) is (a) transported by ψ^V. □

Exercise 13.7.6

For $U, W \leq V$: (a) $(U + W)^{\text{perp}} = U^{\text{perp}} \cap W^{\text{perp}}$;
(b) $(U \cap W)^{\text{perp}} = U^{\text{perp}} + W^{\text{perp}}$.

Solution. (a) is immediate.

(b) $\supseteq$ is clear; the dimensions agree by Theorem 13.7.3, (a) and Corollary 12.4.22: $\dim(U^{\text{perp}} + W^{\text{perp}}) = (n - u) + (n - w) - (n - \dim(U + W)) = n - \dim(U \cap W)$. □

Exercise 13.7.7

For $L \in \mathfrak{L}(V, W)$: (a) $\ker(L^*) = (\operatorname{im} L)^{\text{perp}}$;
(b) $\operatorname{im}(L^*) = (\ker L)^{\text{perp}}$.

Solution. (a) $L^* g = 0 \iff g(Lv) = 0 \; \forall v \iff g \in (\operatorname{im} L)^{\text{perp}}$. (b) $\operatorname{im} L^* \subseteq (\ker L)^{\text{perp}}$: $(g \circ L)(v) = 0$ for $v \in \ker L$. The dimensions agree: $\dim \operatorname{im} L^* = \dim W - \dim \ker L^* = \dim W - (\dim W - \operatorname{rk} L) = \operatorname{rk} L = \dim V - \dim \ker L$. □

(Compare Exercise 10.3.6 and Observation 10.7.9.)

13.8 B-identification

Motivation. A non-degenerate B identifies $V = V^*$ and becomes the pairing ε.

Definition 13.8.1 (φ_B)

For a bilinear form B on V, $\varphi_B^V : V \to V^*$, $\varphi_B^V(v)(w) = B(w, v)$, is a linear map (by bilinearity).

Theorem 13.8.2

The following are equivalent: (1) B is non-degenerate; (2) φ_B^V is a monomorphism; (3) φ_B^V is an isomorphism; (4) every element of V^* is $\varphi_B^V(v)$ for some v.

Proof. $\ker \varphi_B^V = \{v : B(w, v) = 0 \ \forall w\} = V^\perp$. So (1)⇔(2), and (2)⇔(3)⇔(4) by the dimension theorem ($\dim V = \dim V^*$). ■

(Surjectivity fails in infinite dimensions.)

Observation 13.8.3

For a basis $\mathfrak{B}$ of V with dual basis $\mathfrak{B}^*$: $[\varphi_B^V]_{\mathfrak{B}}^{\mathfrak{B}^*} = [B]_{\mathfrak{B}}$.

Proof. With $[\varphi_B^V]_{\mathfrak{B}}^{\mathfrak{B}^*} = (b_{ij})$, $\varphi_B^V(v_j) = \sum_i b_{ij} v_i^*$, so $B(v_k, v_j) = \varphi_B^V(v_j)(v_k) = b_{kj}$. ■

Second proof of Proposition 13.5.10. B is non-degenerate iff φ_B^V is an isomorphism, iff its matrix $[B]_{\mathfrak{B}}$ is invertible. ■

Exercise 13.8.4

Identifying $V = V^{**}$, the dual map $(\varphi_B^V)^* : V^{**} \to V^*$ becomes a map $V \to V^*$. Show that B is symmetric iff $(\varphi_B^V)^* = \varphi_B^V$.

Solution. $(\varphi_B^V)^*(\psi^V v)(w) = \psi^V(v)(\varphi_B^V w) = \varphi_B^V(w)(v) = B(v, w)$, while $\varphi_B^V(v)(w) = B(w, v)$. These agree for all v, w iff B is symmetric. (In matrices: $[B]^t = [B]$.) □

Notation 13.8.5 (B-identification)

If B is non-degenerate, φ_B^V is a natural isomorphism, and we identify $V = V^*$ through it, writing $v = \varphi_v = \varphi_B^V(v)$.

Remark 13.8.6

$B^*(\varphi_v, \varphi_w) = B(w, v)$ is a non-degenerate form on V^*, and the identifications are consistent, $\varphi_{B^*}^{V^*} \circ \varphi_B^V = \psi^V$: $\varphi_{B^*}^{V^*}(\varphi_B^V(v))(\varphi_w) = B^*(\varphi_w, \varphi_v) = B(v, w) = \varphi_w(v) = \psi^V(v)(\varphi_w)$.

Exercise 13.8.7

For $f \in (\mathbb{R}^2)^*$, $f((x,y)^t) = x+y$, find the vector of $\mathbb{R}^2$ corresponding to f under $B^J_{\mathcal{E}}$-identification, for (a) $J = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$,
(b) $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$.

Solution. We need v with $X^tJv = x+y$, i.e. $Jv = (1,1)^t$. (a) $v = (1,-1)^t$.
(b) $v = J^{-1}(1,1)^t = \frac{1}{3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)(1,1)^t = (1,1)^t$. □

Exercise 13.8.8

If B is non-degenerate and $\mathfrak{B} = \{v_i\}$ is a basis, there is a basis $\mathfrak{C} = \{w_j\}$ with $B(v_i, w_j) = \delta_{ij}$, and $[I]^{\mathfrak{B}}_{\mathfrak{C}} = [B]^{-1}_{\mathfrak{B}}$.

Solution. $w_j = (\varphi^V_B)^{-1}(v_j^*)$ gives $B(v_i, w_j) = v_j^*(v_i) = \delta_{ij}$. With $w_j = \sum_k c_{kj} v_k$, $C = [I]^{\mathfrak{B}}_{\mathfrak{C}}$, $\delta_{ij} = \sum_k B(v_i, v_k) c_{kj} = ([B]_{\mathfrak{B}} C)_{ij}$, so $C = [B]^{-1}_{\mathfrak{B}}$. □

Exercise 13.8.9

If V has an orthonormal basis $\{v_i\}$ (so (V,B) is a non-degenerate quadratic space), then $\varphi^V_B(v_i) = v_i^*$.

Solution. $\varphi^V_B(v_i)(v_k) = B(v_k, v_i) = \delta_{ki} = v_i^*(v_k)$. (Compare Observation 13.6.3.) □

Proposition 13.8.10

If B is non-degenerate, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{B} & F \\ {\scriptstyle I \times \varphi^V_B}\downarrow & & \downarrow{\scriptstyle I} \\ V \times V^* & \xrightarrow[\varepsilon]{} & F \end{array}$$

commutes: under B-identification, $B = \varepsilon$. In particular "$\perp$= perp".

Proof. $\varepsilon(v, \varphi^V_B w) = \varphi^V_B(w)(v) = B(v,w)$. ■

Theorem 13.8.11 (Perp theorem, non-degenerate case)

If B is non-degenerate and $W \leq V$, then under B-identification $W^\perp = W^{\text{perp}}$; in particular $\dim W^\perp = \dim V - \dim W$.

Proof. By Proposition 13.8.10, $v \in W^\perp$ iff $\varphi^V_B(v) \in W^{\text{perp}}$, i.e. $\varphi^V_B(W^\perp) = W^{\text{perp}}$; apply Theorem 13.7.3(a). ■

Exercise 13.8.13

Let (V,B) be a non-degenerate quadratic space and $U \leq W \leq V$. Define $\vartheta : W \to (U^\perp/W^\perp)^*$, $\vartheta(w)(\bar{v}) = B(v,w)$ $(w \in W,\ v \in U^\perp)$. (a) ϑ is a well-defined linear map with $\ker\vartheta = U$. (b) There is a natural isomorphism $W/U \approx (U^\perp/W^\perp)^*$; in particular $W \approx (V/W^\perp)^*$. (c) There is a natural isomorphism $W/(W \cap W^\perp) \approx (W/(W \cap W^\perp))^*$.

Solution. (a) $W^\perp \le U^\perp$, and $v-v' \in W^\perp$ gives $B(v-v', w) = 0$: well defined; linear in w. $\vartheta(w) = 0$ iff $w \in (U^\perp)^\perp = U$ (Exercise 13.5.13). (b) $W/U \approx \operatorname{im}\vartheta$, and $\dim(U^\perp/W^\perp) = (n-u)-(n-w) = w-u$ (Theorem 13.5.11), so ϑ is onto. Take $U = 0$ for the special case. (c) $\bar{B}(\bar{w}, \bar{w}') = B(w, w')$ is well defined and non-degenerate on $W/(W \cap W^\perp)$ (the radical of $B|_W$), so $\varphi_{\bar{B}}$ is a natural isomorphism (Theorem 13.8.2). □

13.9 Transpose operators

Motivation. The dual map under B-identification is the transpose: $C(Lv, w) = B(v, L^t w)$.

(V, B) and (W, C) are non-degenerate quadratic spaces.

Definition 13.9.1 (Transpose map)

(a) For $L \in \mathfrak{L}(V, W)$, the **transpose** $L^t \in \mathfrak{L}(W, V)$ (with respect to B, C) is L^* under B- and C-identification, i.e. the unique linear map making

$$\begin{array}{ccc} W & \xrightarrow{L^t} & V \\ {\scriptstyle \varphi_C^W}\downarrow & & \downarrow{\scriptstyle \varphi_B^V} \\ W^* & \xrightarrow[L^*]{} & V^* \end{array}$$

commute: $L^t = (\varphi_B^V)^{-1} \circ L^* \circ \varphi_C^W$.
(b) For $L \in \mathfrak{L}(V, V)$, the **transpose operator** is $L^t = (\varphi_B^V)^{-1} \circ L^* \circ \varphi_B^V$.

(The dual map does not depend on B, C; the transpose does.)

Observation 13.9.2

If $L, M \in \mathfrak{L}(V, V)$ and $B(Lv, w) = B(Mv, w)$ for all v, w, then $L = M$ (likewise in the second variable).

Proof. $B(Lv - Mv, w) = 0$ for all w, so $Lv - Mv \in V^\perp = 0$ (by symmetry of B). ■

Proposition 13.9.3

L^t is the unique linear map with $C(Lv, w) = B(v, L^t w)$ for all $v \in V$, $w \in W$ (equivalently $C(w, Lv) = B(L^t w, v)$).

Proof. Evaluate $L^*(\varphi_C^W(w)) = \varphi_B^V(L^t w)$ at v: $C(Lv, w) = (L^*\varphi_w)(v) = \varphi_{L^t w}(v) = B(v, L^t w)$. Uniqueness: Observation 13.9.2. ■

Proposition 13.9.4

For bases $\mathfrak{B}, \mathfrak{C}$ of V, W with $J_\mathfrak{B} = [B]_\mathfrak{B}$, $J_\mathfrak{C} = [C]_\mathfrak{C}$: (a) $[L^t]_\mathfrak{C}^\mathfrak{B} = (J_\mathfrak{B})^{-1}([L]_\mathfrak{B}^\mathfrak{C})^t J_\mathfrak{C}$; (b) for $L \in \mathfrak{L}(V, V)$ and $J = [B]_\mathfrak{B}$, $[L^t]_\mathfrak{B}^\mathfrak{B} = J^{-1}([L]_\mathfrak{B}^\mathfrak{B})^t J$.

Proof. From $L^t = (\varphi_B^V)^{-1} L^* \varphi_C^W$: $[L^t]_\mathfrak{C}^\mathfrak{B} = [(\varphi_B^V)^{-1}]_{\mathfrak{B}^*}^\mathfrak{B} [L^*]_{\mathfrak{C}^*}^{\mathfrak{B}^*} [\varphi_C^W]_\mathfrak{C}^{\mathfrak{C}^*}$, with $[L^*] = ([L]_\mathfrak{B}^\mathfrak{C})^t$ (Observation 13.6.8) and $[\varphi_B^V] = J_\mathfrak{B}$, $[\varphi_C^W] = J_\mathfrak{C}$ (Observation 13.8.3).

(b) is a special case. ■

(Compare Observation 13.1.8.)

Exercise 13.9.5

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal bases, $[L^t]_{\mathfrak{C}}^{\mathfrak{B}} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^t$.

Solution. $J_{\mathfrak{B}} = J_{\mathfrak{C}} = I$ in Proposition 13.9.4. □

Exercise 13.9.6

(a) For $L(x_1, x_2, x_3, x_4)^t = (x_4, x_1, x_2, x_3)^t$ on $\mathbb{R}^4$ with the dot product, find L^t. (b) For $J = \mathrm{diag}(1, -1)$ and $L(x, y)^t = (x + y, -x + 3y)^t$ on $(\mathbb{R}^2, B_{\mathcal{E}}^J)$, find L^t.

Solution. (a) $[L]$ is a permutation matrix; $L^t = L^{-1}$: $L^t(x_1, x_2, x_3, x_4)^t = (x_2, x_3, x_4, x_1)^t$. (b) $[L] = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$ and $J^{-1}[L]^t J = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$: $L^t = L$, though $[L]$ is not symmetric. □

Proposition 13.9.7

$\mathbf{O}(V, B) = \{L \in \mathrm{GL}(V) : L^t \circ L = I\}$.

Proof. For $L \in \mathrm{GL}(V)$: $B(v, w) = B(Lv, Lw) = B(v, (L^t \circ L)w)$ for all v, w iff $L^t L = I$ (Observation 13.9.2). ■

Remark 13.9.8

Reading $A^t J A = J$ as $L^t M L = M$ for an operator M is a dead end: J is the matrix of a form.

Observation 13.9.9

$\mathbf{t} : \mathfrak{L}(V, W) \to \mathfrak{L}(W, V)$, $L \mapsto L^t$, is an isomorphism; in particular $(L + M)^t = L^t + M^t$ and $(cL)^t = cL^t$.

Proof. It is the composite of the isomorphism $*$ (Exercise 13.6.10) with the identifications. ■

Observation 13.9.10

For $L \in \mathfrak{L}(V, W)$, $M \in \mathfrak{L}(W, U)$: (a) $I^t = I$;
(b) $(M \circ L)^t = L^t \circ M^t$;
(c) if L is invertible, so is L^t, with $(L^t)^{-1} = (L^{-1})^t$;
(d) $(L^t)^t = L$.

Proof. (a), (b) from Exercise 13.6.9. (c) from (a), (b).

(d) By Proposition 13.9.3 twice: $C(Lv, w) = B(v, L^t w) = C((L^t)^t v, w)$; apply Observation 13.9.2. ■

Exercise 13.9.11

Prove Observations 13.9.9 and 13.9.10(d) directly from $L^t = (\varphi_B^V)^{-1} L^* \varphi_C^W$.

Solution. $\mathbf{t}$ is linear because $*$ and composition with fixed isomorphisms are. For (d), $(L^t)^t = (\varphi_C^W)^{-1}(L^t)^* \varphi_B^V$ and $(L^t)^* = (\varphi_C^W)^* L^{**} ((\varphi_B^V)^{-1})^*$. Using $(\varphi_B^V)^* = \varphi_B^V$ (Exercise 13.8.4, B symmetric) and $L^{**} = L$ (Exercise 13.6.12), this gives L. □

Exercise 13.9.12

For $L \in \mathfrak{L}(V,V)$: (a) $\mathrm{tr}L^t = \mathrm{tr}L$;
(b) $\det L^t = \det L$;
(c) $\phi_{L^t} = \phi_L$;
(d) $m_{L^t} = m_L$.

Solution. $[L^t] = J^{-1}[L]^t J$ (Proposition 13.9.4(b)) is similar to $[L]^t$. For (d), $f(L^t) = f(L)^t$ (Observations 13.9.9, 13.9.10). □

Exercise 13.9.13

For $L \in \mathfrak{L}(V,W)$: (a) $\ker(L^t) = (\mathrm{im}\, L)^\perp$;
(b) $\mathrm{im}(L^t) = (\ker L)^\perp$.

Solution. (a) $L^t w = 0 \iff B(v, L^t w) = 0\ \forall v \iff C(Lv, w) = 0\ \forall v \iff w \in (\mathrm{im}\, L)^\perp$. (b) Apply (a) to L^t (using $(L^t)^t = L$) and take $\perp$: $\mathrm{im}\, L^t = ((\mathrm{im}\, L^t)^\perp)^\perp = (\ker L)^\perp$. □

Exercise 13.9.14

Explain the relation between Exercises 13.7.7 and 13.9.13 from the point of view of Proposition 13.8.10.

Solution. Under the identifications $L^t = L^*$ and $\perp$= perp (Proposition 13.8.10, Theorem 13.8.11). □

Chapter 14. Hermitian Forms

This is Chapter 13 with conjugations added; proofs identical up to conjugation are only indicated. $F = \mathbb{C}$ and V is n-dimensional.

14.1 Hermitian forms

Motivation. Sesquilinear forms are matrices via $H(v,w) = [v]^t J\overline{[w]}$.

Definition 14.1.1 (Sesquilinear and Hermitian forms)

$H : V \times V \to \mathbb{C}$ is **sesquilinear** if for all $u, v, w \in V$, $c \in \mathbb{C}$: (H1) $H(u+v,w) = H(u,w) + H(v,w)$, $H(cv,w) = cH(v,w)$; (H2) $H(u,v+w) = H(u,v) + H(u,w)$, $H(v,cw) = \bar{c}H(v,w)$. If moreover (H3) $H(v,w) = \overline{H(w,v)}$, H is a **Hermitian form** and (V,H) a **Hermitian space**.

Example 14.1.2

For a basis $\mathfrak{B} = \{v_i\}$, $[H]_\mathfrak{B} = (H(v_i,v_j))$ is the **matrix of** H, and for $v = \sum a_i v_i$, $w = \sum b_i v_i$:

$$H(v,w) = \sum_{i,j} a_i H(v_i,v_j)\bar{b}_j = [v]_\mathfrak{B}^t [H]_\mathfrak{B} \overline{[w]_\mathfrak{B}}.$$

Exercise 14.1.3

For $J \in \mathfrak{M}_{n,n}(\mathbb{C})$, $H_\mathfrak{B}^J(v,w) = [v]_\mathfrak{B}^t J \overline{[w]_\mathfrak{B}}$ is sesquilinear with $[H_\mathfrak{B}^J]_\mathfrak{B} = J$. (The dot product of $\mathbb{C}^n$ is $H_\mathcal{E}^I$.)

Solution. Linear in $[v]$, conjugate linear in $[w]$; $H_\mathfrak{B}^J(v_i,v_j) = e_i^t J e_j = J_{ij}$. □

Exercise 14.1.4

The sesquilinear forms $\mathfrak{Ses}(V)$ form a $\mathbb{C}$-vector space under pointwise operations.

Solution. As in Exercise 13.1.5. □

Theorem 14.1.5

$\Omega_\mathfrak{B} : \mathfrak{Ses}(V) \to \mathfrak{M}_{n,n}(\mathbb{C})$, $H \mapsto [H]_\mathfrak{B}$, is a vector space isomorphism.

Proof. Injective by Example 14.1.2, surjective by Exercise 14.1.3, linear entrywise. ■

Remark 14.1.6

With $\alpha_\mathfrak{B} : V \to \mathbb{C}^n$ and $J = [H]_\mathfrak{B}$, (V,H) and $(\mathbb{C}^n, H_\mathcal{E}^J)$ are essentially the same (the analogue of Remark 13.1.7).

Exercise 14.1.7

If $J \in \mathrm{GL}_n(\mathbb{C})$ and $A \in \mathfrak{M}_{n,n}(\mathbb{C})$, then $H_\mathcal{E}^J(AX,Y) = H_\mathcal{E}^J(X, (\bar{J}^{-1}A^*\bar{J})Y)$.

Solution. $(AX)^t J\bar{Y} = X^t A^t J\bar{Y}$, and $X^t J\,\overline{\bar{J}^{-1}A^*\bar{J}Y} = X^t JJ^{-1}\overline{A^*}J\bar{Y} = X^t A^t J\bar{Y}$, since $\overline{A^*} = A^t$. □

Exercise 14.1.8

A sesquilinear H is Hermitian iff $J = [H]_{\mathfrak{B}}$ satisfies $J^t = \bar{J}$, i.e. $J^* = J$. Such J are **self-adjoint** (**Hermitian**) matrices.

Solution. (H3) on basis vectors is $J_{ij} = \overline{J_{ji}}$. Conversely, by Example 14.1.2, $\overline{H(w,v)} = \overline{[w]^t J[v]} = \overline{[w]}^t \bar{J}[v] = [v]^t \bar{J}^t[w] = [v]^t J[w]$ when $\bar{J}^t = J$. □

Remark 14.1.9

$\mathfrak{Her}(V)$ is only a real vector space (iH is not Hermitian if $H \neq 0$).

Definition 14.1.10 (Unitary group)

$\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : H(Lv,Lw) = H(v,w)\ \forall v,w\}$, the **unitary group**; its elements are **unitary operators**; this is **unitary geometry**.

Exercise 14.1.11

For a basis $\{v_i\}$, $\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : H(Lv_i,Lv_j) = H(v_i,v_j)\ \forall i,j\}$.

Solution. By sesquilinearity, both sides of $H(Lv,Lw) = H(v,w)$ expand into the same combination of these values. □

Proposition 14.1.12

With $J = [H]_{\mathfrak{B}}$ (self-adjoint), $\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{U}(V,H)\} = \{A \in \mathrm{GL}_n(\mathbb{C}) : A^t J\bar{A} = J\}$.

Proof. As in Proposition 13.3.5: $H(Lv,Lw) = [v]^t([L]^t J\overline{[L]})\overline{[w]}$, which equals $[v]^t J\overline{[w]}$ for all v,w iff $[L]^t J\overline{[L]} = J$ (Exercise 9.3.7, applied with $\overline{[w]}$ ranging over $\mathbb{C}^n$). ■

This settles Remark 10.6.4 in the complex case ($J = I$).

Definition 14.1.13 (Matrix unitary groups)

For self-adjoint J, $\mathbf{U}_n^J(\mathbb{C}) = \{A \in \mathrm{GL}_n(\mathbb{C}) : A^t J\bar{A} = J\}$. (So $\mathbf{U}(n) = \mathbf{U}_n^I(\mathbb{C})$.)

Exercise 14.1.14

If J is invertible self-adjoint and $A \in \mathbf{U}_n^J(\mathbb{C})$, then $|\det A| = 1$.

Solution. $\det A \cdot \det J \cdot \overline{\det A} = \det J \neq 0$, so $|\det A|^2 = 1$. □

14.2 Non-degenerate Hermitian forms

Motivation. The perp theorem and orthogonal bases transfer to Hermitian spaces.

Definition 14.2.1 (Orthogonality)

$v \perp w$ if $H(v,w) = 0$; $S \perp T$ similarly; $S^\perp = \{v : H(w,v) = 0 \ \forall w \in S\}$. For Hermitian H, $H(v,w) = 0 \iff H(w,v) = \overline{H(v,w)} = 0$: orthogonality is symmetric.

Exercise 14.2.2

$S^\perp = \langle S \rangle^\perp \leq V$.

Solution. Conjugate linearity in the second variable still gives a subspace, and linearity in the first gives $S^\perp = \langle S \rangle^\perp$. □

Definition 14.2.3 (Orthogonal bases)

$\mathfrak{B}$ is an **orthogonal basis** if $[H]_\mathfrak{B}$ is diagonal and an **orthonormal basis** if $[H]_\mathfrak{B} = I$.

Definition 14.2.4 (Non-degenerate)

H is **non-degenerate** if $V^\perp = 0$. (E.g. the dot product $H^I_\mathcal{E}$ of $\mathbb{C}^n$.)

Proposition 14.2.5

H is non-degenerate iff $\det [H]_\mathfrak{B} \neq 0$.

Proof. As in the first proof of Proposition 13.5.10, replacing Y by $\bar{Y}$: H is degenerate iff some $\bar{Y} \neq 0$ has $X^t J \bar{Y} = 0$ for all X, iff $J\bar{Y} = 0$ for some $\bar{Y} \neq 0$. ■

Theorem 14.2.6 (Perp theorem, non-degenerate case)

If H is non-degenerate and $W \leq V$, then $\dim W^\perp = \dim V - \dim W$.

Proof. As in Theorem 13.5.11: with K the first m rows of J, $v \in W^\perp$ iff $K\overline{[v]} = 0$, i.e. $\overline{W^\perp}$ (in coordinates) is $\ker L_K$. By Exercise 10.3.10, $\dim W^\perp = \dim \ker L_K = n - m$ (rank and dimension theorems). ■

Exercise 14.2.7

For a non-degenerate Hermitian space and $U, W \leq V$: $(W^\perp)^\perp = W$, $(U+W)^\perp = U^\perp \cap W^\perp$, $U^\perp + W^\perp = (U \cap W)^\perp$.

Solution. As in Exercise 13.5.13, using Theorem 14.2.6. □

Exercise 14.2.8

If H is Hermitian and $H(w,w) \neq 0$, then $V = \langle w \rangle \oplus \langle w \rangle^\perp$.

Solution. $v = \frac{H(v,w)}{H(w,w)} w + \left(v - \frac{H(v,w)}{H(w,w)} w \right)$, and the second term lies in $\langle w \rangle^\perp$ by Hermitian symmetry ($H(w,\cdot) = \overline{H(\cdot,w)}$); $\langle w \rangle \cap \langle w \rangle^\perp = 0$. □

Observation 14.2.9

If H is Hermitian and $H(v,v)=0$ for all v, then $H=0$.

Proof. $H(v+w,v+w)-H(v,v)-H(w,w)=2\operatorname{Re}H(v,w)$, and as in the proof of Observation 10.6.5, $H(v+iw,v+iw)-H(v,v)-H(w,w)=2\operatorname{Im}H(v,w)$. Both vanish, so $H(v,w)=0$. ■

Theorem 14.2.10

Every nonzero Hermitian space (V,H) has an orthogonal basis (H need not be non-degenerate).

Proof [Supplement]. As in Theorem 13.5.16: if $H=0$ any basis works; otherwise pick w with $H(w,w)\neq 0$ (Observation 14.2.9), split $V=\langle w\rangle\oplus\langle w\rangle^{\perp}$ (Exercise 14.2.8) and induct on dimension. ■

14.3 H-identification and adjoint operators

Motivation. The conjugate linear bijection $V\to V^*$ turns the dual map into the adjoint $L^\star$.

Remark 14.3.1

$\varphi_H^V: V\to V^*$, $\varphi_H^V(v)(w)=H(w,v)$, is well defined (linear in w), but it is *not* linear: H is conjugate linear in its second variable.

Exercise 14.3.2

$\varphi_H^V(u+v)=\varphi_H^V(u)+\varphi_H^V(v)$ and $\varphi_H^V(cv)=\bar{c}\varphi_H^V(v)$: φ_H^V is conjugate linear.

Solution. $H(w,u+v)=H(w,u)+H(w,v)$ and $H(w,cv)=\bar{c}H(w,v)$. □

Observation 14.3.3

For $W\leq V$: (a) $\varphi_H^V(W)\leq V^*$;
(b) if H is non-degenerate, $\dim W=\dim\varphi_H^V(W)$.

Proof. (a) Conjugate linear images of subspaces are subspaces ($c\varphi(w)=\varphi(\bar{c}w)$).

(b) If $\{w_j\}$ is a basis of W, then $\{\varphi(w_j)\}$ spans $\varphi(W)$. It is independent: $\sum c_j\varphi(w_j)=\varphi(\sum\bar{c}_jw_j)=0$ forces $\sum\bar{c}_jw_j\in V^{\perp}=0$ (as in Exercise 10.3.10). ■

Theorem 14.3.4

If H is non-degenerate, φ_H^V is a bijection (an isomorphism of abelian groups $(V,+)\to(V^*,+)$); every element of V^* is $\varphi_H^V(v)$ for a unique v.

Proof. Injective: $\varphi(u)=\varphi(v)$ gives $H(w,u-v)=0$ for all w, so $u=v$. Surjective: Observation 14.3.3(b) with $W=V$. ■

(φ_H^V is not linear, so has no matrix.)

Notation 14.3.5 (H-identification)

For non-degenerate H, identify $V = V^*$ via φ_H^V, $v = \varphi_v$ (only conjugate linear; Remark 13.8.6 does not carry over).

Exercise 14.3.6

If H is non-degenerate and $\{v_i\}$ is a basis, there is a basis $\{w_j\}$ with $H(v_i, w_j) = \delta_{ij}$.

Solution. $w_j = (\varphi_H^V)^{-1}(v_j^*)$; these form a basis by Observation 14.3.3(b). □

Exercise 14.3.7

If V has an orthonormal basis $\{v_i\}$, then $\varphi_H^V(v_i) = v_i^*$.

Solution. $\varphi_H^V(v_i)(v_k) = H(v_k, v_i) = \delta_{ki}$. □

Exercise 14.3.8

If H is non-degenerate, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{H} & \mathbb{C} \\ {\scriptstyle I\times\varphi_H^V}\downarrow & & \downarrow{\scriptstyle I} \\ V \times V^* & \xrightarrow[\varepsilon]{} & \mathbb{C} \end{array}$$

commutes: under H-identification, $H = \varepsilon$.

Solution. $\varepsilon(v, \varphi_H^V w) = \varphi_H^V(w)(v) = H(v, w)$. □

Theorem 14.3.9 (Perp theorem, non-degenerate case)

If H is non-degenerate and $W \leq V$, then under H-identification $W^\perp = W^{\text{perp}}$; in particular $\dim W^\perp = \dim V - \dim W$.

Proof. By Exercise 14.3.8, $v \in W^\perp \iff \varphi_v \in W^{\text{perp}}$, i.e. $\varphi_H^V(W^\perp) = W^{\text{perp}}$. By Observation 14.3.3(b) and Theorem 13.7.3, $\dim W^\perp = \dim W^{\text{perp}} = \dim V - \dim W$. ■

From now on (V, H) and (W, K) are non-degenerate Hermitian spaces with bases $\mathfrak{B}, \mathfrak{C}$.

Definition 14.3.10 (Adjoint map)

(a) For $L \in \mathfrak{L}(V, W)$, the **adjoint map** $L^\star \in \mathfrak{L}(W, V)$ (with respect to H, K) is L^* under H- and K-identification:

$$\begin{array}{ccc} W & \xrightarrow{L^\star} & V \\ {\scriptstyle \varphi_K^W}\downarrow & & \downarrow{\scriptstyle \varphi_H^V} \\ W^* & \xrightarrow[L^*]{} & V^* \end{array}$$

i.e. $L^\star = (\varphi_H^V)^{-1} \circ L^* \circ \varphi_K^W$, which is linear (a composite of two conjugate linear maps and a linear one).
(b) For $L \in \mathfrak{L}(V, V)$, the **adjoint operator** is $L^\star = (\varphi_H^V)^{-1} \circ L^* \circ \varphi_H^V$.

Proposition 14.3.11

$L^\star$ is the unique linear map with $K(Lv,w) = H(v,L^\star w)$ for all $v \in V$, $w \in W$ (equivalently $K(w,Lv) = H(L^\star w, v)$).

Proof. As in Proposition 13.9.3: $K(Lv,w) = \varphi_w(Lv) = (L^*\varphi_w)(v) = \varphi_{L^\star w}(v) = H(v,L^\star w)$; uniqueness by non-degeneracy. The second form follows by conjugating. ■

Proposition 14.3.12

(a) With $J_{\mathfrak{B}} = [H]_{\mathfrak{B}}$, $J_{\mathfrak{C}} = [K]_{\mathfrak{C}}$: $[L^\star]_{\mathfrak{C}}^{\mathfrak{B}} = \overline{J_{\mathfrak{B}}}^{-1}([L]_{\mathfrak{B}}^{\mathfrak{C}})^*\overline{J_{\mathfrak{C}}}$. (b) For $L \in \mathfrak{L}(V,V)$ and $J = [H]_{\mathfrak{B}}$: $[L^\star]_{\mathfrak{B}}^{\mathfrak{B}} = \bar{J}^{-1}([L]_{\mathfrak{B}}^{\mathfrak{B}})^*\bar{J}$.

Proof. Proposition 14.3.11 reads $([Lv]_{\mathfrak{C}})^t J_{\mathfrak{C}}\overline{[w]_{\mathfrak{C}}} = [v]_{\mathfrak{B}}^t J_{\mathfrak{B}}\overline{[L^\star w]_{\mathfrak{B}}}$, i.e. $[v]^t([L]_{\mathfrak{B}}^{\mathfrak{C}})^t J_{\mathfrak{C}}\overline{[w]} = [v]^t J_{\mathfrak{B}}\overline{[L^\star]_{\mathfrak{C}}^{\mathfrak{B}}}\,\overline{[w]}$ for all v,w. By Exercise 9.3.7, $([L]_{\mathfrak{B}}^{\mathfrak{C}})^t J_{\mathfrak{C}} = J_{\mathfrak{B}}\overline{[L^\star]_{\mathfrak{C}}^{\mathfrak{B}}}$; conjugate and solve. (Without the bars this is a second proof of Proposition 13.9.4.) ■

Exercise 14.3.13

If $\mathfrak{B},\mathfrak{C}$ are orthonormal, $[L^\star]_{\mathfrak{C}}^{\mathfrak{B}} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^*$.

Solution. $J_{\mathfrak{B}} = J_{\mathfrak{C}} = I$ in Proposition 14.3.12. □

Proposition 14.3.14

$\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : L^\star \circ L = I\}$.

Proof. $H(Lv,Lw) = H(v,L^\star Lw)$ equals $H(v,w)$ for all v,w iff $L^\star L = I$ (non-degeneracy, as in Observation 13.9.2). ■

Observation 14.3.15

For $L,M \in \mathfrak{L}(V,W)$, $c \in \mathbb{C}$: (a) $(L+M)^\star = L^\star + M^\star$;
(b) $(cL)^\star = \bar{c}L^\star$.

Proof. (b) $(cL)^\star = (\varphi_H^V)^{-1}(cL^*)\varphi_K^W = \bar{c}(\varphi_H^V)^{-1}L^*\varphi_K^W$, since $(\varphi_H^V)^{-1}$ is conjugate linear. (a) is similar. ■

Exercise 14.3.16

For $L \in \mathfrak{L}(V,W)$, $M \in \mathfrak{L}(W,U)$: (a) $I^\star = I$;
(b) $(M \circ L)^\star = L^\star \circ M^\star$;
(c) if L is invertible, $(L^\star)^{-1} = (L^{-1})^\star$;
(d) $(L^\star)^\star = L$.

Solution. Use Proposition 14.3.11:

(b) $U(MLv,u) = K(Lv,M^\star u) = H(v,L^\star M^\star u)$.

(d) $K(Lv,w) = H(v,L^\star w) = \overline{H(L^\star w,v)} = \overline{K(w,(L^\star)^\star v)} = K((L^\star)^\star v,w)$. (a), (c) follow. □

Exercise 14.3.17

For $L \in \mathfrak{L}(V,V)$: (a) $\operatorname{tr} L^\star = \overline{\operatorname{tr} L}$;
(b) $\det L^\star = \overline{\det L}$;
(c) $\phi_{L^\star}(t) = \overline{\phi_L}(t)$ (conjugate coefficients);
(d) $m_{L^\star} = \overline{m_L}$.

Solution. $[L^\star] = \bar{J}^{-1}[L]^*\bar{J}$ is similar to $[L]^* = \overline{[L]}^t$, whose trace, determinant and characteristic polynomial are the conjugates of those of $[L]$.

(d) $f(L^\star) = (\bar{f}(L))^\star$ by Observation 14.3.15 and Exercise 14.3.16. □

Exercise 14.3.18

For $L \in \mathfrak{L}(V,W)$: (a) $\ker L^\star = (\operatorname{im} L)^\perp$;
(b) $\operatorname{im} L^\star = (\ker L)^\perp$.

Solution. As in Exercise 13.9.13, using Proposition 14.3.11, Exercise 14.2.7 and $(L^\star)^\star = L$. □

14.4 Selected exercises

Motivation. A sesquilinear form is determined by its values $H(v,v)$.

Exercise 14.4.1 (Polarization)

Let H be a sesquilinear form on the $\mathbb{C}$-vector space V.
(a) Show that $H(v,w) = \frac{1}{4}\sum_{k=0}^{3} i^k\, H(v + i^k w,\, v + i^k w)$ for all $v, w \in V$.
(b) Show that H is Hermitian iff $H(v,v) \in \mathbb{R}$ for all v.
(c) Show that $L \in \mathrm{GL}(V)$ lies in $\mathbf{U}(V,H)$ iff $H(Lv,Lv) = H(v,v)$ for all v.

Solution [Supplement].

(a) For $c \in \mathbb{C}$ with $|c| = 1$, (H1) and (H2) give

$$H(v + cw, v + cw) = H(v,v) + \bar{c}\, H(v,w) + c\, H(w,v) + |c|^2 H(w,w).$$

Take $c = i^k$, multiply by i^k, sum over k: as $i^k\overline{i^k} = 1$ and $\sum_k i^k = \sum_k i^{2k} = 0$, only $4H(v,w)$ survives.

(b) ($\Rightarrow$) by (H3). ($\Leftarrow$) $H'(v,w) = \overline{H(w,v)}$ is sesquilinear ($H'(cv,w) = \overline{\bar{c}H(w,v)} = cH'(v,w)$, $H'(v,cw) = \bar{c}H'(v,w)$) with $H'(v,v) = H(v,v)$, so $H' = H$ by (a).

(c) If L is unitary, take $w = v$. Conversely, $H_L(v,w) = H(Lv,Lw)$ is sesquilinear and agrees with H on all pairs (u,u); by (a), $H_L = H$. (Compare Observation 14.2.9, the special case $H(v,v) = 0$.) □

Chapter 15. The Spectral Theorem

Spectral theorems describe operators by orthonormal eigen-decompositions.

15.1 Notation and terminology

Motivation. One notation T^* covers transposes and adjoints; $T^* = T$, $T^*T = I$, $TT^* = T^*T$ define the special operators.

Notation 15.1.1

"$T \in \mathfrak{LM}$" means $T \in \mathfrak{L}(V, V)$ or $T \in \mathfrak{M}_{n,n}(F)$, with $\dim V = n$; a matrix T is regarded as L_T on $V = F^n$. We assume $V \neq 0$.

Notation 15.1.2

(a) For $F = \mathbb{C}$ and a non-degenerate Hermitian space (V, H), T^* means the adjoint $T^\star$; for a matrix A, $T^* = (L_A)^\star$. (b) For a non-degenerate quadratic space (V, B), T^* means the transpose T^t (usually written T^t).

Remark 15.1.3

(a) Dual maps do not appear again. (b) In $(\mathbb{C}^n, H)$, $[(L_A)^\star]_\mathcal{E} = \bar{J}^{-1}A^*\bar{J}$ ($J = [H]_\mathcal{E}$), which equals the adjoint matrix A^* when $\mathcal{E}$ is orthonormal (e.g. the Hermitian dot product). (c) Likewise $[(L_A)^t]_\mathcal{E} = J^{-1}A^tJ$ in (F^n, B), equal to A^t when $\mathcal{E}$ is orthonormal.

Observation 15.1.4

If $W \leq V$ is both T-invariant and T^*-invariant, then $(T|_W)^* = T^*|_W$.

Proof. Writing $\langle\,,\rangle$ for H or B: $\langle T|_W w, w'\rangle = \langle Tw, w'\rangle = \langle w, T^*w'\rangle = \langle w, T^*|_W w'\rangle$ for $w, w' \in W$; apply Propositions 14.3.11, 13.9.3 (on W, when $\langle\,,\rangle|_W$ is non-degenerate, e.g. for inner products). ■

Definition 15.1.5 (Special operators)

For $T \in \mathfrak{LM}$: (a) if $T^* = T$, T is **self-adjoint** (**Hermitian**) in the case (V, H), and **symmetric** in the case (V, B); (b) if $T^*T = I$, T is **unitary** (case (V, H)) or **orthogonal** (case (V, B)); (c) if $TT^* = T^*T$, T is **normal**. ($A \in \mathfrak{M}_{n,n}(\mathbb{C})$ is a **normal matrix** if $AA^* = A^*A$.)

Exercise 15.1.6

(a) Self-adjoint (symmetric) operators are normal.
(b) Unitary (orthogonal) operators are normal.

Solution. (a) $TT^* = T^2 = T^*T$.

(b) $T^*T = I$ forces T invertible with $T^{-1} = T^*$, so $TT^* = I = T^*T$. □

15.2 Normal operators

Motivation. On a Hermitian inner product space, orthonormally diagonalizable means normal.

Notation 15.2.1

Here $F = \mathbb{C}$ and $(V, \langle\,,\rangle)$ is an n-dimensional Hermitian inner product space.

Observation 15.2.2

If V has an orthonormal basis of eigenvectors of T, then T is normal.

Proof. $[T]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(\lambda_i)$ and $[T^*]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(\bar{\lambda}_i)$ (Proposition 14.3.12, $[H]_{\mathfrak{B}} = I$) commute. ■

Observation 15.2.3

T is normal iff $\langle Tv, Tw\rangle = \langle T^*v, T^*w\rangle$ for all v, w.

Proof. $\langle Tv, Tw\rangle = \langle v, T^*Tw\rangle$ and $\langle T^*v, T^*w\rangle = \langle v, TT^*w\rangle$; these agree for all v, w iff $T^*T = TT^*$. ■

Observation 15.2.4

If T is normal and $\lambda \in \mathbb{C}$, then $T - \lambda I$ is normal.

Proof. $(T - \lambda I)^* = T^* - \bar{\lambda} I$ (Observation 14.3.15, Exercise 14.3.16), and $(T - \lambda I)(T^* - \bar{\lambda} I) = TT^* - \bar{\lambda}T - \lambda T^* + |\lambda|^2 I = (T^* - \bar{\lambda} I)(T - \lambda I)$. ■

Observation 15.2.5

If T is normal and $Tv = \lambda v$, then $T^*v = \bar{\lambda} v$.

Proof. By Observations 15.2.3, 15.2.4: $0 = \langle (T - \lambda I)v, (T - \lambda I)v\rangle = \langle (T^* - \bar{\lambda} I)v, (T^* - \bar{\lambda} I)v\rangle$, so $(T^* - \bar{\lambda} I)v = 0$ by positive definiteness. ■

Theorem 15.2.6 (Spectral theorem, positive definite Hermitian case)

For $T \in \mathfrak{LM}$ (Notation 15.2.1) the following are equivalent: (1) T is normal; (2) V has an orthonormal basis of eigenvectors of T.

Proof. (2)⇒(1): Observation 15.2.2. (1)⇒(2): induct on n. Take a unit eigenvector $Tv = \lambda v$ and $W = \langle v\rangle^{\perp}$, $V = \langle v\rangle \oplus W$.

(i) W is T-invariant: for $u \in W$, $\langle v, Tu\rangle = \langle T^*v, u\rangle = \langle \bar{\lambda} v, u\rangle = \bar{\lambda}\langle v, u\rangle = 0$ (Observation 15.2.5).

(iii) W is T^*-invariant: $\langle v, T^*u\rangle = \langle Tv, u\rangle = \lambda\langle v, u\rangle = 0$.

(ii) By Observation 15.1.4, $(T|_W)^* = T^*|_W$, so $T|_W$ is normal. By induction W has an orthonormal eigenbasis of $T|_W$; add v. ■

Definition 15.2.7 (Maximal tori)

With $\mathbf{D}_n(\mathbb{C})$ the invertible diagonal matrices, $\mathbf{T}_{\mathbf{U}(n)} = \mathbf{U}(n) \cap \mathbf{D}_n(\mathbb{C})$ and $\mathbf{T}_{\mathbf{SU}(n)} = \mathbf{SU}(n) \cap \mathbf{D}_n(\mathbb{C}) = \mathbf{T}_{\mathbf{U}(n)} \cap \mathrm{SL}_n(\mathbb{C})$.

Exercise 15.2.8

(a) $\mathbf{T}_{\mathbf{U}(n)} = \{\mathrm{diag}(\lambda_1, \ldots, \lambda_n) : |\lambda_i| = 1\}$. (b) $\mathbf{T}_{\mathbf{U}(n)} \approx S^1 \times \cdots \times S^1 \approx \mathbf{U}(1) \times \cdots \times \mathbf{U}(1)$ (n factors). (c) $\mathbf{T}_{\mathbf{SU}(n)} \approx S^1 \times \cdots \times S^1$ ($n-1$ factors).

Solution. (a) A diagonal matrix has orthonormal columns iff $|\lambda_i| = 1$.

(b) $\mathrm{diag}(\lambda_i) \mapsto (\lambda_i)$.

(c) The last entry is determined by $\lambda_n = (\lambda_1 \cdots \lambda_{n-1})^{-1}$ (Exercise 11.9.13). □

Corollary 15.2.9 (Spectral theorem for $\mathbf{U}(n)$)

If $A \in \mathbf{U}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{U}(n)}$ for some $U \in \mathbf{U}(n)$. Hence $\mathbf{U}(n) = \bigcup_{U \in \mathbf{U}(n)} U\, \mathbf{T}_{\mathbf{U}(n)}\, U^{-1}$.

Proof. $(L_A)^\star = L_{A^*}$ for the dot product (Remark 15.1.3(b)), so L_A is unitary, hence normal (Exercise 15.1.6), with an orthonormal eigenbasis $\mathfrak{B}$ (Theorem 15.2.6). Then $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} = U^{-1}AU \in \mathbf{T}_{\mathbf{U}(n)}$ (Remark 10.6.4, Proposition 14.1.12) with $U = [I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{U}(n)$ (Exercise 10.6.9(5)). ■

Exercise 15.2.10

If $U \in \mathbf{U}(n)$, then $\alpha U \in \mathbf{SU}(n)$ for some $\alpha \in \mathbb{C}$.

Solution. $\det U = e^{i\theta}$; $\alpha = e^{-i\theta/n}$ gives $\det(\alpha U) = 1$, and $|\alpha| = 1$. □

Corollary 15.2.11 (Spectral theorem for $\mathbf{SU}(n)$)

If $A \in \mathbf{SU}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{SU}(n)}$ for some $U \in \mathbf{SU}(n)$; so $\mathbf{SU}(n) = \bigcup_{U \in \mathbf{SU}(n)} U\, \mathbf{T}_{\mathbf{SU}(n)}\, U^{-1}$.

Proof. Corollary 15.2.9 gives $U \in \mathbf{U}(n)$ with $U^{-1}AU \in \mathbf{T}_{\mathbf{U}(n)}$, of determinant 1; replace U by $\alpha U \in \mathbf{SU}(n)$ (Exercise 15.2.10). ■

Exercise 15.2.12

In Corollary 15.2.9, can U always be chosen in $\mathbf{SU}(n)$, even when $A \notin \mathbf{SU}(n)$?

Solution. Yes: replace U by αU as in the proof of Corollary 15.2.11; the conjugate is unchanged. □

Exercise 15.2.13

If $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ is normal ($AA^* = A^*A$), then $U^{-1}AU \in \mathbf{D}_n(\mathbb{C}) \cup \{\text{singular diagonal}\}$, i.e. $U^{-1}AU$ is diagonal, for some $U \in \mathbf{SU}(n)$.

Solution. L_A is normal for the dot product, so Theorem 15.2.6 gives an orthonormal eigenbasis; proceed as in Corollary 15.2.9 and Exercise 15.2.12. □

Exercise 15.2.14

For $A = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right) \in \mathbf{SU}(2)$, find $U \in \mathbf{SU}(2)$ with $U^{-1}AU \in \mathbf{T}_{\mathbf{SU}(2)}$.

Solution. Eigenvalues $e^{\pm i\pi/4}$, unit eigenvectors $\frac{1}{\sqrt{2}}(1,1)^t$, $\frac{1}{\sqrt{2}}(-1,1)^t$; $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$, $\det U = 1$, $U^{-1}AU = \operatorname{diag}(e^{i\pi/4}, e^{-i\pi/4})$. □

15.3 Symmetric operators

Motivation. Real symmetric operators are orthonormally diagonalizable; this diagonalizes quadratic forms.

Notation 15.3.1

Through Exercise 15.3.11, $F = \mathbb{R}$, $(V, \langle\,,\rangle)$ is an n-dimensional real inner product space, and $T^t = T^*$.

Proposition 15.3.2

(a) A symmetric operator T has a real eigenvalue.
(b) A real symmetric matrix has a real eigenvalue.

Proof. By §10.5 take $T = L_A$ on $(\mathbb{R}^n, \text{dot product})$, $A = A^t$. On $\mathbb{C}^n$ with the Hermitian dot product take $AX = \lambda X$, $X \neq 0$, $\lambda \in \mathbb{C}$. As $A^* = A$, Observation 10.7.1 gives $\lambda\langle X, X\rangle = \langle AX, X\rangle = \langle X, A^*X\rangle = \langle X, \lambda X\rangle = \bar{\lambda}\langle X, X\rangle$, so $\lambda = \bar{\lambda} \in \mathbb{R}$. ■

Observation 15.3.3

If V has an orthonormal basis of eigenvectors of T, then T is symmetric.

Proof. $[T^t]_{\mathfrak{B}}^{\mathfrak{B}} = ([T]_{\mathfrak{B}}^{\mathfrak{B}})^t = [T]_{\mathfrak{B}}^{\mathfrak{B}}$ (Proposition 13.9.4, $[B]_{\mathfrak{B}} = I$, diagonal). ■

Theorem 15.3.4 (Spectral theorem, positive definite real case)

For $T \in \mathfrak{LM}$ (Notation 15.3.1) the following are equivalent: (1) T is symmetric; (2) V has an orthonormal basis of eigenvectors of T.

Proof. (2)⇒(1): Observation 15.3.3. (1)⇒(2): as Theorem 15.2.6 without conjugations; Proposition 15.3.2 gives a real unit eigenvector v; $W = \langle v\rangle^{\perp}$ is T-invariant ($\langle v, Tu\rangle = \langle Tv, u\rangle = \lambda\langle v, u\rangle = 0$); $T|_W$ is symmetric; induct. ■

Corollary 15.3.5 (Spectral theorem for real symmetric matrices)

For $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ (dot product on $\mathbb{R}^n$) the following are equivalent: (1) A is symmetric; (2) $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of A; (3) $U^{-1}AU$ is diagonal for some $U \in \mathbf{O}(n)$.

Proof. Theorem 15.3.4 for L_A, with U the orthogonal matrix whose columns are the orthonormal eigenbasis. ■

Diagonalizing quadratic forms. With symmetric $J = (a_{ij})$, $2Q(X) = X^tJX$. Take $U \in \mathbf{O}(n)$ with $U^{-1}JU = D = \operatorname{diag}(\lambda_i)$; as $U^t = U^{-1}$, $X = UX'$ gives

$$2Q(X) = X^tUDU^{-1}X = (U^{-1}X)^tD(U^{-1}X) = \lambda_1(x_1')^2 + \cdots + \lambda_n(x_n')^2 :$$

the quadratic form is **diagonalized**.

Example 15.3.6

For $J = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $2Q(x,y) = 2x^2 + 2xy + 2y^2$, and $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) J \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right)$. Substituting $x = \frac{1}{\sqrt{2}}x' + \frac{1}{\sqrt{2}}y'$, $y = -\frac{1}{\sqrt{2}}x' + \frac{1}{\sqrt{2}}y'$ gives $2x^2 + 2xy + 2y^2 = (x')^2 + 3(y')^2$.

Example 15.3.7

In the eigenbasis $\{v, w\} = \{\frac{1}{\sqrt{2}}(1,-1)^t, \frac{1}{\sqrt{2}}(1,1)^t\}$, $X = x'v + y'w$, the curve $2x^2+2xy+2y^2 = 4$ is $(x')^2 + 3(y')^2 = 4$: an ellipse rotated clockwise by 45°.

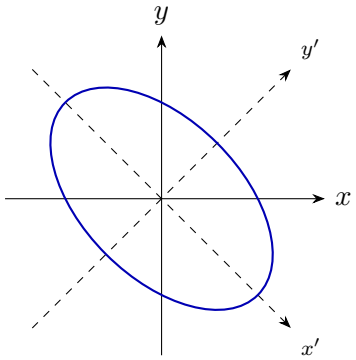


Figure. $2x^2 + 2xy + 2y^2 = 4$ *is the ellipse* $(x')^2 + 3(y')^2 = 4$ *in the rotated frame.*

Exercise 15.3.8

$x^2 + xy + y^2 - x - y + \frac{1}{3} \geq 0$ for all real x, y.

Solution. $x = u + \frac{1}{3}$, $y = v + \frac{1}{3}$ gives $u^2 + uv + v^2$, with matrix eigenvalues $\frac{3}{2}, \frac{1}{2} > 0$: $= \frac{3}{2}(u')^2 + \frac{1}{2}(v')^2 \geq 0$. □

Exercise 15.3.9

Sketch (a) $2x^2 - 2\sqrt{2}xy + 3y^2 = 4$;
(b) $x^2 - 6xy + y^2 = 2$;
(c) $11x^2 + 24xy + y^2 = 15$.

Solution. (a) $J = \left(\begin{smallmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 3 \end{smallmatrix}\right)$ has eigenvalues $1, 4$, with eigenvectors $(\sqrt{2}, 1)$, $(-1, \sqrt{2})$: the ellipse $(x')^2 + 4(y')^2 = 4$. (b) $J = \left(\begin{smallmatrix} 1 & -3 \\ -3 & 1 \end{smallmatrix}\right)$ has eigenvalues -2 (eigenvector $(1,1)$) and 4 (eigenvector $(-1,1)$): the hyperbola $4(y')^2 - 2(x')^2 = 2$. (c) $J = \left(\begin{smallmatrix} 11 & 12 \\ 12 & 1 \end{smallmatrix}\right)$ has eigenvalues 19 (eigenvector $(3,2)$) and -7 (eigenvector $(-2,3)$): the hyperbola $19(x')^2 - 7(y')^2 = 15$. □

Exercise 15.3.10

Why are the high-school conics (with centre at the origin) exactly ellipses (including circles) and hyperbolas, plus parabolas?

Solution. Rotating axes (Corollary 15.3.5), $X^t JX + \ell(X) + c = 0$ has quadratic part $\lambda_1(x')^2 + \lambda_2(y')^2$; completing squares gives an ellipse if $\lambda_1\lambda_2 > 0$, a hyperbola if $\lambda_1\lambda_2 < 0$, a parabola if one $\lambda_i = 0$ (or degenerate cases). □

Exercise 15.3.11

For a finite-dimensional real quadratic space (V, B) with basis $\mathfrak{B}$, the following are equivalent: (1) B is positive definite; (2) all eigenvalues of $J = [B]_{\mathfrak{B}}$ are positive.

Solution. With $X = [v]_{\mathfrak{B}}$ and $J = UDU^t$, $U \in \mathbf{O}(n)$ (Corollary 15.3.5): $B(v,v) = \sum \lambda_i (x_i')^2$, $X' = U^t X$ arbitrary, positive for $X' \neq 0$ iff all $\lambda_i > 0$. (Answers Question 13.1.14.) □

Exercise 15.3.12

For a finite-dimensional Hermitian space (V, H) with basis $\mathfrak{B}$, the following are equivalent: (1) H is positive definite; (2) all eigenvalues of $J = [H]_{\mathfrak{B}}$ are positive reals.

Solution. $H(v,v) = X^t J \bar{X} = \overline{\bar{X}^t \bar{J} X}$, $X = [v]_{\mathfrak{B}}$. $\bar{J}$ is self-adjoint, so unitarily diagonalizable (Theorem 15.2.6) with real eigenvalues (as in Proposition 15.3.2), the same as those of J. So $H(v,v) = \sum \lambda_i |x_i'|^2$, positive for $X \neq 0$ iff all $\lambda_i > 0$. □

15.4 Orthogonal operators

Motivation. A real orthogonal operator splits V into orthogonal invariant lines and planes.

Here $F = \mathbb{R}$.

Proposition 15.4.1

Every $T \in \mathfrak{LM}$ over $\mathbb{R}$ has an invariant subspace W with $\dim W = 1$ or 2.

Proof. Let $m_T = p_1 \cdots p_s$ (monic irreducibles in $\mathbb{R}[t]$, of degree 1 or 2). Some $0 \neq v$ has $p_1(T)v = 0$: any $v \neq 0$ if $s = 1$, else $v = p_2(T) \cdots p_s(T) w \neq 0$ by minimality. If $p_1 = t + c$, $Tv = -cv$; if $p_1 = t^2 + at + b$, $T^2 v = -aTv - bv$. So $W = \langle v, Tv \rangle$ is T-invariant. ■

Theorem 15.4.2 (Spectral theorem for real orthogonal operators)

If V is a finite-dimensional real inner product space and T is orthogonal, there are subspaces $W_1, \ldots, W_s$ of V with (1) $V = W_1 \oplus \cdots \oplus W_s$; (2) each W_i is T-invariant; (3) $W_i \perp W_j$ for $i \neq j$; (4) $\dim W_i \in \{1, 2\}$.

Proof. Induction on $\dim V$. Take W as in Proposition 15.4.1; $V = W \oplus W^{\perp}$. T is injective, so $TW = W$, and for $w' = Tw \in W$, $v \in W^{\perp}$: $\langle w', Tv \rangle = \langle w, T^t T v \rangle = \langle w, v \rangle = 0$. So $T|_{W^{\perp}}$ is orthogonal; induct. ■

Corollary 15.4.3 (Spectral theorem for $\mathbf{O}(n)$)

For $A \in \mathbf{O}(n)$ there are $A_i \in \mathbf{SO}(2)$ and $U \in \mathbf{O}(n)$ with
$A \in \mathbf{SO}(n)$, $n = 2m$: $U^{-1} A U = \operatorname{diag}(A_1, \ldots, A_m)$;
$A \in \mathbf{SO}(n)$, $n = 2m+1$: $U^{-1} A U = \operatorname{diag}(1, A_1, \ldots, A_m)$;
$A \notin \mathbf{SO}(n)$, $n = 2m$: $U^{-1} A U = \operatorname{diag}(-1, 1, A_1, \ldots, A_{m-1})$;
$A \notin \mathbf{SO}(n)$, $n = 2m+1$: $U^{-1} A U = \operatorname{diag}(-1, A_1, \ldots, A_m)$.

Proof. (i) Theorem 15.4.2 for L_A gives an orthonormal basis $\mathfrak{B}$ from the W_i, $[I]_{\mathcal{E}}^{\mathfrak{B}} \in \mathbf{O}(n)$ (Observation 9.3.6(5)), and $[I]_{\mathcal{E}}^{\mathfrak{B}} \, A \, [I]_{\mathfrak{B}}^{\mathcal{E}} = [L_A]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(B_1, \ldots, B_s) \in \mathbf{O}(n)$ with B_i of size 1 or 2.

(ii) $\operatorname{diag}(B_1,\dots,B_s)\in\mathbf{O}(n)$ iff each $B_i\in\mathbf{O}(1)$ or $\mathbf{O}(2)$. So (a) $B_i=\pm1$ if $\dim W_i=1$;

(b) a reflection $B_i\in\mathbf{O}(2)-\mathbf{SO}(2)$ has $U_i^{-1}B_iU_i=\operatorname{diag}(-1,1)$, $U_i\in\mathbf{O}(2)$ (§9.4), so we may take $B_i=\operatorname{diag}(-1,1)$.

(iii) Reordering (a permutation matrix in $\mathbf{O}(n)$) gives $\operatorname{diag}(\pm1,\dots,\pm1,C_1,\dots,C_r)$, $C_j\in\mathbf{SO}(2)$. Pairs of 1's or -1's merge into $R_0=I_2$ or $R_\pi=-I_2$, leaving at most one 1 and one -1; as $\det A=(-1)^{\#(-1)}$, reordering gives the four forms. ■

Definition 15.4.4 (Maximal torus of SO(n))

(a) For $n=2m$: $\mathbf{T}_{\mathbf{SO}(n)}=\{\operatorname{diag}(A_1,\dots,A_m):A_i\in\mathbf{SO}(2)\}$. (b) For $n=2m+1$: $\mathbf{T}_{\mathbf{SO}(n)}=\{\operatorname{diag}(1,A_1,\dots,A_m):A_i\in\mathbf{SO}(2)\}$.

Exercise 15.4.5

(a) $\mathbf{T}_{\mathbf{SO}(n)}\le\mathbf{SO}(n)$.
(b) $\mathbf{T}_{\mathbf{SO}(n)}\approx\mathbf{SO}(2)\times\cdots\times\mathbf{SO}(2)\approx S^1\times\cdots\times S^1$ ($m=\lfloor n/2\rfloor$ factors).

Solution. Blockwise products and inverses, $\det=\prod\det A_i=1$; $\operatorname{diag}(\dots,A_i,\dots)\mapsto(A_i)$, and $\mathbf{SO}(2)\approx S^1$ (Example 11.9.19). □

Corollary 15.4.6 (Spectral theorem for SO(n))

If $A\in\mathbf{SO}(n)$, then $U^{-1}AU\in\mathbf{T}_{\mathbf{SO}(n)}$ for some $U\in\mathbf{SO}(n)$. Hence $\mathbf{SO}(n)=\bigcup_{U\in\mathbf{SO}(n)}U\,\mathbf{T}_{\mathbf{SO}(n)}\,U^{-1}$.

Proof. Corollary 15.4.3 gives $U\in\mathbf{O}(n)$ with $U^{-1}AU=\operatorname{diag}(1,R_{\theta_1},\dots,R_{\theta_m})$ (no leading 1 for even n; $n\ge2$). If $\det U=-1$, let $S=\operatorname{diag}(1,I_2,\dots,I_2,\operatorname{diag}(-1,1))$; since $\operatorname{diag}(-1,1)R_\theta\operatorname{diag}(-1,1)=R_{-\theta}$ (Observation 9.5.9), $(US)^{-1}A(US)=\operatorname{diag}(1,R_{\theta_1},\dots,R_{\theta_{m-1}},R_{-\theta_m})\in\mathbf{T}_{\mathbf{SO}(n)}$, and $US\in\mathbf{SO}(n)$. ■

Exercise 15.4.7

In Corollary 15.4.3, can U be chosen in $\mathbf{SO}(n)$ even when $A\notin\mathbf{SO}(n)$?

Solution. Yes: replacing the basis vector u of the block -1 by $-u$ flips $\det U$ and leaves $U^{-1}AU$ unchanged. □

So in a suitable orthonormal frame an orthogonal operator rotates planes and flips at most one axis; Examples 9.6.7 and 9.6.10 are the typical cases.

Exercise 15.4.8

For $A\in\mathbf{SO}(n)$ below, find $U\in\mathbf{SO}(n)$ with $U^{-1}AU\in\mathbf{T}_{\mathbf{SO}(n)}$: (a) $A=\left(\begin{smallmatrix}0&1&0\\1&0&0\\0&0&-1\end{smallmatrix}\right)$;
(b) $A=\left(\begin{smallmatrix}0&1&0&0\\1&0&0&0\\0&0&0&-1\\0&0&-1&0\end{smallmatrix}\right)$.

Solution. (a) $Au_1=u_1$ for $u_1=\frac{1}{\sqrt2}(1,1,0)^t$, and $A=-I$ on $\langle u_2,u_3\rangle$, $u_2=\frac{1}{\sqrt2}(-1,1,0)^t$, $u_3=e_3$. $U=(u_1,u_2,u_3)$ has $\det U=1$ and $U^{-1}AU=\operatorname{diag}(1,R_\pi)$. (b) Eigenvalue 1: $u_1=\frac{1}{\sqrt2}(1,1,0,0)^t$, $u_2=\frac{1}{\sqrt2}(0,0,1,-1)^t$; eigenvalue -1: $u_3=\frac{1}{\sqrt2}(1,-1,0,0)^t$, $u_4=\frac{1}{\sqrt2}(0,0,1,1)^t$. With $U=(u_1,u_2,u_3,u_4)$, $U^{-1}AU=\operatorname{diag}(R_0,R_\pi)$; $\det U=1$ (if it were -1, replace u_4 by $-u_4$, which keeps $R_\pi=-I_2$). □

15.5 Epilogue

Motivation. Without positive definiteness the spectral theorems fail; Schur and the perp theorem survive.

§§15.2–15.4 needed Gram–Schmidt: if $Tv = \lambda v$ with $B(v,v) = 0$, $V = \langle v\rangle \oplus \langle v\rangle^\perp$ fails.

A false proof. "For normal T on non-degenerate (V,H), induct via $\bar{V} = V/\langle v\rangle$ with $\bar{H}(\bar{u},\bar{w}) = H(u,w)$." But $\bar{H}$ is well defined only if $v \perp V$. The statement is false:

Example 15.5.1

For $J = \operatorname{diag}(1,-1)$ and $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$ in $(\mathbb{R}^2, B_{\mathcal{E}}^J)$, L_A is symmetric (Exercise 13.9.6(b)), but $\phi_A(t) = (t-2)^2$ and $A \neq 2I$, so L_A is not diagonalizable.

Example 15.5.2

For $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ in $(\mathbb{C}^2, H_{\mathcal{E}}^J)$: $H_{\mathcal{E}}^J(A(x,y)^t,(z,w)^t) = (x+y)\bar{w} + y\bar{z} = H_{\mathcal{E}}^J((x,y)^t, A(z,w)^t)$, so L_A is self-adjoint, yet not diagonalizable.

Theorem 15.5.3 (Schur)

If $F = \mathbb{C}$ and $(V, \langle\,,\rangle)$ is an n-dimensional Hermitian inner product space, then for every $T \in \mathfrak{LM}$ there is an orthonormal basis $\mathfrak{B}$ with $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ upper triangular.

Proof. Take a unit eigenvector $T^*v = \mu v$ (for $T = L_A$, Observation 10.7.1 gives T^*). $\langle v\rangle^\perp$ is T-invariant: $\langle v, Tw\rangle = \langle T^*v, w\rangle = \mu\langle v,w\rangle = 0$. By induction it has an orthonormal basis $\mathfrak{C}$ triangularizing T; put $\mathfrak{B} = \mathfrak{C} \cup \{v\}$, v last. ∎

(The proofs of §7.3, §12.5 work over any algebraically closed field.)

Coding theory. $\mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$ is a field ($\bar{1}\cdot\bar{1} = \bar{1}$).

Definition 15.5.5 (Binary codes)

A $\mathbb{Z}_2$-subspace C of $(\mathbb{Z}_2)^n$ is a **linear binary code of length** n; with the ordinary inner product $B_{\mathcal{E}}^I$ on $(\mathbb{Z}_2)^n$, $C^\perp$ is the **dual code**.

Theorem 15.5.6

If C is a linear code in $(\mathbb{Z}_2)^n$, then $\dim C + \dim C^\perp = n$; so if $\dim C = m$, $|C| = 2^m$ and $|C^\perp| = 2^{n-m}$.

Proof. $B_{\mathcal{E}}^I$ is non-degenerate ($\det I = 1$): the perp theorem (Theorem 13.5.11 or 13.8.11) holds over any field, though Gram–Schmidt fails ($B_{\mathcal{E}}^I((\bar{1},\bar{1})^t,(\bar{1},\bar{1})^t) = \bar{0}$). ∎

Chapter 16. A Taste of Topology

Identify $\mathfrak{M}_{n,n}(\mathbb{R}) = \mathbb{R}^{n^2}$ and $\mathfrak{M}_{n,n}(\mathbb{C}) = \mathbb{C}^{n^2} = \mathbb{R}^{2n^2}$ (coordinates in any fixed order), with the Euclidean norm, distance and continuity.

16.1 Matrix group isomorphisms

Motivation. Matrix groups live in Euclidean space, so they have shapes as well as group structure.

Exercise 16.1.1

Under $\mathbb{C}^n = \mathbb{R}^{2n}$, the Hermitian dot product of $\mathbb{C}^n$ and the dot product of $\mathbb{R}^{2n}$ give the same norm (and metric).

Solution. $|a+ib|^2 = a^2 + b^2$ coordinatewise (compare Remark 10.2.20). □

Example 16.1.2

$\det : \mathfrak{M}_{n,n}(\mathbb{R}) \to \mathbb{R}$ is a polynomial map, hence continuous, so $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} - \{0\})$ is an open subset of $\mathbb{R}^{n^2}$; likewise $\mathrm{GL}_n(\mathbb{C})$ is open in $\mathbb{R}^{2n^2}$.

Example 16.1.3

$\mathrm{SL}_n(\mathbb{R}) = \det^{-1}(1)$ is a closed subset of $\mathbb{R}^{n^2}$; it is a **closed subgroup** of $\mathrm{GL}_n(\mathbb{R})$. Likewise $\mathrm{SL}_n(\mathbb{C})$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$.

Example 16.1.4

$\mathbf{O}(n) = \{A : A^tA - I = 0\}$ is the common zero set of n^2 polynomials in n^2 variables, hence closed; it is a closed subgroup of $\mathrm{GL}_n(\mathbb{R})$ (and a closed subset of $\mathfrak{M}_{n,n}(\mathbb{R})$).

Exercise 16.1.5

$\mathbf{U}(n)$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$.

Solution. $A^*A = I$ is a system of polynomial equations in the real and imaginary parts of the entries. □

Exercise 16.1.6

(a) $\mathbf{SO}(n)$ is a closed subgroup of $\mathbf{O}(n)$.
(b) $\mathbf{SU}(n)$ is a closed subgroup of $\mathbf{U}(n)$.

Solution. Each is cut out by an additional polynomial equation $\det A = 1$ (real and imaginary parts). □

Question 16.1.7

For $F = \mathbb{R}$ in Exercise 12.4.12, is $\operatorname{im}\psi$ a closed subgroup of $\mathrm{SL}_3(\mathbb{R})$?

Definition 16.1.8 ("Same shape")

For $X \subseteq \mathbb{R}^r$, $Y \subseteq \mathbb{R}^s$, a bijection $f : X \to Y$ with f and f^{-1} continuous is a **homeomorphism** ("same-shape map"); then X and Y have the **same shape** (are homeomorphic).

Exercise 16.1.9

"Same shape" is an equivalence relation.

Solution. Identities, inverses and composites of homeomorphisms are homeomorphisms. □

Exercise 16.1.10

If $f : X \to Y$ is a homeomorphism and $Z \subseteq X$: (a) Z open in $X \Rightarrow f(Z)$ open in Y; (b) Z closed in $X \Rightarrow f(Z)$ closed in Y.

Solution. $f(Z) = (f^{-1})^{-1}(Z)$, and preimages of (relatively) open/closed sets under the continuous map f^{-1} are open/closed. □

Example 16.1.11

$(a, b) \subseteq \mathbb{R}$ and $X = \{(x, 1)^t : a < x < b\} \subseteq \mathbb{R}^2$ have the same shape, although (a, b) is open in $\mathbb{R}$ and X is not open in $\mathbb{R}^2$. Any two open intervals have the same shape.

Example 16.1.12

$\exp : \mathbb{R} \to (0, \infty)$ is a homeomorphism (Example 11.5.7).

Example 16.1.13

$\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ is a homeomorphism.

Exercise 16.1.14

$[a, b)$ and $[a, \infty)$ have the same shape.

Solution. $x \mapsto a + \tan\left(\frac{\pi}{2} \cdot \frac{x-a}{b-a}\right)$ is a continuous bijection $[a, b) \to [a, \infty)$ with continuous inverse. □

So openness, closedness and boundedness are not shape invariants.

Exercise 16.1.15

A circle and an ellipse have the same shape. What about a circle and a square?

Solution. $(x, y) \mapsto (ax, by)$ maps the unit circle onto $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The square (its boundary) also has the same shape as the circle, via radial projection $v \mapsto v/\|v\|$ from the centre. □

Exercise 16.1.16

Explain how a topologist turns a soccer ball with a needle hole inside out.

Solution. A sphere with a small hole has the shape of a disk: stretch the hole, flip the disk, shrink the hole back. □

Exercise 16.1.17

(a) Do $\mathbb{R}$ and $\mathbb{R}^2$ have the same shape?
(b) $\mathbb{R}$ and S^1?

Solution. [Supplement] (a) No: removing one point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$, and a homeomorphism restricts to a homeomorphism of the complements.

(b) No: S^1 is compact and $\mathbb{R}$ is not (compactness is preserved, Proposition 16.2.1). □

Exercise 16.1.18

For a matrix group $G \leq \mathrm{GL}_n(F)$: (a) left multiplication $\lambda_g : G \to G$ is a homeomorphism (inverse $\lambda_{g^{-1}}$); likewise right multiplication; (b) inversion $x \mapsto x^{-1}$ is a homeomorphism of G; (c) for fixed g, $\mathrm{Int}(g) : G \to G$ is a homeomorphism.

Solution. Matrix multiplication is polynomial in the entries, hence continuous. Inversion is continuous by the adjugate formula $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$ and is its own inverse. $\mathrm{Int}(g) = \lambda_g \circ \rho_{g^{-1}}$. □

Observation 16.1.19

If H is a subgroup of a matrix group G and $g \in G$, then H and the coset gH have the same shape.

Proof. $\lambda_g(H) = gH$, and λ_g restricts to a homeomorphism $H \to gH$ (Exercise 16.1.18(a)). ■

Definition 16.1.20 (Matrix group isomorphism)

For matrix groups $G \leq \mathrm{GL}_n(F) \subseteq \mathbb{R}^N$ and $H \leq \mathrm{GL}_m(F') \subseteq \mathbb{R}^M$ ($F, F' \in \{\mathbb{R}, \mathbb{C}\}$, possibly different), $\varphi : G \to H$ is a **matrix group isomorphism** if it is a group isomorphism and a homeomorphism; then G and H are **isomorphic as matrix groups**, $G \approx_{\mathrm{mat}} H$.

Example 16.1.21

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a matrix group isomorphism.

Exercise 16.1.22

$\mathbf{SO}(2) \approx_{\mathrm{mat}} S^1 \approx_{\mathrm{mat}} \mathbf{U}(1)$.

Solution. $R_\theta \mapsto e^{i\theta}$, i.e. $\left(\begin{smallmatrix} x & -y \\ y & x \end{smallmatrix}\right) \mapsto x + iy$, is a group isomorphism (Exercise 11.9.20) and restricts a linear map with linear inverse; $S^1 = \mathbf{U}(1)$. □

Observation 16.1.23

If $H \leq G \leq \mathrm{GL}_n(F)$ and $g \in G$, then $H \approx_{\mathrm{mat}} gHg^{-1}$.

Proof. $\mathrm{Int}(g)$ restricts to a group isomorphism $H \to gHg^{-1}$, which is a homeomorphism by Exercise 16.1.18(c). ■

16.2 Compactness and connectedness

Motivation. $\mathbf{O}(n)$ and $\mathbf{U}(n)$ are compact, and the spectral theorems make $\mathbf{SO}(n)$, $\mathbf{U}(n)$ connected.

In $\mathbb{R}^N$, X is **compact** iff it is closed and bounded, and (for the sets considered here) **connected** means arcwise connected.

Proposition 16.2.1

If $f : X \to Y$ is continuous ($X \subseteq \mathbb{R}^N$, $Y \subseteq \mathbb{R}^M$): (a) X compact $\Rightarrow$ $f(X)$ compact; (b) X connected $\Rightarrow$ $f(X)$ connected.

Proof. Well known from analysis. ■

So compactness and connectedness are shape invariants.

Example 16.2.2

$\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{SL}_n(\mathbb{R})$ ($n \geq 2$) are not bounded in $\mathbb{R}^{n^2}$ (e.g. $\mathrm{diag}(t, t^{-1}, 1, \dots)$), hence not compact; likewise $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{SL}_n(\mathbb{C})$.

Example 16.2.3

For $A \in \mathbf{O}(n) \subseteq \mathbb{R}^{n^2}$, $\|A\| = \sqrt{n}$ (its columns are unit vectors, Observation 9.3.6(5)), so $\mathbf{O}(n)$ is bounded; it is closed (Example 16.1.4), hence a compact group.

Example 16.2.4

$\mathbf{SO}(n)$, a closed subgroup of $\mathbf{O}(n)$, is compact.

Exercise 16.2.5

$\mathbf{U}(n)$ and $\mathbf{SU}(n)$ are compact.

Solution. As in Examples 16.2.3, 16.2.4: $\|A\| = \sqrt{n}$ for $A \in \mathbf{U}(n)$ (Exercise 10.6.9(5), Exercise 16.1.1), and they are closed (Exercises 16.1.5, 16.1.6). □

Example 16.2.6

(a) $\mathbf{SO}^\circ(1,1)$ is unbounded in $\mathbb{R}^4$ ($\cosh t \to \infty$), so $\mathbf{SO}^\circ(1,1)$ and $\mathbf{O}(1,1)$ are not compact. (b) Alternatively, $\mathbf{SO}^\circ(1,1) \approx_{\mathrm{mat}} (\mathbb{R}, +)$ (Exercise 13.4.5; the inverse $\left(\begin{smallmatrix} x & y \\ y & x \end{smallmatrix}\right) \mapsto \operatorname{arsinh} y$ is continuous), and $\mathbb{R}$ is not compact. (c) So $\mathbf{SO}(2) \approx_{\mathrm{mat}} S^1$ (compact) and $\mathbf{SO}^\circ(1,1)$ are not isomorphic as matrix groups, nor even as groups: $\mathbf{SO}(2)$ has elements of every finite order, $\mathbf{SO}^\circ(1,1) \approx \mathbb{R}$ none (cf. Exercise 13.4.6).

Example 16.2.7

$\det : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times = \mathbb{R} - \{0\}$ is a continuous surjection and $\mathbb{R}^\times$ is not connected; so $\mathrm{GL}_n(\mathbb{R})$ is not connected.

Exercise 16.2.8

$\mathbf{O}(n)$ and $\mathbf{O}(1,1)$ are not connected.

Solution. det maps each continuously onto $\{\pm 1\}$, which is not connected. □

Proposition 16.2.9

For $X_i \subseteq \mathbb{R}^{N_i}$, identify $X_1 \times \cdots \times X_r \subseteq \mathbb{R}^{N_1+\cdots+N_r}$. (a) If all X_i are compact, so is the product.
(b) If all X_i are connected, so is the product.

Proof. Well known. ■

Definition 16.2.10 (Torus)

$\mathbf{T} \subseteq \mathbb{R}^N$ is a **torus** if it has the same shape as $S^1 \times \cdots \times S^1$. A torus is compact and connected; $S^1 \times S^1$ has the shape of a doughnut.

Exercise 16.2.11

$\mathbf{T}_{\mathbf{U}(n)}$, $\mathbf{T}_{\mathbf{SU}(n)}$ (Definition 15.2.7) and $\mathbf{T}_{\mathbf{SO}(n)}$ (Definition 15.4.4) are tori.

Solution. The isomorphisms of Exercises 15.2.8, 15.4.5 are coordinate maps with continuous inverses. □

Corollary 16.2.12 (Corollary of the spectral theorems)

$\mathbf{U}(n)$, $\mathbf{SU}(n)$ and $\mathbf{SO}(n)$ are connected groups.

Proof. For $\mathbf{SO}(n)$ (the others likewise via Corollaries 15.2.9, 15.2.11): $\mathbf{SO}(n) = \bigcup_U U\,\mathbf{T}_{\mathbf{SO}(n)}\,U^{-1}$ (Corollary 15.4.6), a union of connected sets (Observation 16.1.23, Exercise 16.2.11) all containing I. ■

Every $X \subseteq \mathbb{R}^N$ is uniquely the disjoint union of its **connected components** (maximal connected subsets), its **component decomposition**.

Observation 16.2.13

(a) $\mathbf{SO}(n)$ is a connected component of $\mathbf{O}(n)$.
(b) The coset decomposition $\mathbf{O}(n) = \mathbf{SO}(n) \amalg (\mathbf{O}(n) - \mathbf{SO}(n))$ is the component decomposition of $\mathbf{O}(n)$.

Proof. $\mathbf{SO}(n)$ is connected (Corollary 16.2.12), and so is $\mathbf{O}(n) - \mathbf{SO}(n) = S \cdot \mathbf{SO}(n)$ (Observation 16.1.19). No connected subset can meet both, since det would map it continuously onto $\{\pm 1\}$. ■

Observation 16.2.14

(a) $\mathbf{SO}^\circ(1,1)$ is a connected component of $\mathbf{O}(1,1)$.
(b) The coset decomposition of $\mathbf{O}(1,1)$ modulo $\mathbf{SO}^\circ(1,1)$ (§13.4) is its component decomposition.

Proof [Supplement]. $\mathbf{SO}^\circ(1,1) \approx \mathbb{R}$ and its cosets are connected (Observation 16.1.19), and the continuous det and $\operatorname{sign}(A_{11})$ ($|A_{11}| \geq 1$) separate the four cosets. ■

(The identity component is always a subgroup; proof postponed.) So $\mathbf{O}(2)$ is two circles and $\mathbf{O}(1,1)$ four lines.